Category Theory and the Injective Generation Condition

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Acknowledgement of Sources

For all ideas taken from other sources (books, articles, internet), the source of the ideas is mentioned in the main text and fully referenced at the end of the report.

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This project compiles with the guidelines in "School of Mathematics: Undergraduate Project Handbook 2023/24".

Abstract

This report provides an overview of key concepts in category theory and uses them to investigate some notions in noncommutative, finite-dimensional algebras. This includes an accessible account of derived categories and quiver representations. In particular, we introduce the "injective generate" condition and show injective generates for the algebra in [GH96, Example 2.8], using techniques similar from Rickard [Ric19, §6]

Acknowledgement

Finishing this report means a lot to me personally. It signifies my four-year journey of mathematics exploration: connecting the dots and crosses, having moments of delight and despair, and trying my best to pen down the essence of it. This would have been impossible without so many people who had walked alongside me and provided me with massive support.

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Introduction

Category Theory is a common language in modern mathematical research to investigate different structures. Popularized by the famous book of Mac Lane [Mac78], it has various applications in vastly different branches of mathematics. In this project, we provide a concise overview of some of the key topics and apply them to the field of abstract algebra.

Part I of the paper will be about basic category theory. We will go through basic definitions and various examples. The purpose of this part is to familiarize readers with the above and invite readers to *think categorically*. In category theory, various tools are used to simplify proof-writing, including commutative diagrams and universal properties. Learning these techniques is very useful for reasoning in other fields of mathematics. This part will follow [Lei14] very closely.

In Part II, we focus on algebra, where the categorical definition will reappear to streamline various definitions in Algebra. In particular, we focus on a result of Rickard [Ric19], which says if the derived category of an algebra satisfies the "injective generates" condition, then it satisfies the "finitistic dimensional conjecture". We show an algebra defined in [GH96] that satisfies the "injective generates" condition. It applies the methods in [Ric19, §6] to a new algebra. We will use two chapters (namely Chapter 4 and 5) to cover all of the basics needed to state the example minimally in Chapter 6.

Readers of this paper are expected to have a first course in abstract algebra and linear algebra. In particular, we will assume basic working knowledge of some constructions in vector spaces and ring theory, such as direct sums, quotient spaces, dual spaces, vector subspaces, and ideals. By a ring R, we always mean the quintuple $(R, +, \times, 0, 1)$ with the basic distributive and associative properties. Ring homomorphisms will send 0 to 0 and 1 to 1.

Note that we do *not* require a ring to be commutative, so there will be a distinction between *left* ideals and *right* ideals. Given a set I, we say I is a **left ideal** of R if for all $r \in R$, $i \in I$, we must have $ri \in I$. Similarly, I is a **right ideal** if for all $r \in R$, $i \in I$, we must have $ir \in I$. By an *ideal* of R we mean a subset of R that is both a left and a right ideal.

Throughout the paper, we will have some technical assumptions or notation choices that are highlighted. For example:

Important Working Assumption

We assume the axiom of choice.

This asserts every ring has a maximal ideal, and it ensures every vector space has a basis, and will be useful in some constructions in which we will not go into detail. The end of proofs is marked with a $\square$, while the end of examples and remarks are marked with a $\triangle$ and $\Diamond$ respectively.

Part I

An Overview of Basic Category Theory

Chapter 1

Building Blocks of Categories

1.1 Categories

The definition of a category is rather long and terse. Still, we would start by introducing it and some very basic concepts.

Definition 1.1.1. A **(locally small) category** $\mathscr{C}$ consist of the following data:

- A 'collection' of **objects**, denoted by $\mathrm{Ob}(\mathscr{C})$.
- For any two object A, B in $\mathscr{C}$, a set $\mathrm{Hom}_{\mathscr{C}}(A, B)$ called the set of **morphisms from** A **to** B.
- For any object A, B, C in $\mathscr{C}$, a set-function called **composition rule** denoted by

$$\circ : \mathrm{Hom}_{\mathscr{C}}(B, C) \times \mathrm{Hom}_{\mathscr{C}}(A, B) \to \mathrm{Hom}_{\mathscr{C}}(A, C).$$

  More succinctly, if $g \in \mathrm{Hom}_{\mathscr{C}}(B, C)$, $f \in \mathrm{Hom}_{\mathscr{C}}(A, B)$, then we denote the composition rule send (g, f) to $g \circ f$.
- For any object A in $\mathscr{C}$, a morphism in $\mathrm{Hom}_{\mathscr{C}}(A, A)$ called **identity**, denoted by 1_A.

We further require the data to satisfy the following conditions:

- (Distinct Morphisms.) For any distinct pairs (A, B), (C, D) of objects in $\mathscr{C}$, $\mathrm{Hom}_{\mathscr{C}}(A, B) \cap \mathrm{Hom}_{\mathscr{C}}(C, D) = \emptyset$.
- (Associative Law.) For every morphism f, g, h from any object to any object, we have $h \circ (g \circ f) = (h \circ g) \circ f$, (where the composition rule is defined).
- (Identity Law.) For any object A in $\mathscr{C}$ and its identity 1_A, the following is satisfied:
  - For any object B in $\mathscr{C}$, and for any $f \in \mathrm{Hom}_{\mathscr{C}}(A, B)$, we have $f \circ 1_A = f$.
  - For any object C in $\mathscr{C}$, and for any $g \in \mathrm{Hom}_{\mathscr{C}}(B, A)$, we have $1_A \circ g = g$.

Remark.

- The reason why the word 'collection' is in quotes, is because $\mathrm{Ob}(\mathscr{C})$ is generally not a set, as we will see in the category $\mathbf{Set}$ in Example 1.1.4. That being said, we will treat everything as a set. That is, we are happy to discuss memberships, and elements in the collection of objects, and define usual set-theoretic constructions on it.

Notation	Objects	Morphisms
$\mathbf{Grp}$	Groups	Group Homomorphisms
$\mathbf{Ab}$	Abelian Groups	Group Homomorphisms
$\mathbf{Vect}_k$	Vector spaces over a fixed field k	Linear Maps
$\mathbf{FDVect}_k$	Finite-dimensional vector spaces over a fixed field k	Linear Maps
$\mathbf{Top}$	Topological Spaces	Continuous Maps
$\mathbf{Ring}$	Commutative Rings with unity	Ring Homomorphisms
$\mathbf{Fld}$	Fields	Field Homomorphisms
R-$\mathbf{Mod}$	Right R-modules, for a fixed ring R	R-Module Homomorphisms

Table 1.1: A table of common categories. In all categories above, the composition rules are function compositions, and the identity morphisms are the identity functions.

- We will simplify notation a lot in subsequent writings. For instance, we will often omit the $\mathscr{C}$ in $\mathrm{Hom}_{\mathscr{C}}(A, B)$, if it arises no confusion about what category $\mathscr{C}$ has A, B as objects. Moreover, we will write $A \in \mathscr{C}$ as a shorthand for $A \in \mathrm{Ob}(\mathscr{C})$.
- A morphism $f \in \mathrm{Hom}_{\mathscr{C}}(A, B)$ is often (interchangeably) called an *arrow*, it serves as an intuitive term consistent with commutative diagrams and building finite categories. However, note that the word *function* (or *set-function*) will always mean function from a set to another set.
- Sometimes we will denote $f \in \mathrm{Hom}_{\mathscr{C}}(A, B)$ as $f \colon A \to B$, consistent with functions. Note that in this notation we understood the objects are in $\mathscr{C}$ implicitly.

$\Diamond$

When we are working with a general category $\mathscr{C}$, we will always assume it is locally small in this paper (i.e. $\mathrm{Hom}_{\mathscr{C}}(A, B)$ is a set for all $A, B \in \mathrm{Ob}(\mathscr{C})$) unless otherwise specified. There is also a useful definition:

Definition 1.1.2. A category $\mathscr{C}$ is **small** if $\mathrm{Ob}(\mathscr{C})$ is a set.

There is an immediate consequence from the definition.

Proposition 1.1.3. In any category $\mathscr{C}$, for any object A in $\mathscr{C}$, the morphism 1_A is always unique.

Proof. Suppose there are two identity morphisms 1_A and $\widetilde{1_A}$, by following the two bullet points of the identity law in $\mathrm{Hom}(A, A)$ we have $1_A \circ \widetilde{1_A} = 1_A$ and $1_A \circ \widetilde{1_A} = \widetilde{1_A}$, hence $1_A = \widetilde{1_A}$. $\square$

As promised, we have various examples of categories that we know of.

Example 1.1.4. There is a category $\mathbf{Set}$ where the objects are sets, morphisms of A to B are the set of all functions $A \to B$, the identity 1_A for a set (object) A is the usual identity function. The composition rule is the usual composition of functions. Note that, $\mathrm{Ob}(\mathbf{Set})$ is the collection of all sets, which is not a set by the famous 'Russell's Paradox'. So $\mathbf{Set}$ is locally small but not small.

Similarly, we can introduce the category of finite set $\mathbf{FinSet}$, where all the data are the same except the objects are finite sets. $\triangle$

Example 1.1.5. Similar to $\mathbf{Set}$, other categories have objects as special sets, morphisms, and structure-preserving functions as mentioned in the introduction, we now provide some common categories and their notation in Table 1.1. $\triangle$

With the Table 1.1, one might expect objects and morphisms are always sets and functions. The following example shows this need not be the case, which is one of the strengths that makes category theory versatile.

Example 1.1.6. Consider the following category which we denote as $\mathbb{R}_{\leqslant}$: its objects are real numbers. For any real number $x, y \in \mathbb{R} = \mathrm{Ob}(\mathbb{R}_{\leqslant})$, define

$$\mathrm{Hom}_{\mathbb{R}_{\leqslant}}(x,y) = \begin{cases} \{(x,y)\} & \text{if } x \leqslant y \\ \emptyset & \text{otherwise} \end{cases}.$$

That is, for two objects (real numbers), there is exactly one morphism if one is not larger than the other. It is easy to check that this is a category. In particular, there is only one way of defining composition. The well-definedness of it and the identity condition is equivalent to $\leqslant$ being transitive and reflexive. This leads to a general definition. △

Definition 1.1.7. Given a set X and a partial order $\leqslant$ (a relation that is reflexive, anti-symmetric and transitive) on X, we can define the **poset category** $X_{\leqslant}$ to be $X = \mathrm{Ob}(X_{\leqslant})$, and

$$\mathrm{Hom}_{X_{\leqslant}}(x,y) = \begin{cases} \{(x,y)\} & \text{if } x \leqslant y \\ \emptyset & \text{otherwise} \end{cases}.$$

This definition applies to many examples, with the most notable one as power sets with inclusion.

In the poset category, each distinct pair of elements has at most one morphism. On the contrary, there is a category where there is only one object and a lot of morphisms, the answer is so fundamental that it might seem surprising.

Example 1.1.8. Any group G is a category in the following way. There is only one object $\{\bullet\} = \mathrm{Ob}(G)$, and the morphisms are elements in G (i.e. $\mathrm{Hom}_G(\bullet,\bullet) = G$). The composition rules are defined as the group operation. This is because we only need to define one composition rule $\mathrm{Hom}_G(\bullet,\bullet) \times \mathrm{Hom}_G(\bullet,\bullet) \to \mathrm{Hom}_G(\bullet,\bullet)$, which just translates to a function $G \times G \to G$. This composition rule satisfies all the axioms (while the distinct morphism condition is vacuously true). △

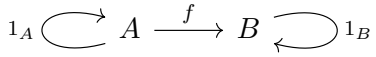
Example 1.1.9. Fix a field k. Define a category $\mathscr{C}$ with the following data.

- $\mathrm{Ob}(\mathscr{C}) = \{0, 1, 2, \cdots\}$.
- Given two integer m, n, we define $\mathrm{Hom}(m,n)$ to be all the $n \times m$ matrices with entries in field k.
- The composition rule is defined as follows: given $G \in \mathrm{Hom}(n,\ell)$, $F \in \mathrm{Hom}(m,n)$, then the composition $G \circ F$ is defined as the matrix multiplication product $GF \in \mathrm{Hom}(m,\ell)$.

We can check this is indeed a category, as the associative law is satisfied since matrix multiplication is associative. The identity is the $n \times n$ identity matrix in $\mathrm{Hom}(n,n)$ and satisfies the identity law. We will see this category contains the same information as $\mathbf{FDVect}_k$ in a precise way in Example 2.2.2. △

With enough concrete examples in mind, we move on to more abstract examples.

Example 1.1.10. Given objects A, B (just in an abstract sense), how can we build a category $\mathscr{C}$ out of them? Firstly by definition, there must be identity morphisms $1_A \in \mathrm{Hom}_{\mathscr{C}}(A,A), 1_B \in \mathrm{Hom}_{\mathscr{C}}(B,B)$. Now say there is one morphism $f\colon A \to B$ from A to B, we can visualize all the information in a diagram below.



This diagram commutes (since $1_A \circ f = f, f \circ 1_B = 1_B$ by the definition of identity functors). By convention, since the identity functors $1_A, 1_B$ always exist, when we depict a category as a diagram, we will often omit them. thus, the category $\mathscr{C}$ describe above will just be denoted simply as $A \xrightarrow{f} B$. This is not the only way we can turn two objects into a category, the diagrams below denote three more such possibilities.

$$A \underset{g}{\overset{f}{\rightleftarrows}} B \qquad\qquad A \underset{f_2}{\overset{f_1}{\rightrightarrows}} B \qquad\qquad A \qquad\qquad B$$

All the categories outlined here will be very useful in later discussions. In particular, the rightmost category, where there are no morphisms except the identities, is called the **discrete category** (with two objects in this case). △

Example 1.1.11. As with sets, if we are given (small) category $\mathscr{C}, \mathscr{D}$ we can form the **product category** $\mathscr{C} \times \mathscr{D}$, for which objects are ordered pairs in the set $\mathrm{Ob}(\mathscr{C}) \times \mathrm{Ob}(\mathscr{D})$, with morphisms being the set $\mathrm{Hom}_{\mathscr{C}}(A, B) \times \mathrm{Hom}_{\mathscr{D}}(A', B')$ for any two objects $(A, A'), (B, B') \in \mathrm{Ob}(\mathscr{C} \times \mathscr{D})$. The identity is $(1_A, 1_B)$ for $(A, B) \in \mathscr{C} \times \mathscr{D}$, and the composition is the composition in each entry. △

Example 1.1.12. Given a category $\mathscr{C}$, we can construct the **opposite category** $\mathscr{C}^{\mathrm{op}}$, where the objects and the identity morphisms are the same, but $\mathrm{Hom}_{\mathscr{C}^{\mathrm{op}}}(A, B) := \mathrm{Hom}_{\mathscr{C}}(B, A)$ for all objects A, B. For the composition rule, fix three objects $X, Y, Z \in \mathrm{Ob}(\mathscr{C}) = \mathrm{Ob}(\mathscr{C}^{op})$ given the composition rule $g \circ f$ where $g \in \mathrm{Hom}_{\mathscr{C}}(Y, Z), f \in \mathrm{Hom}_{\mathscr{C}}(X, Y)$, we define the composition rule in C^{op} to be

$$\mathrm{Hom}_{\mathscr{C}^{op}}(Y, Z) \times \mathrm{Hom}_{\mathscr{C}^{op}}(X, Y) \qquad (\widetilde{f}, \widetilde{g}) = \widetilde{g} \circ \widetilde{f}.$$

△

Informally, we can see the opposite category as "reversing arrows" for all morphisms in the category.

Example 1.1.13. In Example 1.1.8 we see a single group G as a category with group operation $\circ$, the opposite category G^{op} has a new definition of composition, hence a new group operation. Tracing the definition in Example 1.1.12 we can find the new operation $\bullet$ is defined as $g \bullet h = h \circ g$. Note that these two groups are isomorphic (as in usual group theory, that is, isomorphism in the category $\mathbf{Grp}$) because there is an isomorphism $\phi(g) = g^{-1}$ from G to G^{op}. △

The opposite category also introduces a powerful idea, informally named "**dualizing**". The idea is as follows: if a statement is applicable for any Category $\mathscr{C}$, applying the result to the category $\mathscr{C}^{\mathrm{op}}$ produces another statement in terms of $\mathscr{C}$ by repeatedly applying $\mathrm{Hom}_{\mathscr{C}^{\mathrm{op}}}(A, B) = \mathrm{Hom}_{\mathscr{C}}(B, A)$.

In group theory, we treat isomorphisms as "the same": that is, abstract groups like $\{e, g\}$ with $g^2 = e$ and $\{1, -1\}$ on real numbers with multiplication is the "same" group, although taken in a very formal sense they are not the same object in $\mathbf{Grp}$, because they are different set, but there is an isomorphism between the two sets. We want to formulate this idea in category theory in general, and the definition would not be surprising:

Definition 1.1.14. Let A, B be objects in $\mathscr{C}$, we say A, B are **isomorphic**, denoted by $A \cong B$, if there are morphisms $f \in \mathrm{Hom}(A, B), g \in \mathrm{Hom}(B, A)$ such that $gf = 1_A, fg = 1_B$. We will also refer to f or g in this definition as an **isomorphism**.

Sometimes, we will denote g as f^{-1}, again consistent with the setting with functions.

Example 1.1.15. In the category $\mathbf{Set}$, the isomorphisms are exactly the bijections. △

Example 1.1.16. Consider Table 1.1 of common categories, the isomorphisms are exactly the isomorphisms in their respective definition. For example, isomorphisms in $\mathbf{Grp}$ are exactly group isomorphisms. In particular, note that isomorphisms in $\mathbf{Top}$ are homeomorphisms, not simply bijections. This is because we need the inverse f^{-1} to be continuous (an element in the set of morphisms) in the definition of an isomorphism. △

Here are some obvious facts on $\cong$ that will help us.

Proposition 1.1.17. The inverse of isomorphisms is unique. That is, given A, B as isomorphic objects in a category $\mathscr{C}$ and exists $f : A \to B$, $g : B \to A$ such that $g \circ f = 1_A, f \circ g = 1_B$, then g is unique (if f is fixed).

Proof. Suppose we have morphisms $g, \widetilde{g} \in \mathrm{Hom}_{\mathscr{C}}(B, A)$ such that $g \circ f = \widetilde{g} \circ f = 1_A$, then

$$\widetilde{g} \circ f \circ g = 1_A \circ g, \text{ which means } \widetilde{g} = g$$

by the associativity and property of 1_A in $\mathscr{C}$. □

One can also easily check that $\cong$ is an equivalence relation.

1.2 Subcategories and Functors

When defining a new mathematical structure, oftentimes times a sensible question is to investigate substructures and nice maps between two such structures. This is no exception for categories too.

Definition 1.2.1. Given a Category $\mathscr{C}$, a **subcategory** $\mathscr{S}$ of $\mathscr{C}$ is a category such that it has morphisms and objects in $\mathscr{C}$, and is closed under identity, compositions, and isomorphisms. That is, the following three conditions hold for $\mathscr{S}$:

1. For any object $A \in \mathscr{S}$, $1_A \in \mathrm{Hom}_{\mathscr{S}}(A, A) \in \mathscr{S}$.
2. If f, g are morphisms in $\mathscr{S}$, then $f \circ g \in \mathscr{S}$ (whenever it is defined),
3. If X, Y are isomorphic objects in $\mathscr{C}$, then $X \in \mathscr{S}$ if and only if $Y \in \mathscr{S}$

Definition 1.2.2. Let $\mathscr{S}$ be a subcategory of a category $\mathscr{C}$. If for all $X, Y \in \mathscr{S}$, we have $\mathrm{Hom}_{\mathscr{S}}(X, Y) = \mathrm{Hom}_{\mathscr{C}}(X, Y)$, then $\mathscr{S}$ is called a **full subcategory** of $\mathscr{C}$

So to define a full subcategory $\mathscr{S}$ given the original category $\mathscr{C}$, it suffices to define what objects are in $\mathscr{S}$.

Example 1.2.3. We can define $\mathbf{FDVect}_k$ to be the full subcategory of $\mathbf{Vect}_k$ consisting only of finite-dimensional vector spaces. △

Definition 1.2.4. Given two categories $\mathscr{C}, \mathscr{D}$. A **(covariant) functor** F from $\mathscr{C}$ to $\mathscr{D}$, written as $F : \mathscr{C} \to \mathscr{D}$, is a function from object to object, morphisms to morphisms that satisfies certain conditions. Formally, a functor consists of:

- For each object A in $\mathrm{Ob}(\mathscr{C})$, assign to a object $F(A)$ in $\mathrm{Ob}(\mathscr{D})$, and
- For each morphism $f \in \mathrm{Hom}_{\mathscr{C}}(A, B)$, a morphism $F(f) \in \mathrm{Hom}_{\mathscr{D}}(F(A), F(B))$.

Moreover, the following conditions must be satisfied:

- $F(1_A) = 1_{F(A)}$ for all objects A in $\mathscr{C}$, and

- $F(gf) = F(g)F(f)$ for all morphisms g, f (where $g \circ f$ is defined).

Remark. Note the difference in composition in the second condition as we start to suppress the composition notation. The composition on the left hand side is in $\mathscr{C}$, while the composition on the right hand side is in $\mathscr{D}$. ◊

Dualizing the above definition, we have:

Definition 1.2.5. A **contravariant functor** F **from** $\mathscr{C}$ **to** $\mathscr{D}$ is a covariant functor $F : \mathscr{C}^{\mathrm{op}} \to \mathscr{D}$, where $\mathscr{C}^{\mathrm{op}}$ is the opposite category of $\mathscr{C}$. In other words, a morphism $f : A \to B$ is sent to $F(f) : F(B) \to F(A)$ and $F(1_A) = 1_{F(A)}$ with $F(gf) = F(f)F(g)$

As with the previous section, we start with some concrete examples of functors.

Example 1.2.6. Here is an example of a functor $F : \mathbf{Set} \to \mathbf{Set}$, for each set A, define $F(A) = A \times A$, the Cartesian product. Also, for each $f \in \mathrm{Hom}(X, Y)$, that is, when $f : X \to Y$ is a function, define $F(f) : X \times X \to Y \times Y$ as

$$F(f)(x_1, x_2) := (f(x_1), f(x_2)).$$

△

Example 1.2.7. Here we introduce a pair of functors $F : \mathbf{Set} \to \mathbf{Vect}_k, U : \mathbf{Vect}_k \to \mathbf{Set}$. We first define F on sets S by $F(S)$ being the vector space over k with basis S. That is,

$$F(S) = \left\{ \sum_{s \in S} \alpha_s s : \alpha_s \in k \text{ for any } s \in S \right\}$$

note that the sum $\sum_{s \in S}$ is purely formal: that is, one can have an infinite sum (and addition is defined) if S is an infinite set. Now, given $f : S \to T$ as a function, we define $F(f) : F(S) \to F(T)$ to be the linear function that sends s to $f(s)$, note that this determines $F(f)$ uniquely since to define a linear function between vector spaces, it suffices to define the function on the basis elements of the domain. F is sometimes informally referred as the **free functor**.

Now we define $U : \mathbf{Vect}_k \to \mathbf{Set}$, given a vector spaces V, define $U(V)$ to be the *set* V, and for any linear map $f : V \to W$, $U(f)$ is the function f, considered as a function from the *set* V to the *set* W. Colloquially, we say the functor U "forgets" V is a vector space, and hence U is sometimes informally referred to as the **forgetful functor**.

This pair of functor F, U is not a random choice: F "gives" a structure of vector spaces for any set S, and U "forgets" the structure. There is a relationship called *adjointness* between these functors, introduced in Chapter 3. △

Example 1.2.8. We look at yet another important example. consider the following functor $\mathbf{FDVect}_k \to \mathbf{FDVect}_k$: it sends a vector space X to its double dual X^{**}, and sends a morphism of vector spaces (linear map) $f : X \to Y$ to $f^{**} : X^{**} \to Y^{**}$ by

$$f^{**}(\gamma) := \text{The map } \eta \mapsto \gamma(\eta \circ f).$$

(where $\gamma \in X^{**}, \eta \in Y^*$, one can check f^{**} is linear.) One can check that this is a functor. In particular, note that this definition does not require having a choice of basis in X or Y. This is related to the concept of *naturality* which will be discussed in Section 1.3. This functor is usually denoted by $(-)^{**}$. Note the use of "$-$" as a placeholder, as this is a common notation that will play an important role in later definitions. △

Example 1.2.9. In Example 1.1.8 we see a group G as a category, and we can ask what information a functor $F : G \to \mathbf{Set}$ entails. Firstly, note that G has only one object $\bullet$, so F applied to $\bullet$ is a set, which we will call X. Now, for each morphism in G, we need to assign a morphism in $\mathbf{Set}$. In other words, for every group element $g : \bullet \to \bullet$, $F(g)$ is a set function $f_g : X \to X$.

Now, we trace the definition for F to be a functor. The first condition $F(1_A) = 1_{F(A)}$ means the identity transformation in G (i.e. the identity element), is mapped to the identity transformation of X. The second condition is $F(hg) = F(h)F(g)$ for any $g, h \in G$. Putting it all together, since we have a collection of functions $f_g : X \to X$ depend on G, we have a function $F : G \times X \to X$ (denoting $f_g(x) = F(g, x)$), which:

- If e is the identity element of G, then $F(e, x) = x$ for all $x \in X$, and:
- $f_h \circ f_g = f_{hg}$, that is, $F(h, F(g, x)) = F(hg, x)$ for any $x \in X, g, h \in G$.

Write $F(g, x) = g \cdot x$, this is exactly the definition of a group G acts on a set X. In other words, a functor $F : G \to \mathbf{Set}$ specifies a group action. △

Example 1.2.10. By the similar reasoning above, a functor $\rho : G \to \mathbf{Vect}_k$ will capture how G acts linearly on a vector space X. ρ is what called a **representation** of G. An introduction to representation theory can be found in [Ste12]. △

Example 1.2.11. Recall the poset category $X_\leqslant$ in Definition 1.1.7. What is a functor between two poset categories $F : X_\leqslant \to Y_\ll$? Tracing the definition of functors, it assigns any element (objects in $X_\leqslant$) in X to an element in Y (objects in $Y_\ll$), hence it can be regarded as a function. But the fact that $F(f)$ *is an arrow in* $Y_\ll$ is exactly the fact that $x \leqslant y$ implies $F(x) \ll F(y)$. This means F can be viewed as an *ordering preserving function*. △

Having seen some examples of functors, we move on to building functors from an abstract category $\mathscr{C}$, the following will be the most important functor throughout this project, but for some reason, it does not have a name.

Definition 1.2.12. Given a category $\mathscr{C}$, we define the following functor $\mathrm{Hom}_\mathscr{C}(A, -) : \mathscr{C} \to \mathbf{Set}$ for any $A \in \mathscr{C}$.

- Given any object $B \in \mathscr{C}$, the functor sends it to the set $\mathrm{Hom}_\mathscr{C}(A, B)$,
- Given an arrow $f : B \to B'$, the functor sends f to a set function $f^\#$ defined as follows:

$$f^\# : \mathrm{Hom}_\mathscr{C}(A, B) \to \mathrm{Hom}_\mathscr{C}(A, B'), \qquad f^\#(g) = f \circ g$$

  , where $\circ$ is composition in $\mathscr{C}$.

Dualize the above definition, we got a contravariant functor $\mathscr{C} \to \mathbf{Set}$ as follows:

Definition 1.2.13. Given a category $\mathscr{C}$, we define the following functor $\mathrm{Hom}_\mathscr{C}(-, A) : \mathscr{C}^{op} \to \mathbf{Set}$ for any $A \in \mathscr{C}$.

- Given any object $B \in \mathscr{C}$, the functor sends it to the set $\mathrm{Hom}_\mathscr{C}(B, A)$,
- Given an arrow $\widetilde{f} : B' \to B$, the functor sends f to a set function $\widetilde{f}^\#$ defined as follows:

$$\widetilde{f}^\# : \mathrm{Hom}_\mathscr{C}(B, A) \to \mathrm{Hom}_\mathscr{C}(B', A), \qquad \widetilde{f}^\#(g) = g \circ \widetilde{f}$$

  , where $\circ$ is composition in $\mathscr{C}$.

Note that if we trace the definition, we have $\mathrm{Hom}_{\mathscr{C}}(A,-) = \mathrm{Hom}_{\mathscr{C}^{op}}(-,A)$ exactly as functors $\mathscr{C} \to \mathbf{Set}$. Both functors play a fundamental role in proving statements regarding isomorphisms, as one will see later in Chapter 2.

The two definitions above are what is informally called *functorial*: that is, it behaves well with functors. We can see it in the following sense:

Definition 1.2.14. Given $F : \mathscr{C} \to \mathscr{D}$ a functor, we can define the following functor $\mathrm{Hom}_{\mathscr{D}}(A, F-) : \mathscr{C} \to \mathbf{Set}$ for any $A \in \mathscr{D}$.

- Given any object $B \in \mathscr{C}$, the functor sends it to the set $\mathrm{Hom}_{\mathscr{D}}(A, FB)$,
- Given an arrow $f : B \to B'$, the functor sends f to a set function $f^{\#}$ defined as follows:

$$f^{\#} : \mathrm{Hom}_{\mathscr{D}}(A, FB) \to \mathrm{Hom}_{\mathscr{D}}(A, FB'), \qquad f^{\#}(g) = F(f) \circ g$$

  , where $\circ$ is composition in $\mathscr{C}$.

At the beginning of the section, we introduce functors as "transformations" between categories, this word choice motivates us to see functors as morphisms of certain categories, and indeed this is true.

Definition 1.2.15. We define the category $\mathbf{Cat}$ with $\mathrm{Ob}(\mathbf{Cat})$ as (the class of) all categories, and the morphisms $\mathrm{Hom}_{\mathbf{Cat}}(\mathscr{C}, \mathscr{D})$ to be the set of all functors between $\mathscr{C}$ and $\mathscr{D}$.

Important Working Assumption

Similar to the remark at Definition 1.1.1, we will assume this category exists without set-theory concerns. From now on, we will not worry about any set or classes being "too big".

To check that $\mathbf{Cat}$ is indeed a category, we need to define the identity functor 1_F for any functor $F : \mathscr{C} \to \mathscr{D}$ and a composition rule of functors. There is essentially one way to define it. Namely:

- Given a category $\mathscr{C}$, we can define the identity functor $1_{\mathscr{C}} : \mathscr{C} \to \mathscr{C}$ such that it takes any object $A \in \mathscr{C}$ to A itself, and any arrow $f : A \to B$ to f itself. In symbols, we can write $1_{\mathscr{C}}(A) = A$ and $1_{\mathscr{C}}(f) = f$.
- Given functors $F : \mathscr{C}, \mathscr{D}, G : \mathscr{D}, \mathscr{E}$, we can write down the composition of two functors GF by sending object A to $G(F(A))$, arrows F to $G(Ff)$.

As with the previous section, we now investigate how functors play with isomorphisms.

Proposition 1.2.16 (Functors preserve isomorphisms)**.** Given any category $\mathscr{C}, \mathscr{D}$ and any functor $F : \mathscr{C} \to \mathscr{D}$. If $A, B \in \mathscr{C}$ are isomorphic objects, then so is $F(A), F(B) \in \mathscr{D}$.

Proof. If $f : A \to B$, $g : B \to A$ is the isomorphisms between A, B, then

$$F(g)F(f) = F(gf) = F(1_A) = 1_{F(A)}$$
$$F(f)F(g) = F(fg) = F(1_B) = 1_{F(B)}$$

with each equality established by property of functor, property of isomorphism, and property of functor again, in that order. But the above two lines are exactly the definition of $F(A)$, $F(B)$ being isomorphic with isomorphisms $F(f), F(g)$, as required. □

Also, with functors being the morphisms, we now have a notion of isomorphism of category.

Definition 1.2.17. Two categories $\mathscr{C}, \mathscr{D}$ are **isomorphic** if they are isomorphic objects in $\mathbf{Cat}$.

One might think this is a very useful definition, but it turns out to be too strong. In a (second course) of linear algebra, one might be familiar with the slogan "$X = X^{**}$ naturally", but if we think of the picture $\mathbf{FDVect}_k \underset{1_{\mathbf{FDVect}_k}}{\overset{(-)^{**}}{\rightleftarrows}} \mathbf{FDVect}_k$, this does not form an isomorphism pair in $\mathbf{FDVect}_k$. This is because by Proposition 1.1.17, since $(1_{\mathbf{FDVect}_k}, 1_{\mathbf{FDVect}_k})$ is an isomorphism pair, then this is a unique pair with $1_{\mathbf{FDVect}_k}$ as one of the functors for isomorphism.

The reason for the too-strong concept above is that we require the composition of two functors GF to be *exactly* the identity. We will meet a much more useful definition of two categories being *equivalent* in Section 2.2, where we require GF to be "isomorphic" to the identity functor.

Recall in the definition of a category, we insist between any two objects, the morphisms always form a set. Therefore, the part where a functor assigns morphisms to morphisms (for any two fixed objects) is always a function. In particular, the functor that does this bijectively deserves special attention.

Definition 1.2.18. Given a functor $F : \mathscr{C} \to \mathscr{D}$, F is **faithful** if for any objects $A, B \in \mathscr{C}$, the set-function

$$\mathrm{Hom}_{\mathscr{C}}(A, B) \to \mathrm{Hom}_{\mathscr{D}}(F(A), F(B)) \qquad f \mapsto F(f) \tag{1.1}$$

is an injection. F is **full** if the function (1.1) is surjective.

Faithful and full functors not only preserve isomorphisms but also the converse of Proposition 1.2.16 is also true:

Proposition 1.2.19. Let $\mathscr{C}, \mathscr{D}$ be categories, if $F : \mathscr{C} \to \mathscr{D}$ is a full and faithful functor, then for any $A, B \in \mathscr{C}$:

1. $f : A \to B$ is an isomorphism if and only if $F(f) : F(A) \to F(B)$ is an isomorphism.
2. A and B are isomorphic if and only if $F(A)$, $F(B)$ are isomorphic.

Proof. Note that (2) is a restatement of (1) by the definition of isomorphism 1.1.14. By Proposition 1.2.16, It suffices to prove that $F(f)$ is an isomorphism, then f is an isomorphism. Assume $F(f) : F(A) \to F(B)$ is an isomorphism, then there exists an arrow $h \in \mathrm{Hom}_{\mathscr{D}}(F(B), F(A))$ such that $h \circ F(f) = 1_{F(A)}$ and $F(f) \circ h = 1_{F(B)}$. But since the map (1.1) is surjective, there is $g \in \mathrm{Hom}_{\mathscr{C}}(B, A)$ such that $F(g) = h$. Then we have

$$h \circ F(f) = F(g) \circ F(f) = F(g \circ f) = 1_{F(A)} = F(1_A),$$

where the second and last equality is by functor properties. Hence $F(g \circ f) = F(1_A)$, but by injectivity of the map (1.1) we have $g \circ f = 1_A$. A similar argument on $F(f) \circ h = 1_B$ shows $f \circ g = 1_B$. Hence f is an isomorphism by definition. □

Perhaps a surprising fact is that there is a useful notion of functors being isomorphic as well, as we will introduce in the section below.

1.3 Natural transformations

From here, unless otherwise specified, we will fix $\mathscr{C}, \mathscr{D}$ to be categories.

Definition 1.3.1. Given two functors $F : \mathscr{C} \to \mathscr{D}$, $G : \mathscr{C} \to \mathscr{D}$, we say α is a **natural transformation from** F **to** G if α assigns an object $X \in C$ to a morphism $\alpha(X) \in \mathrm{Hom}_{\mathscr{D}}(F(X), G(X))$, such that for any $X, Y \in \mathscr{C}$, for any $f \in \mathrm{Hom}_{\mathscr{C}}(X, Y)$, we have $\alpha(Y)\, F(f) = G(f)\, \alpha(X)$. We also write $\alpha : F \to G$ for this.

Remark. If $\alpha : F \to G$, sometimes we will write α_X to denote the arrow $\alpha(X) : F(X) \to G(X)$. $\Diamond$

So far we have met all the definitions promised in the title of this chapter. We use more diagrams to make the idea more intuitive. Note that our writing is similar to writing functions, such as writing $F : \mathscr{C} \to \mathscr{D}$ for functors. If we have two functors $F, G : \mathscr{C} \to \mathscr{D}$, we can draw $\mathscr{C} \underset{G}{\overset{F}{\rightrightarrows}} \mathscr{D}$ for clarity.

Then Natural transformations are transformations between functors and can be depicted with a double arrow. If $\alpha : F \to G$ is a natural transformation. The usual way to indicate α is

$$\mathscr{C} \underset{G}{\overset{F}{\Downarrow \alpha}} \mathscr{D} \ .$$

The requirement $\alpha(Y)F(f) = G(f)\alpha(X)$ is usually presented by requiring the following diagram to be commutative:

$$\begin{array}{ccc} F(X) & \xrightarrow[F(f)]{} & F(Y) \\ \Big\downarrow{\scriptstyle \alpha(X)} & & \Big\downarrow{\scriptstyle \alpha(Y)} \\ G(X) & \xrightarrow{G(f)} & G(Y) \end{array} \tag{1.2}$$

for all such X, Y, f.

Example 1.3.2. Consider the functor $1_{\mathbf{FDVect}_k}, (-)^{**} : \mathbf{FDVect}_k \to \mathbf{FDVect}_k$ introduced in Section 1.2, we claim that there is a natural transformation $\alpha : 1_{\mathbf{FDVect}_k} \to (-)^{**}$. To show this, we need to define a linear map $\alpha(V) : V \to V^{**}$ for any vector space $V \in \mathbf{FDVect}_k$. One choice is the evaluation map ev_V:

$$ev_V(\vec{v}) := \text{The linear functional } ev_{\vec{v}} \text{ defined by} \eta \mapsto \eta(\vec{v}),$$

where $\eta \in V^*$. Now, to check that it is a natural transformation, we check for any $f : V \to W$ linear map between two spaces, for any $v \in V$:

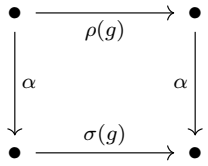
$$\begin{aligned} (-)^{**}(f)\alpha(V)(\vec{v}) &= f^{**} \circ ev_V(\vec{v}) = f^{**}(ev_{\vec{v}}) \\ &= \text{The map } \eta \mapsto ev_{\vec{v}}(\eta \circ f) = \text{The map } \eta \mapsto \eta \circ f(\vec{v}) \\ &= ev_{f(\vec{v})} = ev_W \circ f(\vec{v}) = \alpha(W) \circ f \end{aligned}$$

Hence the naturality requirement is satisfied. $\triangle$

Example 1.3.3. We have seen functors from $G \to \mathbf{Set}$ is a group action on an arbitrary set. What are the natural transformations α between two actions $\rho : G \times X \to X$ and $\sigma : G \times Y \to Y$?

Note that G has one object only, so α assign a single object $\bullet$ to a morphism $\alpha(\bullet) : \rho(\bullet) \to \sigma(\bullet)$. $\rho(\bullet)$ and $\sigma(\bullet)$ picks out the set X and Y respectively, and hence $\alpha(\bullet)$ really can be thought of a set function from X to Y. We abuse the notation here and refer to this set function as α. Now the commutative diagram in

Definition 1.3.1 reads

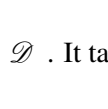


holds for all $g \in \mathrm{Hom}_G(\bullet, \bullet)$, that is, all group elements $g \in G$. Tracing diagrams, this is the condition $\alpha \circ \rho(g) = \sigma(g) \circ \alpha$ for any $g \in G$. α is referred to as G**-equivariant maps** between X and Y in group theory. Informally, this is a set function that respects the group action. $\triangle$

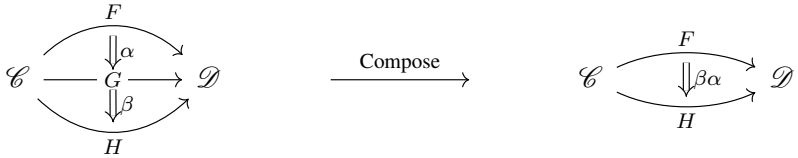
Similar to functors, the natural transformation can be thought of as morphisms as well, this time in another category.

Definition 1.3.4. The **functor category from** $\mathscr{C}$ **to** $\mathscr{D}$, denoted by $[\mathscr{C}, \mathscr{D}]$, is the category whose objects are functors from $\mathscr{C}$ to $\mathscr{D}$, and given two functors $F, G : \mathscr{C} \to \mathscr{D}$, the morphisms $\mathrm{Hom}_{[\mathscr{C},\mathscr{D}]}(F, G)$ is the collection of all natural transformations from $\mathscr{C}$ to $\mathscr{D}$.

Remark. As with the definition of functor categories, we also need an identity and a composition rule. But this time, we will think about the appropriate objects using commutative diagrams.

The identity should be a natural transformation between the same functor (objects in $[\mathscr{C}, \mathscr{D}]$) F, we denote it as 1_F  . It takes any object $X \in \mathscr{C}$ and outputs $1_{F(X)} \in \mathrm{Hom}_{\mathscr{D}}(FX, FX)$.

For composition between two natural transformations $\alpha : F \to G, \beta : G \to H$, one can think of the following picuture:



and we define $\beta\alpha(X) = \beta(X) \circ \alpha(X)$. Note that this is well defined, as $\alpha(X) \in \mathrm{Hom}_{\mathscr{D}}(FX, GX), \beta(X) = \mathrm{Hom}_{\mathscr{D}}(GX, HX)$, hence $\beta(X) \circ \alpha(X) \in \mathrm{Hom}_D(FX, HX)$. $\Diamond$

In Section 1.2, we have the notion of categories being isomorphic using the category $\mathbf{Cat}$, now we have a notion of functors being isomorphic.

Definition 1.3.5. Two functors $F, G : \mathscr{C} \to \mathscr{D}$ are **isomorphic** if they are isomorphic in the functor category $[\mathscr{C}, \mathscr{D}]$. Not only will we write $F \cong G$ in definition, but we will also write $F(A) \cong G(A)$ **naturally in** $A \in \mathscr{C}$ to mean exactly $F \cong G$.

Using the definition, if we were to check some pairs of functor $F, G : \mathscr{C} \to \mathscr{D}$ is isomorphic, there are a lot of steps to follow:

- Construct $\alpha : F \to G$ and check it is a natural transformation.
- Construct $\beta : G \to F$ and check it is a natural transformation.
- Show that $\alpha \circ \beta = 1_G$ and $\beta \circ \alpha = 1_F$ using composition of natural transformation.

The following theorem makes checking functor isomorphism slightly less cumbersome:

Theorem 1.3.6. Given functors any pairs of functor $F, G : \mathscr{C} \to \mathscr{D}$, then $F \cong G$ if and only if the following holds:

> There is a natural transformation $\alpha : F \to G$ such that for any $A \in \mathscr{C}$, $\alpha(A) : F(A) \to G(A)$ is an isomorphism (in $\mathscr{D}$).

Proof. Assume $F \cong G$, then there are natural transformation α, β with $\mathscr{C} \overset{F}{\underset{G}{\Downarrow \alpha}} \mathscr{D}$, $\mathscr{C} \overset{F}{\underset{G}{\beta \Uparrow}} \mathscr{D}$ such that $\alpha\beta = 1_G, \beta\alpha = 1_F$. We want to show that $\alpha(A)$ for any $A \in \mathscr{C}$ is an isomorphism. To show this, note that

$$\begin{aligned} \beta(A) \circ \alpha(A) &= \beta\alpha(A) && \text{definition of composition} \\ &= 1_F(A) && \text{assumption of } \beta\alpha = 1_F \\ &= 1_{F(A)} \end{aligned}$$

and similarly $\alpha(A) \circ \beta(A) = 1_{G(A)}$. So $\beta(A)$ is an inverse for $\alpha(A)$, hence $\alpha(A)$ is an isomorphism.

Conversely, suppose α is a natural transformation such that $\alpha(A)$ are isomorphisms for any $A \in \mathscr{C}$, then by Proposition 1.1.17, there is a unique $\beta(A) \in \mathrm{Hom}_{\mathscr{D}}(GA, FA)$ such that $\beta(A) \circ \alpha(A) = 1_{F(A)}$ and $\alpha(A) \circ \beta(A) = 1_{G(A)}$. So if β is indeed a natural transformation, then $\beta\alpha = 1_F, \alpha\beta = 1_G$ by definition of 1_F and 1_G. Now we indeed check $\beta : G \to F$ is a natural transformation. Since α is a natural transformation, for any $f : X \to Y$:

$$\begin{aligned} G(f)\alpha(X) &= \alpha(Y)F(f) \\ G(f)\alpha(X)\beta(X) &= \alpha(Y)F(f)\beta(X) \\ G(f)1_{G(X)} = G(f) &= \alpha(Y)F(f)\beta(X) \\ \beta(Y)G(f) &= \beta(Y)\alpha(Y)F(f)\beta(X) \\ &= 1_{F(Y)}F(f)\beta(X) = F(f)\beta(X) \end{aligned}$$

so β is a natural transformation by definition as required. □

Theorem 1.3.6 gives a reason why we use the phrase "$F(A) \cong G(A)$ naturally".

Remark. Note that given categories $\mathscr{C}, \mathscr{D}$ and functors $F, G : \mathscr{C} \to \mathscr{D}$, the fact that $F(A) \cong G(A)$ for all A is not enough to conclude F, G are isomorphic. See [Ric13]. Consider the diagram $\mathscr{C}$ as a category with two distinct morphisms on an object $\bullet$ with $f^2 = 1_\bullet$ (where $1_\bullet$ is the identity).



And let F be the functor that send $\bullet$ to $\bullet$, $1_\bullet$ to$1_\bullet$ and f to $1_\bullet$. Although $F(\bullet) = 1_{\mathscr{C}}(\bullet) = \bullet$, we cannot conclude $F \cong 1_{\mathscr{C}}$. Indeed, if there is a natural transformation $\alpha : F \to 1_{\mathscr{C}}$, the following diagram must commute:

$$\begin{array}{ccc} \bullet & \xrightarrow{F(f)=1_\bullet} & \bullet \\ {\scriptstyle \alpha(\bullet)}\downarrow & & \downarrow{\scriptstyle \alpha(\bullet)} \\ \bullet & \xrightarrow[1_{\mathscr{C}}(f)=f]{} & \bullet \end{array}$$

but both choice $\alpha(\bullet) = f$ or $1_\bullet$ cannot make this diagram commute. $\Diamond$

Example 1.3.7. The $\alpha : 1_{\mathbf{FDVect}_k} \to (-)^{**}$ we introduced in Example 1.3.2 is actually a natural isomorphism. To see this, we only need to realize the map ev_V is an isomorphism of vector spaces, note that this is a theorem that only holds if V is finite-dimensional. For infinite dimensional vector space, ev_V is injective but might not be surjective. We therefore write "$1_{\mathbf{FDVect}_k}(V) \cong (-)^{**}(V)$ naturally", or the more common slogan: "$V \cong V^{**}$ naturally for finite dimensional vector spaces V" . In general, if one is working with the category $\mathbf{Vect}_k$, then α is still a natural transformation, but not necessarily a natural isomorphism. $\triangle$

Chapter 2

The Categorical Perspective

2.1 Universal Properties

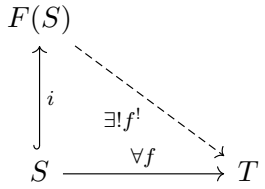
Category Theory language provides a paradigm shift in mathematics. Instead of looking at special sets and their properties by defining them and proofing their properties explicitly, we define an object by declaring how it behaves under a collection of maps and prove such an object is unique up to isomorphism if it exists. This streamlines the proof of many properties and provides an efficient way for generalizations. In this section, we will briefly look at examples of such constructions, which are informally referred to as *universal properties*, and comment on some generalization or properties that motivates this definition.

Example 2.1.1. Our first example is the free vector space $F(S)$ given a set S, introduced in Example 1.2.7. Recall the definition specifies a basis the vector space should have. Having a *basis* is a property of the vector space, is it possible to create a definition without referring to bases? Here is a candidate.

Candidate definition. Given a set S, define the space $F(S)$ such that it satisfies the following two properties:

- There is an injective function $i : S \to F(S)$, and
- For any vector space T, if there is a function $f : S \to T$, then there exists a unique linear map $f^! : F(S) \to T$ such that $i \circ f^! = f$.

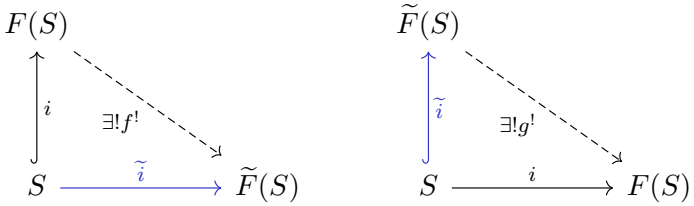
In pictures, we can denote the situation as:



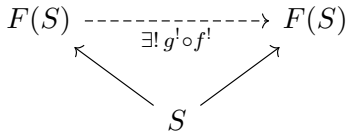
Intuitively, any map $S \to T$ has a "lift" to $F(S) \to T$

Note that this at first glance does not seem like a definition, but if the $F(S)$ *exists* (which the candidate definition above does not specify), then it must be unique up to an isomorphism. The reason being, if another "candidate" $\widetilde{F}(S)$ exists, one can build two commutative diagrams by lifting $i : S \to F(S)$ and

$\widetilde{i} : S \to \widetilde{F}(S)$:



Then we can "stick" the commutative diagram together along the coloured side and yield the commutative diagram:



which mean there is only one map from $F(S) \to F(S)$, and it is $g \circ f$, but the map must be identity $1_{F(S)}$. Hence $g^! \circ f^! = 1_{F(S)}$, similar reasoning shows $f^! \circ g^! = 1_{\widetilde{F}(S)}$, hence $(f^!, g^!)$ is an isomorphism pair. $\triangle$

In general, this type of definition of defining an object $G(S)$ related to (object) S by specifying a map and the statement "for XXX map from S, there is a map from $G(S)$..." is called a *universal property*. If the second part of the statement is "there is a *unique* map...", then in most cases, using techniques from above, one can show that any object defined must be isomorphic. We will heavily use this type of definition later on, here are more examples.

Definition 2.1.2. Given a commutative ring R, a multiplicative set S, the **localization** of R by S, denoted by $S^{-1}R$, has the following property:

- There is an injection $R \to S^{-1}R$, and
- For all ring homomorphism $f : R \to T$ that sends elements of S into units in T, there exists a unique map $f^! : S^{-1}R \to T$ such that $f^! \circ i = f$.

Remark. It is an easy exercise to check the common construction of localization:

$$S^{-1}R := \left\{ \frac{r}{s} : r \in R, s \in S \right\} \subseteq \mathrm{Frac}(R)$$

where $\mathrm{Frac}(R)$ is the fraction field of R, and the injection given by $r \to \frac{r}{1}$satisfies the above relation.

Also, note that it is essential to have R as a commutative ring, for localization of a non-commutative ring, see a construction $S^{-1}R$ in [Coh91, Chapter 9.1]. $\Diamond$

We will see a generalization of this construction in Chapter 5.

Definition 2.1.3. The **tensor product** of two vector spaces V, W over a fixed field k, is the space denoted by $V \otimes W$ having the following property:

- There is a bilinear map $i : V \times W \to V \otimes W$
- For any bilinear map $f : V \times W \to T$, there is a unique map $f^! : V \otimes W \to T$ such that $f^! \circ i = f$.

Remark. A standard construction of tensor products is $V \otimes W = \mathrm{Hom}_{\mathbf{Vect}_k}(V^*, W)$ with the bilinear map i given by

$$(v, w) \mapsto \text{The linear function } \phi : V^* \to W \text{ such that} \phi(\eta) = \eta(v)w.$$

$\Diamond$

2.2 Equivalent Categories

Recall in Section 1.2 we commented the notion of categories being isomorphic is too strong, we now present a "correct working definition" of category, namely *equivalence*.

Definition 2.2.1. Categories $\mathscr{C}, \mathscr{D}$ are **equivalent** if there exists functors $F : \mathscr{C} \to \mathscr{D}, G : \mathscr{D} \to \mathscr{C}$ such that $GF \cong 1_{\mathscr{C}}, FG \cong 1_{\mathscr{D}}$, where $1_{\mathscr{C}}, 1_{\mathscr{D}}$ are as defined in the remark of Definition 1.2.15. We write $\mathscr{C} \simeq \mathscr{D}$ to denote equivalence.

The standard example is the following, with the slogan "finite dimensional vector spaces can be understood via matrices".

Example 2.2.2 ([Lei14])**.** Recall in Example 1.1.9 we defined the category $\mathscr{C}$, we now show $\mathscr{C} \simeq \mathbf{FDVect}_k$. We consider a functor $F : \mathscr{C} \to \mathbf{FDVect}_k$ which sends $i \in \mathscr{C}$ to k^i, and for a matrix (as a morphism in $\mathscr{C}$) $M \in \mathrm{Hom}_{\mathscr{C}}(m, n)$ we send to the linear map $k^m \to k^n$ *induced by the standard basis*. That is, given a $n \times m$ matrix M, computing $Me_1, Me_2 \cdots Me_m$ for the standard basis in k^m, specifies the linear map on $k^m \to k^n$.

We also define the functor $G : \mathbf{FDVect}_k \to \mathscr{C}$ by sending a vector space V to its dimension $\dim(V)$, and any vector space morphism $f : V \to W$ to be the matrix representing the morphism. That is, if we fix the bases $\{v^i\}_{i=1}^n, \{w_j\}_{j=1}^m$ of V and W respectively, the coefficient α_i^j that have

$$f(v_i) = \sum_{j=1}^{m} \alpha_i^j w_j,$$

then $M_{ji} = a_i^j$ will be the corresponding matrix is $\mathrm{Hom}_C(n, m)$.

Then, it is routine to check that $G \circ F$ gives the identity functor on $\mathscr{C}$. However, note that $F \circ G$ will send a vector space $V \in \mathbf{FDVect}_k$ to k^n, and a transformation $V \to W$ to a transformation $k^n \to k^m$. However, we claim that $G \circ F \cong 1_{\mathbf{Vect}_k}$. To see this, we define a natural transformation $\alpha : G \circ F \to 1_{\mathbf{Vect}_k}$ by $\alpha(V) : k^n \to V$, as the linear map sending the n-element standard basis of k^n to a choice of a n-element basis of V bijectively. This need not be the identity map (since there is no guarantee that $V = k^n$), but it is a vector space isomorphism since it sends basis to basis bijectively. Hence by Theorem 1.3.6 it follows that α is a natural isomorphism. △

In the above example, we constructed an isomorphism between k^n, V by a natural transformation for any V. Also, we know that every matrix corresponds to one and only one linear map. This can be generalized to the following proposition, which works for all categories.

Proposition 2.2.3. A functor $F : \mathscr{C} \to \mathscr{D}$ is an equivalence of categories if and only if F is fully faithful, and satisfies the condition that for every object $A \in \mathscr{C}$, there exists an object $B \in \mathscr{D}$ such that $F(A) \cong B$ (this is sometimes called *essentially surjective*).

Therefore, equivalent categories provide a way to "translate" what is known for a category to another. This is a very powerful idea we will explore in Part II of the report.

2.3 The Yoneda Lemma

This section will be devoted to proving an important lemma (that, in the author's opinion, deserves all the credit to be called a theorem) in Category Theory. Although we have only developed very basic machinery in Chapter 1, it will be enough to state the lemma and prove it. One might not immediately see how important

the Yoneda Lemma is. But as we introduce more concepts in subsequence chapters, we will see various applications of this lemma and its corollaries. Here is a definition that will help to reduce the clutter in notation.

Definition 2.3.1. The **Yoneda Embedding** is the following functor $H_\bullet : \mathscr{C} \to [\mathscr{C}^{op}, \mathbf{Set}]$:

- Given a object $A \in \mathscr{C}$, $H_\bullet(A)$ is the functor $\mathrm{Hom}_{\mathscr{C}}(-, A) : \mathscr{C}^{op} \to \mathbf{Set}$,
- Given a morphism $f : A \to A'$ in $\mathscr{C}$, $H_\bullet(f)$ is the natural transformation $\alpha_f : \mathrm{Hom}_{\mathscr{C}}(-, A) \to \mathrm{Hom}_{\mathscr{C}}(-, A')$, defined by $\alpha_f(X) : \mathrm{Hom}_{\mathscr{C}}(X, A) \to \mathrm{Hom}_{\mathscr{C}}(X, A')$ by $\alpha_f(X)(g) = f \circ g$.

The reason for it to be called an embedding is because of a consequence of the Yoneda Lemma (See Corollary 2.3.4)

Theorem 2.3.2 (Yoneda)**.** For any category $\mathscr{C}$, any functor $X : \mathscr{C}^{\mathrm{op}} \to \mathbf{Set}$ and any object $A \in \mathscr{C}$, we have:

$$X(A) \cong \mathrm{Hom}_{[\mathscr{C}^{\mathrm{op}}, \mathbf{Set}]}(\mathrm{Hom}_{\mathscr{C}}(-, A), X) \tag{2.1}$$

naturally in $A \in \mathscr{C}$ and $X \in [\mathscr{C}^{op}, \mathbf{Set}]$.

One can dualize (2.1) to obtain the dualized version of the Yoneda Lemma.

Theorem 2.3.3 (Yoneda, Dualized Version)**.** For any category $\mathscr{C}$, any functor $X : \mathscr{C} \to \mathbf{Set}$ and any object $A \in \mathscr{C}$, we have:

$$X(A) \cong \mathrm{Hom}_{[\mathscr{C}, \mathbf{Set}]}(\mathrm{Hom}_{\mathscr{C}}(A, -), X) \tag{2.2}$$

naturally in $A \in \mathscr{C}$ and $X \in [\mathscr{C}, \mathbf{Set}]$.

Of course, they are logically equivalent so it is a matter of style of proving which version. We choose to prove Theorem 2.3.2.

Proof. Note that there are two naturality statements, so we have to prove them separately, however, the proof will be very similar.

Naturality in A: Fix any $X \in [\mathscr{C}^{op}, \mathbf{Set}]$. Note that to show (2.1) is natural in A we need to establish a natural isomorphism between the functor X and $\mathrm{Hom}_{[\mathscr{C}^{op}, \mathbf{Set}]}(\mathrm{Hom}_{\mathscr{C}}(-, \bullet), X)$. By Theorem 1.3.6 it suffices to create a natural transformation $\alpha : X \to \mathrm{Hom}_{[\mathscr{C}^{op}, \mathbf{Set}]}(\mathrm{Hom}_{\mathscr{C}}(-, \bullet), X)$ such that the set function $\alpha(U) : X(U) \to \mathrm{Hom}_{[\mathscr{C}^{op}, \mathbf{Set}]}(\mathrm{Hom}_{\mathscr{C}}(-, U), X)$ is a bijection for any $U \in \mathscr{C}$.

We first define $\alpha(U)$. It sends an element $a \in X(U)$ to a natural transformation $\gamma_a : \mathrm{Hom}_{\mathscr{C}}(-, U) \to X$. That is $\alpha(U)(a) = \gamma_a$. Now, we define γ_a to finish the definition. For any $V \in \mathscr{C}$, $\gamma_a(V)$ is a set-function $\mathrm{Hom}_{\mathscr{C}}(V, U) \to X(V)$, and it is defined as follows: $\gamma_a(V)(f) = (X(f))(a)$. Note that in this definition, $f \in \mathrm{Hom}_{\mathscr{C}}(V, U) = \mathrm{Hom}_{\mathscr{C}^{op}}(U, V)$, hence $X(f) \in \mathrm{Hom}_{\mathbf{Set}}(X(U), X(V))$ hence $X(f)(a)$ is well-defined since $a \in X(U)$.

Now we show that for all U, $\alpha(U)$ is bijective. To do this, we define the set-function inverse $\alpha(U)^{-1} : \mathrm{Hom}_{[\mathscr{C}^{op}, \mathbf{Set}]}(\mathrm{Hom}_{\mathscr{C}}(-, U), X) \to X(U)$ as follows. Given $\eta : \mathrm{Hom}_{\mathscr{C}}(-, U) \to X$ a natural transformation, we define as $\alpha(U)^{-1}(\eta) = \eta(U)(1_U)$. We first check that this is a well-defined function. For any $\eta : \mathrm{Hom}_{\mathscr{C}}(-, U) \to X$ a natural transformation, $\eta(V)$ is a set-function $\mathrm{Hom}_{\mathscr{C}}(V, U) \to X(V)$ for any $V \in \mathscr{C}$. So $\eta(U)$ is a set-function $\mathrm{Hom}_{\mathscr{C}}(U, U) \to X(U)$, so $\eta(U)(1_U)$ is an element in $X(U)$ as required.

Now we check $\alpha(U)^{-1}\circ\alpha(U)$ is the identity function on $X(U)$. One compute as follows: for any $a \in X(U)$:

$$\begin{aligned}\alpha(U)^{-1}\alpha(U)(a) = \alpha(U)^{-1}(\gamma_a) = \gamma_a(U)(1_U) &\qquad \text{by definition of } \alpha(U) \text{ and } \alpha^-1(U)\\ = X(1_u)(a) &\qquad \text{by definition of } \gamma_a\\ = 1_{X(U)}(a) &\qquad \text{by } X \text{ being a functor}\end{aligned}$$

Since $a \in X(U)$ we indeed see $\alpha(U)^{-1} \circ \alpha(U)$ is the identity on $X(U)$ as required.

Now we check $\alpha(U) \circ \alpha(U)^{-1}$ is the identity function on $\mathrm{Hom}_{[\mathscr{C}^{op},\mathbf{Set}]}(\mathrm{Hom}_{\mathscr{C}}(-,U),X)$. To do this, we compute $\alpha(U)(\alpha(U)^{-1}(\eta))$, where $\eta : \mathrm{Hom}_{\mathscr{C}}(-,U) \to X$ is a natural transformation. Note that the naturality square 1.2 of U, V on η reads as below:

$$\begin{array}{ccc}\mathrm{Hom}(U,U) & \xrightarrow[\mathrm{Hom}(-,U)(f)]{} & \mathrm{Hom}(V,U)\\ \Big\downarrow{\scriptstyle \eta(U)} & & \Big\downarrow{\scriptstyle \eta(V)}\\ X(U) & \xrightarrow{\quad X(f)\quad} & X(V)\end{array} \tag{2.3}$$

Now we compute, fixing $\eta : \mathrm{Hom}_{\mathscr{C}}(-,U) \to X$ a natural transformation

$$\alpha(U) \circ \alpha(U)^{-1}(\eta) = \alpha(U)(\eta(U)(1_U)) = \gamma_{\eta(U)(1_U)} \qquad \text{by definition of } \alpha(U) \text{ and } \alpha^{-1}(U)$$

So, given any $V \in \mathscr{C}, f \in \mathrm{Hom}_{\mathscr{C}}(V,U)$:

$$\begin{aligned}\gamma_{\eta(U)(1_U)}(V)(f) &= X(f)(\eta(U)(1_U))\\ \text{By the commutative square (2.3),} \qquad &= \eta(V)(\mathrm{Hom}_{\mathscr{C}}(-,U)(f))(1_U)\\ &= \eta(V)(1_U \circ f) = \eta(V)(f)\end{aligned}$$

so $\gamma_{\eta(U)(1_U)}(V)(f) = \eta(V)(f)$, hence $\gamma_{\eta(U)(1_U)} = \eta$, hence $\alpha(U)\circ\alpha(U)^{-1}(\eta) = \eta$ for any $\eta : \mathrm{Hom}_{\mathscr{C}}(-,U) \to X$ a natural transformation. This concludes that $\alpha(U)$ is indeed a bijection, hence isomorphism, hence naturality in A is proved.

Naturality in X. This time, fix $A \in \mathscr{C}$, again by Theorem 1.3.6 it suffices to construct set function $\beta(Y) : Y(A) \to \mathrm{Hom}_{[\mathscr{C}^{op},\mathbf{Set}]}(\mathrm{Hom}_{\mathscr{C}}(-,A),Y)$ such that it is a bijection for all $Y \in [\mathscr{C}^{op},\mathbf{Set}]$. We define $\beta(Y)$ as $\beta(Y)(\widetilde{a}) = \gamma_{\widetilde{a}}$, where $\widetilde{\gamma_{\widetilde{a}}}(V)(f) = Y(f)(a)$ for any $V \in \mathscr{C}$, also define the set-function inverse $\beta(Y)^{-1} : \mathrm{Hom}_{[\mathscr{C}^{op},\mathbf{Set}]}(\mathrm{Hom}_{\mathscr{C}}(-,A),Y) \to Y(A)$ by $\beta(Y)^{-1}(\widetilde{\eta}) = \widetilde{\eta}(A)(1_A)$. Note that $\beta(Y), \beta(Y)^{-1}$ is exactly the definition of $\alpha(U)$ and $\alpha^{-1}(U)$, except with renaming of symbols. So it is clear that $\beta(Y)^{-1} \circ \beta(Y)$ and $\beta(Y) \circ \beta(Y)^{-1}$ is the respective identity. □

The statement of Yoneda lemma might seem ad-hoc, but the corollaries of it often yield the most powerful result. We introduce a few of them below.

Corollary 2.3.4. The Yoneda Embedding is a full and faithful functor.

Proof. We have to show the set function $f \to \alpha_f$ defined in Definition 2.3.1 is bijective for any $f : A \to A'$. We can apply the Yoneda lemma 2.3.2 with A as above and X be the functor $\mathrm{Hom}_{\mathscr{C}}(-,A') : \mathscr{C}^{op} \to \mathbf{Set}$, then (2.1) reads:

$$\mathrm{Hom}_{\mathscr{C}}(A,A') \cong \mathrm{Hom}_{[\mathscr{C}^{\mathrm{op}},\mathbf{Set}]}(\mathrm{Hom}_{\mathscr{C}}(-,A),\mathrm{Hom}_{\mathscr{C}}(-,A'))$$

Tracing the definition we can show that the bijection is exactly $f \to \alpha_f$ following the proof of the Yoneda lemma. □

Corollary 2.3.5. If $\mathscr{C}$ is any category, then the following are equivalent:

1. $\mathrm{Hom}_{\mathscr{C}}(A, -) \cong \mathrm{Hom}_{\mathscr{C}}(A', -)$, that is, $\mathrm{Hom}_{\mathscr{C}}(A, B) \cong \mathrm{Hom}_{\mathscr{C}}(A', B)$ *naturally* for all $B \in \mathscr{C}$;
2. $\mathrm{Hom}_{\mathscr{C}}(-, A) \cong \mathrm{Hom}_{\mathscr{C}}(-, A')$;
3. $A \cong A'$

Proof. (1) if and only if (2) follows by dualizing and considering $\mathrm{Hom}_{\mathscr{C}}(A, -) = \mathrm{Hom}_{\mathscr{C}^{op}}(-, A)$. Now since $H^{\bullet}(A) = \mathrm{Hom}_{\mathscr{C}}(-, A)$, and $H^{\bullet}$ is full and faithful by Corollary 2.3.4, so (2) if and only if (3) by Proposition 1.2.19. □

Corollary 2.3.5, as we will see in the next section, provides an important way to check two objects are isomorphic, therefore establishing most of the constructions are unique up to isomorphism.

Chapter 3

Special Objects and Functors

Again, we fix two categories $\mathscr{C}, \mathscr{D}$.

3.1 Adjoints

Adjoints capture the idea of turning objects of one category into another in a "nice" way. What is meant by nice is it preserves a lot of structures, most notably *limits* as one will see in Section 3.2.

Definition 3.1.1. Given two functors $L : \mathscr{C} \to \mathscr{D}$, $R : \mathscr{D} \to \mathscr{C}$ we say L is a **left adjoint** (to R) and R is a **right adjoint** (to L) if the functors

$$\mathrm{Hom}_{\mathscr{C}}(-, R-), \mathrm{Hom}_{\mathscr{D}}(L-, -) : \mathscr{C}^{op} \times \mathscr{D} \to \mathbf{Set}$$

are naturally isomorphic. In other words,

$$\mathrm{Hom}_{\mathscr{D}}(LA, B) \cong \mathrm{Hom}_{\mathscr{C}}(A, RB) \tag{3.1}$$

naturally in $A \in \mathscr{C}, B \in \mathscr{D}$. We write $L \dashv R$ in notation. We also present this pictorially as $\mathscr{C} \underset{R}{\overset{L}{\underset{\longleftarrow}{\overset{\longrightarrow}{\bot}}}} \mathscr{D}$.

Remark.

- Sometimes, we simply say L *is a left adjoint* (resp. R *is a right adjoint*) meaning that the other functor R (resp. L) in definition 3.1.1 exists.
- To check a functor L is a left adjoint by definition, we need to define the functor R, then construct natural isomorphism α, β for which $\alpha : \mathrm{Hom}_{\mathscr{D}}(L-, B) \to \mathrm{Hom}_{C}(-, RB), \beta : \mathrm{Hom}_{\mathscr{D}}(LA, -) \to \mathrm{Hom}_{\mathscr{C}}(A, R-)$. Again by Theorem 1.3.6, it suffices to check $\alpha(A), \beta(B)$ are bijections for any $A \in \mathscr{C}, B \in \mathscr{D}$.

$\Diamond$

Writing out this argument, especially checking the natural isomorphism part, is very cumbersome. However, this definition is still useful in some cases to think about what adjoints are:

Example 3.1.2. What are adjoint functors between two poset categories? Given $X_{\leqslant} \underset{g}{\overset{f}{\underset{\longleftarrow}{\overset{\longrightarrow}{\bot}}}} Y_{\leqslant}$ where f, g are adjoint functors (as order-preserving functions by Example 1.2.11), by (3.1) we have $\mathrm{Hom}_{X_{\leqslant}}(f(A), B) \cong$

$\mathrm{Hom}_{Y_\leqslant}(A, g(B))$ for any $A \in X, B \in Y$. But using the fact that

$$\mathrm{Hom}_{X_\leqslant}(A,B) = \begin{cases} \{\bullet\} & \text{when } A \leqslant B \\ \emptyset & \text{otherwise} \end{cases}$$

We can see since the two sets $\mathrm{Hom}_{Y_\leqslant}(f(A), B), \mathrm{Hom}_{X_\leqslant}(A, g(B))$ can only be singleton or empty (and if they are different cardinalities they cannot be isomorphic), it is easily seen that the $\mathrm{Hom}_{X_\leqslant}(f(A), B) \cong \mathrm{Hom}_{Y_\leqslant}(A, g(B))$ if and only if $f(A) \leqslant B \Leftrightarrow A \leqslant g(B)$ for all $A \in X, B \in Y$. △

This definition has a huge advantage in proving general facts about adjoints. Here is an example: one might ask can a functor, say F, has other right adjoints, we answer this question for the negative up to isomorphisms.

Proposition 3.1.3 (Uniqueness of Left Adjoints)**.** If $L : \mathscr{C} \to \mathscr{D}$, $R : \mathscr{D} \to \mathscr{C}$ are functors such that $L \dashv R$, if $L' : \mathscr{C} \to \mathscr{D}$ is also a functor such that $L' \dashv R$, then $L \cong L'$.

Proof. Assume $L \dashv R$ and $L' \dashv R$ as above, then $\mathrm{Hom}_{\mathscr{D}}(LA, B) \cong \mathrm{Hom}_{\mathscr{C}}(A, RB)$ and $\mathrm{Hom}_{\mathscr{D}}(L'A, B) \cong \mathrm{Hom}_{\mathscr{C}}(A, RB)$ naturally in A and B. But we know $\cong$ is an equivalence relation, so $\mathrm{Hom}_{\mathscr{D}}(LA, B) \cong \mathrm{Hom}_{\mathscr{D}}(L'A, B)$ naturally in A and B. But by definition of naturality, this means the functors $\mathrm{Hom}_{\mathscr{D}}(LA, -)$, $\mathrm{Hom}_D(L'A, -)$ are naturally isomorphic in A. As a consequence of the Yoneda Lemma - Corollary 2.3.5, we have $LA \cong L'A$ naturally in A, which is $L \cong L'$ by definition. □

We now provide an alternate characterization of adjoint functors, which speeds up the check of two functors being adjoints, and we use them to introduce more adjoint functors.[1]

Theorem 3.1.4. Given a functor $R : \mathscr{D} \to \mathscr{C}$, the following are equivalent:

(1) R is a right adjoint (or has a left adjoint). That is, There exists L such that $L \dashv R$.

(2) For any $A \in \mathscr{C}$ there is an $LA \in \mathscr{D}$ and $\eta_A : A \to RLA$ such that for any $f : A \to RB$, there is a unique $f^\# : LA \to B$ such that $R(f^\#) \circ \eta_A = f$. That is, the diagram

$$\begin{array}{ccc} A & \xrightarrow{\eta_A} & RLA \\ & \searrow^{f} & \downarrow{\scriptstyle R(f^\#)} \\ & & RB \end{array} \tag{3.2}$$

commutes,

(3) There is a functor $L : \mathscr{C} \to \mathscr{D}$, natural transformations $\eta : 1_{\mathscr{C}} \to RL, \varepsilon : LR \to 1_{\mathscr{D}}$ such that the following diagrams

$$\begin{array}{ccc} L(A) & \xrightarrow{L\eta_A} & LRL(A) \\ & \searrow^{1_{L(A)}} & \downarrow{\scriptstyle \varepsilon_{L(A)}} \\ & & L(A) \end{array} \qquad \begin{array}{ccc} R(B) & \xrightarrow{\eta_{R(B)}} & RLR(B) \\ & \searrow^{1_{R(B)}} & \downarrow{\scriptstyle R\varepsilon_B} \\ & & R(B) \end{array} \tag{3.3}$$

commutes for any $A \in \mathscr{C}, B \in \mathscr{D}$.

[1] The author wants to credit Dr Nicola Gambino for introducing this theorem in the MAGIC009 Category Theory Online Course in the MAGIC (Mathematics Access Grid Instruction and Collaboration) Mathematics Consortium.

Remark. The diagrams (3.3) are called **triangle equalities**. η is usually called the **unit**, and ε the **counit**. $\Diamond$

Proof. Omitted. See [Lei14, p.53] for a proof of $(1) \Leftrightarrow (3)$. $\square$

Note that in particular, the theorem asserts the existence of the left adjoint *given* the right adjoint R. There is a "dual" version, which we record below:

Corollary 3.1.5. Given a functor $L : \mathscr{C} \to \mathscr{D}$, then $L \dashv R$ if and only if for any $A \in \mathscr{D}$ there is an $RA \in \mathscr{C}$ and $\varepsilon_A : LRA \to A$ such that for any $f : LB \to A$, there is a unique $f^\flat : B \to RA$ such that $\varepsilon_A \circ L(f^\flat) = f$.

We can now provide more examples of adjoint functors, and verify they are adjoints using the equivalent condition in Theorem 3.1.4. First we examine condition (2).

Example 3.1.6. Given X a set, consider the following pair of functor $\mathbf{Set} \to \mathbf{Set}$:

- The functor $- \times X$: Sent a set A to the cartesian product $A \times X$, and a set-function $f : A \to A'$ to $f' : A \times X \to A' \times X$ as $f'(a, x) = (f(a), x)$.
- The functor $\mathrm{Hom}_{\mathbf{Set}}(X, -)$ in Definition 1.2.12.

We claim that $- \times X \dashv \mathrm{Hom}_{\mathbf{Set}}(X, -)$. To show this, let $\circ$ be the composition of function, we construct $\eta_A : A \to \mathrm{Hom}(X, A \times X)$ by $a \mapsto (x \to (a, x))$, and we verify the property. Given any $f : A \to \mathrm{Hom}(X, B)$, define $f^\# : A \times X \to B$ by $(a, x) \mapsto (f(a))(x)$. Tracing the definition in $\mathrm{Hom}_{\mathbf{Set}}(X, -)$ we know that $R(f^\#)$ is a function that takes $g \in \mathrm{Hom}_{\mathbf{Set}}(X, A \times X)$ to $f^\# \circ g \in \mathrm{Hom}_{\mathbf{Set}}(X, B)$. Hence $(\mathrm{Hom}_{\mathbf{Set}}(X, -)(f^\#) \circ \eta_A)(a) = (f^\# \circ -)(x \to (a, x)) = f^\# \circ (x \to (a, x)) = f(a)$. So the diagram (3.2) commutes. $\triangle$

Example 3.1.7. We now consider the category $\mathbf{Top}$ and we exhibit two adjoint pairs

$$\mathbf{Set} \underset{\longleftarrow U \longrightarrow}{\overset{\longrightarrow D \longleftarrow}{\overset{\longleftarrow I \longrightarrow}{}}} \mathbf{Toph}$$

Where D represents the discrete topology, U is the forgetful functor, and I is the indiscrete topology. All the functor sends objects and morphisms to itself. But note that D, I are different functors, equipping any set X with discrete and indiscrete topology respectively. Note that this relies on the fact that any function is continuous in both the discrete and indiscrete topology. Then, we use 3.1.4 (2) to show (D, U) is an adjoint pair. We define $\eta_A : A \to A$ to be the identity function. Then for any topological space B, a function $f : A \to B$, we need to find *continuous function* $f^\# : DA \to B$, note that a morphism $DA \to B$ means a set-function $A \to B$ with A equipped the discrete topology. This can be done just by defining $f^\# = f$ (in the set-function level, then $f^\#$ is automatically continuous), so clearly $U(f^\#) \circ \eta_A = f$ Since $U(f^\#) = f^\# = f$ and η_A is identity. A very similar reason shows (U, I) is an adjoint pair. As we will see in Example 3.3.8, it is not generally the case that the forgetful functor U has a left adjoint. $\triangle$

Now we use condition (3) to check for an adjoint.

Example 3.1.8. We return to the Example 1.2.7 and claim that the free functor F is a left adjoint to the forgetful functor U. We note that there are counit $\eta : 1_{\mathbf{Set}} \to UF$ by $\eta(A) : A \to F(A)$ by sending

elements $a \in A$ to a as (basis) element in the vector space $F(A)$. It is clearly a natural transformation as

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ {\scriptstyle \eta(A)}\downarrow & & \downarrow{\scriptstyle \eta(B)} \\ F(A) & \xrightarrow{F(f)} & F(B) \end{array}$$

commutes, since the map $F(f)$ is determined by specifying the target of the basis using f, and η act as identity on them. Similarly, one can define $\varepsilon : FU \to 1$ by $\varepsilon(V) : F(V) \to V$ by defining $F(V)$ to be the linear map defined *on basis*, sending the basis element $v \in F(V)$ to itself v in any vector spaces V. Similarly one can check it is a natural transformation, and for all set A, vector spaces V, the triangle equalities

$$\begin{array}{ccc} F(A) & \xrightarrow{F\eta(A)} & FF(A) \\ & {\scriptstyle 1_{F(A)}}\searrow & \downarrow{\scriptstyle \varepsilon(F(A))} \\ & & F(A) \end{array} \qquad \begin{array}{ccc} V & \xrightarrow{\eta(V)} & F(V) \\ & {\scriptstyle 1_B}\searrow & \downarrow{\scriptstyle \varepsilon(V)} \\ & & V \end{array}$$

clearly holds. Note that we have suppressed the forgetful functor U. △

3.2 Limits (and colimits)

So far, we have been dealing with special functors and their natural transformations, but we have not said much about how to build special *objects* from other objects in a category $\mathscr{C}$, this section aims to introduce a very general way using the notion of a *limit*. In turn, we will see a lot of the notions introduced in the previous section do "play well" with limits, in a sense that they *preserve* them.

As a motivation, what are the special objects that can be built from other objects in the Category $\mathbf{Set}$? One thing we can immediately think of is the cartesian product $X \times Y$, and in pure mathematics, there are a lot of products that use the cartesian products as their "Base Set": the product of groups $G \times H$, the product topology, the direct sum of vector spaces... etc. This indeed comes from the categorical definition of a *product*, which makes the above example a special case using the category $\mathbf{Set}, \mathbf{Grp}, \mathbf{Top}, \mathbf{Vect}_k$... etc.

We can think about how to define the product in full generality. A good observation comes from proving this lemma in topology:

Proposition 3.2.1. Given topological spaces T_1, T_2, and let X be any topological spaces. The function $F : X \to T_1 \times T_2$ is continuous if and only $p_1 \circ F : X \to T_1, p_2 \circ F : X \to T_2$ is continuous, where $p_1 : T_1 \times T_2 \to T_1, p_2 : T_1 \times T_2 \to T_2$ are the projections (that is, $p_1(t_1, t_2) = t_1, p_2(t_1, t_2) = t_2$).

So a continuous map *to* the product space can be found by composing it with the projections. It turns out that any map that has this property *defines* the product space up to isomorphism (homeomorphism in $\mathbf{Top}$). To see this, we introduce the general notion of a limit.

Definition 3.2.2. Given a category I, a **diagram of shape I in** $\mathscr{C}$ is a functor $D : I \to \mathscr{C}$.

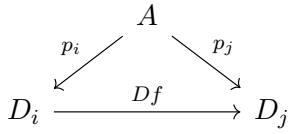
One might wonder why we define a rather convoluted term to represent a functor. The reason is we will often think of I as an *indexing category*: in particular, if I is a "small" category with very few objects and morphisms, then the functor D can be thought of as "picking out" elements of C. For example, if I is the discrete category with two objects (introduced in Example 1.1.10), then a functor $D : I \to \mathscr{C}$ sends object

$A \in I$ to an object $D_1 \in \mathscr{C}$, $B \in I$ to an object $D_2 \in \mathscr{C}$, and it sends identity functors to identity functors. That is all D has to entail.

In light of this, when we consider a diagram $D : I \to C$ instead of writing $D(i) \in \mathscr{C}$ for a fixed $i \in I$, we will write $D_i \in \mathscr{C}$ to mean the same thing. Sometimes, we will refer D_i as **entries** of D.

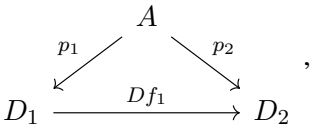
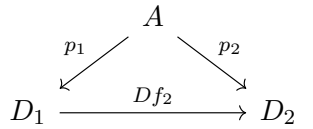
Remark. We have an unfortunate coincidence by calling *commutative diagrams* as *diagrams*. We will trust the reader to differentiate between the two uses of the same word. $\Diamond$

Definition 3.2.3. A **cone** of a diagram D (of shape I in $\mathscr{C}$) is a pair $(A, \{p_i\}_{i\in I})$ with the following properties:

- A is an object in $\mathscr{C}$ and it is called the **vertex** of the cone.
- $\{p_i\}_{i\in I}$ is a collection of morphisms $p_i : A \to D_i$ such that, for all $f : i \to j$ morphisms in i, the diagram  commutes. $\{p_i\}_{i\in I}$ are sometimes called the collection of **projections**.

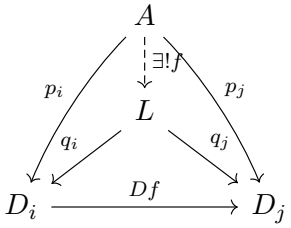
Looking at I being the discrete category with two objects again, Note that there are no morphisms between the two objects, so a cone of a diagram D which "picks out" object $D_1, D_2 \in \mathscr{C}$ with shape I is just a choice of an element $A \in \mathscr{C}$, and a choice of two morphisms $p_1 : A \to D_1, p_2 : A \to D_2$.

However, things get more complicated when we investigate categories with non-identity morphisms.

Example 3.2.4. Recall in example 1.1.10 we write down the category I : $\bullet \underset{f_2}{\overset{f_1}{\rightrightarrows}} \bullet$ We can ask what is a cone of this diagram (of shape I). Just like the discrete category case, a cone consists of the vertex $A \in \mathscr{C}$ and the projections $p_1 : A \to D_1, p_2 : A \to D_2$. But this time, there are two diagrams  ,  that needs to commute. We can see those two diagrams are really saying $p_2 = Df_1 \circ p_1 = Df_2 \circ p_1$. Note that Df_1, Df_2 is given by the diagram $D : I \to C$. So p_2 is completely determined by p_1, Df_1, Df_2. So we can, in principle define a cone as (A, p_1) for which $Df_1 \circ p_1 = Df_2 \circ p_1$. $\triangle$

Now we are ready to introduce what a limit is.

Definition 3.2.5. A **limit cone** of a diagram $D : I \to \mathscr{C}$ is a cone of D $(L, \{q_i\}_{i\in I})$ with the following property: For any $(A, \{p_i\}_{i\in I})$ any cone of D, there exists a unique morphism $f : A \to L$ such that the following diagram commutes:

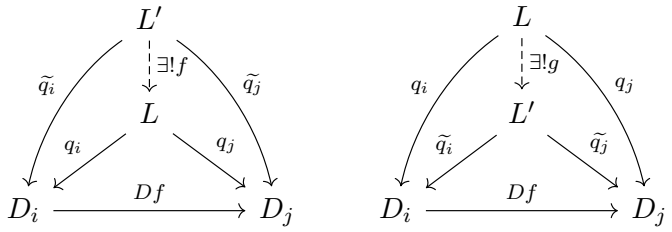


for any $i, j \in I$. We call L a **limit** of the diagram $D : I \to \mathscr{C}$, and write $L = \varprojlim D$ or $\varprojlim D_i$.

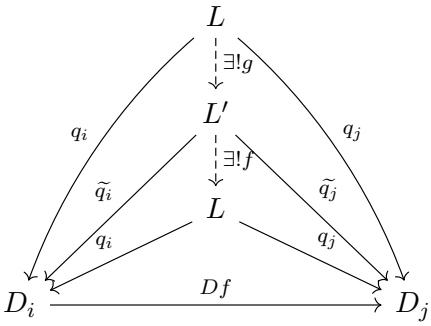
Note that we do not assert the existence of L and indeed sometimes it might not exist - see Example 3.2.12. However, in the case that it did, L will be *the* limit:

Proposition 3.2.6. If a limit exists, then it is unique up to isomorphism of objects in $\mathscr{C}$.

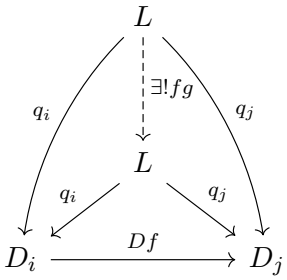
Proof. Suppose $(L, \{q_i\}_{i\in I}), (\widetilde{L}, \{\widetilde{q_i}\}_{i\in I})$ are limit cones of L, then just by the commutative diagrams in the definition we can assert the unique existence of two functions $f : L' \to L, g : L \to L'$ as follows:



Now we can "glue" the blue part and yield the commutative diagram:



, which simplifies to



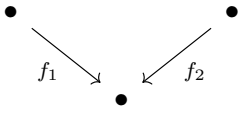
Note that in the dotted arrow of the right hand side diagram, the map 1_L also commutes with the q_is. But the property states such a map is unique, so one must have $fg = 1_L$. A similar argument yields $gf = 1_{L'}$, hence $L \cong L'$ by definition of isomorphism. □

Due to this, when we write expressions like $\varprojlim D = A$, what we mean secretly is $\varprojlim D \cong A$. In this case, we will interchange the use of $=$ and $\cong$ frequently.

This proof technique is similar to Example 2.1.1 where we showed the free vector space, defined via a universal property, must be isomorphic. And seeing the free functor F has a (right) adjoint, one might expect the limit is some kind of adjoint too. Indeed, we will investigate this phenomenon in Section 3.3.

For some important index categories I, there are special names to the limit.

Definition 3.2.7. Let $D : I \to \mathscr{C}$ be a diagram. Then if I is:

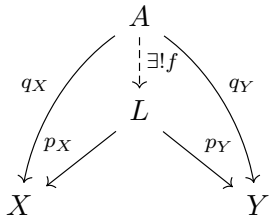
- the empty category, then the limit of D is called a **terminal object**.
- the discrete category with two objects • •, then the limit of D is called a **binary product**.
- the category • $\overset{f_1}{\underset{f_2}{\rightrightarrows}}$ •, then the limit of D is called an **equalizer**.
- the category 

, then the limit of D is called a **pullback**

(of the diagram D).

Example 3.2.8. We trace the definition of L being a terminal object, noting that if I is the empty category, then there will be no projections in the cone, so any object $A \in \mathscr{C}$ will be a cone. The property that L satisfies can just be simply stated as "for any object A, there is a unique morphism $f : A \to L$", which is a common definition of terminal objects. △

Example 3.2.9. Let I be the discrete category with two objects. If $\mathscr{C} = \mathbf{Set}$, any diagram $D : I \to \mathbf{Set}$ picks out two sets X, Y. We claim that the limit cone to this diagram is the Cartesian Product with the projections $(X \times Y, \{p_X, p_Y\})$, with $p_X(x, y) = x, p_Y(x, y) = y$. For any cone $(A, \{q_X, q_Y\})$, we need to find function $f : A \to X \times Y$ such the diagram



commutes. That is, $p_X \circ f = q_X, p_Y \circ f = q_Y$. So writing $f(a) = (f_1(a), f_2(a))$ (for any $a \in A$), the commuting requirement means $f_1(a) = q_X(a), f_2(a) = q_Y(a)$ that determines the function $f(a) = (f_1(a), f_2(a))$, and hence the uniqueness and existence is proved. △

Remark. The above Example 3.2.9 can be generalized to any discrete category, and the limit is the (potentially) infinite Cartesian product of sets.

Also note that the above example immediately generalized to vector spaces with the product of V, W (where the discrete category picks out) is the direct sum of vector spaces $V \oplus W$. Note that in this example, we have the projections $V \oplus W \to V$, which are in the definition of a *external* direct sum in linear algebra. ◊

Example 3.2.10. Let us look at an example of an equalizer. Consider $\mathscr{C} = \mathbf{Grp}$, I is the index category for the equalizer, and D is defined in figure 3.1. where θ is any group homomorphism $G \to H$, and ε is the

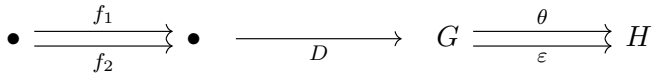
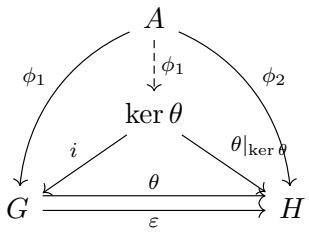


Figure 3.1: The diagram D in equalizers.

trivial homomorphism that sends every element in the group G to the identity element e_H in H. We claim the equalizer $\lim D = \ker \theta$, more concretely, we claim the limit of the above diagram presented in figure 3.1 is $(\ker \theta, \{i, \theta|_{\ker \theta}\})$, where $i : \ker \theta \to G$ is the inclusion map. To see this, note that we investigated the cones of a diagram of shape I in Example 3.2.4, and a cone consists of a pair (A, ϕ_1) such that $\theta \circ \phi_1 = \varepsilon \circ \phi_1$, that is, $\theta(\phi_1(a)) = e_H$ for all $a \in A$, that is, $\phi_1(a) \in \ker \theta$ for all $a \in A$. So, ϕ_1 defines a map from $A \to \ker \theta$.

It is clear then the commutative diagram



indeed commutes. △

Example 3.2.11. Here is an example of pullbacks. Let us work in $\mathscr{C} = \mathbf{Set}$ and consider the diagram D where a limit is a pullback in figure 3.2 (bear in mind that if we are given I, then X, Y, Z, s, t are given). We

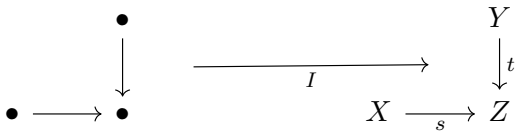
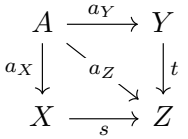


Figure 3.2: A diagram of pullback.

claim that in this case, the limit exists and $\lim D = \{(x, y) : s(x) = t(y)\}$. To see this, consider what a cone of D would entail. It asserts the existence of projections p_X, p_Y, p_Z such that the diagram



commutes. With the same vein in Example 3.2.4, a_Z is determined by a_X, a_Y. Now we claim the limit is $(L := \{(x, y) : s(x) = t(y)\}, \{p_X, p_Y, p_Z\})$ with $p_X(x, y) = x, p_Y(x, y) = y, p_Z = s \circ p_X$. Again, to show that it is a limit, we need to define a function $f : A \to L$ such that the limit diagram 3.3 commutes. Since one

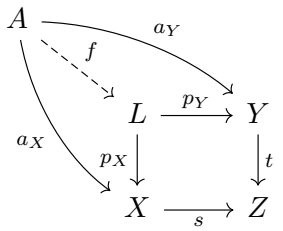


Figure 3.3: The commutative diagram for verifying L is a limit.

must have $p_X \circ f = a_X$ and $p_Y \circ f = a_Y$, by similar argument to 3.2.9 we must have $f(a) = (a_X(a), a_Y(a))$. We can check that f is well defined since $s \circ a_X = t \circ a_Y$ by the commutative diagram. So f is uniquely determined as required. △

Note that the pullback above generalized a lot of useful concepts:

- Let $f : X \to Y$ be a set-function where and take $Z \subseteq Y$, then the preimage $f^{-1}(Z)$ is the pullback

of the diagram $\begin{array}{ccc} & & Z \\ & & \downarrow i \\ X & \xrightarrow[f]{} & Y \end{array}$, where i is the inclusion. Note that following the definition above we have the limit as $\{(x,z) \in X \times Z : f(x) = z \text{ for some } z \in Z\}$, but there is a bijection with the set $f^{-1}(Z)$. This shows limit is only defined up to an isomorphism.

- Given two sets $X \subseteq Z, Y \subseteq Z$, the set $X \cap Y$ is $i_X^{-1}(Y)$, where $i_X : X \to Z$ is the inclusion, so it is also a pullback since it is a preimage).

Example 3.2.12. We now meet an example of a category where a limit might not exist. Consider $\mathscr{C} = \mathbf{Fld}$ the category of fields and field homomorphisms. Consider Example 3.2.9 where the functor D picks out $\mathbb{F}_2$ and $\mathbb{F}_3$, the field of two elements and three elements. Note that a cone consists of a field A with maps $A \to \mathbb{F}_2$ and $A \to \mathbb{F}_3$. But every field homomorphism preserves the characteristics of the field, so A, if exists, does not even have a well-defined characteristic. This contradicts the fact that every field has a characteristic, hence no cones, let alone limit cones, exist. △

Definition 3.2.13. A category $\mathscr{C}$ **has limits of shape** I if for any diagram $D : I \to \mathscr{C}$, $\varprojlim D$ exists.

We put some of our extra attention to $\mathbf{Set}$. It turns out that it has limits of shape I for every (small) category I.

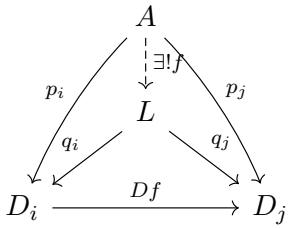
Proposition 3.2.14. If $D : I \to \mathbf{Set}$ is a diagram, then

$$\varprojlim D = \{(x_i)_{i \in I} : \text{If } f : i \to j \text{ is a morphism in } I, \text{ then } Df(x_i) = x_j\}.$$

where $(x_i)_{i \in I}$ is an element in the set $\prod_{i \in I} X_i$ (the cartesian product of the family $\{X_i\}_{i \in I}$) Note that the set L have natural projection maps $q_j : L \to D_j$ by $(x_i)_{i \in I} \mapsto x_j$.

Remark. The existence of $\prod_{i \in I} X_i$ for any family I relies on the axiom of choice. ◊

Proof. Let $(A, \{p_i\}_{i \in I})$ any cone of D with vertex A. To have

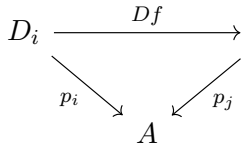


this diagram commutes, $q_i(f(a)) = p_i(a)$, where the p_is are given, so the only definition for $f : A \to L$ is $f(a) = (p_i(a))_{i \in I}$ and the diagram clearly commutes. □

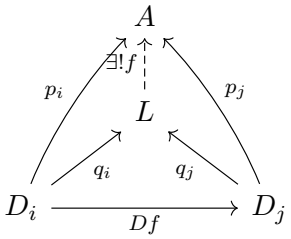
One might think of dualizing the definition of limit. This is in turn quite useful as it produces another class of special object called *colimit*:

Definition 3.2.15. A **cocone** of a diagram D (of shape I in $\mathscr{C}$) is a pair $(A, \{p_i\}_{i \in I})$ with the following properties:

- A is an object in $\mathscr{C}$ and it is called the **vertex** of the cone.

- $\{p_i\}_{i\in I}$ is a collection of morphisms $p_i : A \to D_i$ such that, for all $f : i \to j$ morphisms in I, the diagram  commutes. $\{p_i\}_{i\in I}$ are sometimes called the collection of **coprojections**.

Definition 3.2.16. A **colimit cocone** of a diagram $D : I \to \mathscr{C}$ is a cocone of D $(L, \{q_i\}_{i\in I})$ with the following property: For any $(A, \{p_i\}_{i\in I})$ any cocone of D, there exists a unique morphism $f : L \to A$ such that the following diagram commutes:



for any $i, j \in I$. We call L a **colimit** of the diagram $D : I \to \mathscr{C}$, and write $L = \varinjlim D$.

Example 3.2.17. Using similar techniques as above, one can find colimits in different categories.

- In $\mathbf{Set}$, binary coproducts are a disjoint union of two sets.
- In $\mathbf{Vect}_k$, binary coproducts are also the direct sum of two vector spaces. Recall the theorem in linear algebra that the internal and external direct sum are isomorphic. The coprojection $V \to V \oplus W$ is in the definition of *internal* direct sum.

△

Remark. One useful intuition to distinguish between product and coproduct is as follows. Given a random object A in a category $\mathscr{C}$, finding the limit of a diagram D is similar to asking the question:

> Knowing the (projection) maps $A \to D_i$ for any i, what is the object L that the above determines a map $A \to L$?

While finding the coproduct ask the "dual" question:

> Knowing the (coprojection) maps $D_i \to A$ for any i, what is the object L that the above determines a map $L \to A$?

We practice this intuition in the example below, leaving the details for readers to check. ◊

Example 3.2.18 (Differences between Infinite Product and Coproduct)**.** Consider the category $\mathbf{Vect}_k$, and I the discrete category of countable many objects labelled by the set $\{1, 2, \ldots\}$. Suppose we have a diagram $D : I \to \mathbf{FDVect}_k$ and write $D_i := V_i$, what is its limit (i.e. the infinite product) $\prod_{i=1}^{\infty} V_i$ and colimit (i.e. the infinite coproduct) $\oplus_{i=1}^{\infty} V_i$? To see the limit, suppose we have a vector space T and we know all the maps $T \to V_i$ for all i, then we can actually determine a map from T to the space of all sequences:

$$\{(x_i)_{i=1}^{\infty} : x_i \in V_i \text{ for all } i \geqslant 1\},$$

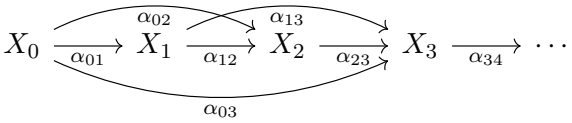
which we claim will be $\prod_{i=1}^{\infty} V_i$ (another name of this space is the *direct product*).

Conversely, if we are given all the maps $V_i \to T$, note that this only can determine a map to T where we extend the above by adding *finitely many* linear combinations. Then, the space we seek for is:

$$\{(x_i)_{i=1}^{\infty} : x_i \in V_i \text{ for all } i \geqslant 1 \text{ and } x_i = 0 \text{ for all but finitely many } i \geqslant 1\}$$

, and we claim this is $\oplus_{i=1}^{\infty} V_i$ (another name of this space is the *direct sum*). In particular, note that in this case $\oplus_{i=1}^{\infty} V_i$ is a subspace of $\prod_{i=1}^{\infty} V_i$ but they are not equal. △

Example 3.2.19 (Direct Limit)**.** We pay special attention to the colimit associated with the poset category $\mathbb{N}_{\leqslant}$ ($\{0, 1, 2 \cdots\}$ with the usual ordering) in the category $\mathbf{Vect}_k$. Hence, the diagram "assigns" vector spaces and morphisms:



As with the previous examples, since the diagram is commutative, so for example the arrow α_{02} will not be drawn as $\alpha_{02} = \alpha_{12}\alpha_{01}$. Therefore, it is usual to de-clutter the above picture and just write $X_0 \xrightarrow[\alpha_0]{} X_1 \xrightarrow[\alpha_1]{} X_2 \xrightarrow[\alpha_2]{} X_3 \xrightarrow[\alpha_3]{} \cdots$. This diagram is called a *direct system* in older literature.

In $\mathbf{Vect}_k$, the colimit can be realized as the quotient vector space $W := \oplus_{i=0}^{\infty} X_i / \sim$, where $\sim$ is the following equivalence relation between two elements $x_i, y_j \in \oplus_{i=0}^{\infty} X_i$, where $x_i \in X_i, y_j \in X_j$:

$$x_i \sim y_j \qquad \text{if and only if} \begin{cases} i = j \text{ and } x_i = y_j, \textbf{ OR} \\ x_i = \alpha_{ji}(y_j), \textbf{ OR} \\ y_j = \alpha_{ij}(x_i). \end{cases}$$

Then one can check that it is the colimit as required. △

3.3 Limit Preserving

Now we have a brief look at how adjoints and limits interact. A more detailed treatment and the main reference of this section is [Lei14, Chapter 6].

To motivate this section, we can think about the case where $F(S)$ is the free vector space, in Example 1.2.7. How can we show $F(S) \oplus F(T) = F(S \times T)$? We can write down a proof with the constructions, basis, and the "normal" definition of the two vector spaces, but a quicker solution is to realize that $\oplus$ and $\times$ are binary products on the categories $\mathbf{Vect}_k$ and $\mathbf{Set}$ respectively, and argue that we can "move out" the limit. This is the perspective we want to take.

Definition 3.3.1. Let $D : I \to \mathscr{C}$ a diagram. A functor F is **continuous / preserve limits** if $\varprojlim F(X_i) = F(\varprojlim(X_i))$.

Remark. Sometimes the word "preserves limit" means the *limit cones* are preserved by F, which is a stronger condition than defined above. ◊

Definition 3.3.2. Given a category I, the **diagonal functor** $\Delta : \mathscr{C} \to [I, \mathscr{C}]$ is specified by:

- An object $A \in \mathscr{C}$ is send to the functor ΔA that sends any $i \in I$ to A and any $f : i \to j$ morphism in I to the identity moprhism in A.

- A morphism $f : A \to B$ in $\mathscr{C}$ is sent to a natural transformation $\Delta f : \Delta A \to \Delta B$ which is defined by $f(i) := f$ for any $i \in I$.

Note that the diagonal functor gives us another way to look at limit cones of a fixed diagram D with vertex at A:

Proposition 3.3.3. Let D be a diagram, then

$$\mathrm{Hom}_{[I,\mathscr{C}]}(\Delta A, D) = \mathrm{Cone}(A, D). \tag{3.4}$$

Proof. This is by tracing the definition of $\mathrm{Hom}_{[I,\mathscr{C}]}(\Delta A, D)$. An element in this set is a natural transformation specified by $\alpha(i) : \Delta(A)(i) \to D(i)$, which is $\alpha(i) : A \to D_i$, with the requirement that for all $f : i \to j$ morphism in I:

$$\begin{array}{ccc} \Delta A & \xrightarrow{1_A} & \Delta A \\ {\scriptstyle \alpha(i)}\downarrow & & \downarrow{\scriptstyle \alpha(j)} \\ D_i & \xrightarrow[D(f)]{} & D_j \end{array}$$

commutes, which is exactly the commuting requirement for a cone. □

Note that for any diagram $D : I \to \mathscr{C}$ one forms the limit (as an object in $\mathscr{C}$) $\varprojlim D$,with projection maps $\{p_i\}_{i\in I}$. So $\varprojlim$ can be thought of as a functor $[I, \mathscr{C}] \to \mathscr{C}$ as follows:

- A diagram $D : I \to \mathscr{C}$ is sent to $\varprojlim D$, and
- Given a natural transformation $\{\alpha_i\}_{i\in I}$ as a morphism in $[I, \mathscr{C}]$ of two diagrams D, D' (where $\alpha_i : D(i) \to D'(i)$) to the unique map induced by the cone $\alpha_i p_i : \varprojlim D \to D'(i)$.

Proposition 3.3.4. $(\Delta, \varprojlim)$ is an adjoint pair. So by the definition of adjoint, we can write:

$$\mathrm{Hom}_{[I,\mathscr{C}]}(\Delta A, D) \cong \mathrm{Hom}_{\mathscr{C}}(A, \varprojlim D) \tag{3.5}$$

natural in A and D.

Proof. Note that the left adjoint Δ is given, hence we find the right adjoint R by rewriting the condition in Corollary3.1.5. The condition reads: for any $D \in [I, \mathscr{C}]$, there is RD and $\varepsilon_D : \Delta RD \to D$ such that for any $f : \Delta B \to D$, there is a unique $f^\flat : B \to RD$ with $\varepsilon_A \circ \Delta(f^\flat) = f$. But □

Proposition 3.3.5. Let D be a diagram with entries D_i, then

$$\mathrm{Cone}(A, D) \cong \varprojlim \mathrm{Hom}_{\mathscr{C}}(A, D_i). \tag{3.6}$$

Proof. Note that $\mathrm{Hom}_{\mathscr{C}}(A, D_i)$ is a set, then this immediately follows from Proposition 3.2.14 by reading the definition of that limit. □

Equation (3.4) to (3.6) concludes

$$\mathrm{Hom}_{\mathscr{C}}(A, \varprojlim D_i) \cong \varprojlim \mathrm{Hom}_{\mathscr{C}}(A, D_i),$$

which means the functor $\mathrm{Hom}_{\mathscr{C}}(A, -)$ preserves limits, and the isomorphism is natural in $A \in \mathscr{C}$.

Theorem 3.3.6. If $R : \mathscr{C} \to \mathscr{D}$ is a right adjoint, then it preserves limits.

Proof. Fix any $A \in \mathscr{C}$ with a diagram $D \in [I, \mathscr{D}]$ with entries D_i, From here it is a routine calculation with adjointness and the fact that $\mathrm{Hom}_{\mathscr{C}}(A, -)$ preserves limits.

$$\begin{aligned}
\mathrm{Hom}_{\mathscr{C}}(A, R(\varprojlim D_i)) &\cong \mathrm{Hom}_{\mathscr{D}}(LA, \varprojlim D_i) && \text{by adjointness} \\
&\cong \varprojlim \mathrm{Hom}_{\mathscr{D}}(LA, D_i) && \text{by limit preserving } \mathrm{Hom}_{\mathscr{D}}(LA, -) \\
&\cong \varprojlim \mathrm{Hom}_{\mathscr{C}}(A, RD_i) && \text{by adjointness} \\
&\cong \mathrm{Hom}_{\mathscr{C}}(A, \varprojlim RD_i) && \text{by limit preserving } \mathrm{Hom}_{\mathscr{C}}(A, -)
\end{aligned}$$

So $\mathrm{Hom}_{\mathscr{C}}(A, R(\varprojlim D_i)) \cong \mathrm{Hom}_{\mathscr{C}}(A, \varprojlim RD_i)$ for all $A \in \mathscr{C}$, and all of those isomorphism are natural in A. Hence $R(\varprojlim D_i) \cong \varprojlim RD_i$ by Corollary 2.3.5 of Yoneda Lemma. □

By duality one would get

Corollary 3.3.7. If $L : \mathscr{D} \to \mathscr{C}$ is a left adjoint, then it preserves colimits.

Example 3.3.8 (Non-existence of adjoints)**.** (co)Limit preserving is a very useful criterion to check the non-existence of adjoints. Consider the forgetful functor $U : \mathbf{Grp} \to \mathbf{Set}$, we claim that there is no functor $F : \mathbf{Set} \to \mathbf{Grp}$ such that $U \dashv F$. If there is, U would preserve colimits since it is a left adjoint. So given two groups G, H, we would have $U(G \otimes_{\mathbf{Grp}} H) = U(G) \otimes_{\mathbf{Set}} U(H)$, where we temporarily denote $\otimes_{\mathscr{C}}$ as the colimit in a category $\mathscr{C}$ of two objects. But it turns out that the colimit of a group is the *free product*. For example, if $G = \mathbb{Z}/3\mathbb{Z} = \langle x | x^3 = 0 \rangle$, $H = \mathbb{Z}/2\mathbb{Z} = \langle y | y^2 = 0 \rangle$, the free product $G \otimes_{\mathbf{Grp}} H$ is actually $\langle x, y | x^3 = y^2 = 0 \rangle$. This is an infinite set, so it is not isomorphic to $G \otimes_{\mathbf{Set}} H$ as a set (i.e. $U(G) \otimes_{\mathbf{Set}} U(H)$), as $\otimes_{\mathbf{Set}}$ is only the disjoint union, so the disjoint union has 5 elements. △

Part II

Category Theory in Non-commutative Algebras

Chapter 4

Algebra, Modules, and Quiver Representations

4.1 Modules over Algebras

In this section, we will introduce algebras and modules over them as a way to combine the study of rings and vector spaces. These modules will be building blocks for us to do homological algebra, which is a powerful tool as introduced in Chapter 5. Our main reference is Chapters I to III of [ASS06].

Definition 4.1.1. If R is a ring, recall a **right R-module** is a set $(M, +, \cdot)$ where $\cdot$ is a "multiplication on the right by R" map $\cdot : M \times R \to M$ such that for all $r, s \in A$ and $x, y \in M$ we have

- $(x + y) \cdot r = x \cdot r + y \cdot r$
- $x \cdot (r + s) = x \cdot r + x \cdot s$
- $x \cdot (rs) = (x \cdot r) \cdot s$
- $x \cdot 1 = x$.

The notion of a **left R-module** is defined similarly with the multiplication map defined to attain the type $R \times M \to M$.

Definition 4.1.2. A **k-algebra** A is a ring that is also a k-vector space. Moreover, scalar multiplication and elementwise multiplication agree in any entries. That is, for any $a, b \in A, \lambda \in k$, we have $\lambda(ab) = (a\lambda)b = a(\lambda b) = (ab)\lambda$

Note that k is always a subset of a k-algebra by having λ as $\lambda(1)$, so multiplying an element of the ring and a "scalar" in the above definition makes sense. Here are more definitions inherited from the basic theory of rings and vector spaces.

Definition 4.1.3. Given A a k-algebra,

- A is **finite-dimensional** if it is finite dimensional as a vector space over k.
- I is an **left** (resp. **right**) ideal of A if I is a left (resp. right) ideal of A considered as a ring, and it is a vector subspace of A considered as a vector space.
- A **k-algebra homomorphism** is a k-linear map that is also a ring homomorphism.

Echoing with Part I, we present a way to look at a k-algebra A using commutative diagrams. First, we assert A is an object in $\mathbf{Vect}_k$. Then, note that the additional ring structure of A induces an elementwise multiplication function $m : A \times A \to A$, and the function is bilinear. Now recall in Definition 2.1.3 where bilinear maps are in one-to-one correspondence with linear maps in tensor products. That is, the multiplication specifies a linear map $m : A \otimes A \to A$. m satisfies additional properties. For example, the associativity property can be written in a commutative diagram below:

$$\begin{array}{ccc} A \otimes A \otimes A & \xrightarrow{id \otimes m} & A \otimes A \\ \downarrow{\scriptstyle m \otimes id} & & \downarrow{\scriptstyle m} \\ A \otimes A & \xrightarrow{\quad m \quad} & A \end{array}$$

where maps like $id \otimes m$ are understood as the map where a (basis) element $a \otimes b \otimes c$ is sent to $a \otimes bc$.

For the existence of additive identity 1_A, one can think of the existence of an element in A as the existence of a linear map $\eta : k \to A$ (since η is completely determined by $\eta(1)$), then the existence of identity can be thought of as the following commutative diagram:

$$\begin{array}{ccccc} k \otimes A & \xleftrightarrow{iso} & A & \xleftrightarrow{iso} & A \otimes k \\ \downarrow{\scriptstyle \eta \otimes id} & & \downarrow{\scriptstyle id} & & \downarrow{\scriptstyle id \otimes \eta} \\ A \otimes A & \xrightarrow[m]{} & A & \xleftarrow[m]{} & A \end{array}$$

where we assert there are isomorphisms between $k \otimes A$, $A \otimes k$ and A.

Our focus in the whole Part II will be studying (right) modules over a finite-dimensional k-algebra A. That is, the category $\mathbf{Mod}$-A.

Definition 4.1.4. Let A be a k-algebra, a **right A-module** is a vector space $(M, +)$ over k with a "multiplication on the right by A" map $\cdot : M \times A \to M$ such that for all $r, s \in A$ and $x, y \in M$, $\lambda \in k$ we have

- $(x + y) \cdot r = x \cdot r + y \cdot r$
- $x \cdot (r + s) = x \cdot r + x \cdot s$
- $x \cdot (rs) = (x \cdot r) \cdot s$
- $x \cdot 1 = x$.
- $(x\lambda)r = x(\lambda r) = (xr)\lambda$.

where rs is the multiplication of the algebra. The notion of a **left A-module** is defined similarly with the multiplication map defined to attain the type $A \times M \to M$.

An **A-module homomorphism** is a k-linear map between two right A-modules $h : M \to N$ such that $h(ma) = h(m)a$ for all $m \in M, a \in A$.

Remark. Again, the requirement can be packed into a commutative diagram, note that right multiplication specifies a linear map $r : M \otimes A \to M$, and it satisfies two commutative diagrams:

$$\begin{array}{ccc} M \otimes A \otimes A & \xrightarrow{r \otimes id} & M \otimes A \\ \downarrow{\scriptstyle id \otimes m} & & \downarrow{\scriptstyle r} \\ M \otimes A & \xrightarrow[r]{} & M \end{array} \qquad \begin{array}{ccc} M \otimes k & \xrightarrow{id \otimes \eta} & M \otimes A \\ & {\scriptstyle iso} \searrow \quad \swarrow {\scriptstyle r} & \\ & M & \end{array} ,$$

where $\eta : K \to A$ is the existence of unit element 1 in A. For example, the diagram on the right is the requirement $x \cdot 1 = x$. ◊

All of those setups will lead us to:

Definition 4.1.5. There is a **category of k-algebras** with objects as k-algebras and morphisms as k-algebra homomorphism, we denote it by k-$\mathbf{Alg}$. Fix an k-algebra A, then there is also a **category of right A-modules**, denoted by A-$\mathbf{Mod}$, with objects as right A-modules, and morphisms as A-module homomorphisms.

Important Working Assumptions (2)

From now on, modules always mean *right* modules unless otherwise specified. Also, we fix an algebraically closed field (say $\mathbb{C}$) k and fix a k-algebra A.

As with the study of groups, rings, and vector spaces, one would expect the same basic notation for A-modules. The most important construction, as usual, will be forming a quotient. In groups we quotient through normal subgroups, in rings through ideals, in vector spaces through vector subspaces, and in modules, we quotient through submodules.

Definition 4.1.6. N is a **submodule** of an A-module M if N is a vector subspace of M such that $n \cdot a \in N$ for all $a \in A$.

Definition 4.1.7. Let N be a submodule of M, the **quotient module** M/N is the quotient vector space M/N, with multiplication given by: $[m] \cdot r = [mr]$ as equivalence classes.

Of course, one has the first isomorphism theorem.

Proposition 4.1.8. Let $f : M \to N$ be a A-module homomorphism, then $M/\ker f \cong \operatorname{im} f$.

Those constructions have analogous proofs in the case of vector spaces, so we will not repeat them here.

In Part I, we introduced a lot of examples using the category $\mathbf{Vect}_k$, one should be aware that they all work in the category A-$\mathbf{Mod}$ with the same reason, as long as we define the multiplication map $M \times A \to M$ carefully. For example, the direct sum of two A-modules $M_1 \oplus M_2$ has multiplication defined as $(m_1, m_2)a := (m_1 a, m_2 a)$. This also works for finitely many A-modules, and one can check, similar to Example 3.2.9, we can see finite direct sum is both a (finite) product and a (finite) coproduct in the category $\mathbf{Mod}$-A.

One basic example of an A-module is the algebra A itself. It might seem too trivial to consider it as a module, but in turn, a lot of important information can be extracted from thinking about this module.

Example 4.1.9. Any algebra A is both a left A-module and a right A-module. We write ${}_A A$ and A_A respectively when we consider A with the respective module structure. This is also referred to the **left** and **right regular A-module** respectively. △

Example 4.1.10 (Duals)**.** Given a right A-module M, M is a k-vector space, so we are interested in the dual space M^*, it turns out there is a *left* A-module structure by defining left multiplication $a\eta$ for $a \in A, \eta \in M^*$ by $a\eta(m) := \eta(ma)$. One can check this gives a left A-module. Similarly, if N is a left A-module, N^* will be a right A-module.

In the same vein, from a right A-module M, one can also define a left A-module structure from the set $\operatorname{Hom}_{A\text{-}\mathbf{Mod}}(M, A)$, i.e. the set of all A-module homomorphism from M to A, also from the same formula in the definition above. △

Example 4.1.11. Recall A is also a ring, any left (resp. right) ideal will be a left (resp. right) A-module, and in fact is a submodule of ${}_AA$ (resp. A_A). $\triangle$

There are some features in A-modules that are very special in understanding them.

Definition 4.1.12. An A-module M is **simple** if the only submodules of M are $\{0\}$ and M itself.

Definition 4.1.13. Given an A-module M, we say M is **indecomposable** if it cannot be written in the form of $M \cong M_1 \oplus M_2$ (as A-modules) for some nontrivial (that is, not 0 or M) A-modules M_1, M_2.

Remark. Clearly, a simple module is indecomposable. However, the converse need not hold. We will see this in later sections. $\Diamond$

Recall a n-dimensional vector space over k can always be seen as a direct sum of k (that is, there is an isomorphism with it and k^n), and k itself is indecomposable and simple. A similar phenomenon happens in finite-dimensional k-algebras.

Proposition 4.1.14 (Krull-Schmidt Theorem)**.** Let A be a finite dimensional k-algebra, M is an A-module that is a finite-dimensional k-vector space, then if k is algebraically complete (say $k = \mathbb{C}$),

(a) M can be written in the form $M \cong M_1 \oplus \cdots \oplus M_m$, where $M_1, \ldots, M_m$ are indecomposable A-modules, and

(b) If $M \cong \bigoplus_{i=1}^m M_i \cong \bigoplus_{j=1}^n N_j$, where M_i and N_j are indecomposable, then $m = n$ and there exists a permutation σ of $\{1, \ldots, n\}$ such that $M_i \cong N_{\sigma(i)}$ as A-modules for each $i = 1, \ldots, n$.

Remark. Note that we generally cannot require the M_is in this theorem to be both indecomposable and simple. $\Diamond$

We omit the proof here. This asserts every A-module has a unique decomposition into indecomposable modules up to ordering. As an important example, we want to look at the decomposition of the regular right A-module.

Definition 4.1.15. Given a k-algebra A, and $e \in A$:

- e is an **idempotent** if $e^2 = e$. Note that $0, 1$ are always idempotents, so if $e \neq 0, 1$, it is called a n**non-trivial idempotent**.
- e is **orthogonal** to e' if $ee' = e'e = 0$, and
- e is **primitive** if there does not exists non-zero orthogonal idempotent $e_1, e_2 \in A$ such that $e_1 + e_2 = e$.

Proposition 4.1.16. Let $e_1, e_2 ... e_n$ be non-trivial, pairwise orthogonal idempotents of a finite-dimensional k-algebra A such that $\sum_{i=1}^n e_i = 1$. We have the decomposition

$$A_A = e_1A \oplus e_2A ... \oplus e_nA. \tag{4.1}$$

as right A-modules, Where $e_iA = \{e_ia | a \in A\}$.

Proof. Fix $1 \leqslant i \leqslant n$. Since e_iA is a right ideal of A, hence it is a submodule of A_A. Pick a basis $\{a_j\}_{j=1}^m$ of A as a k vector space, and notice the set $\{e_ia_j\}_{j=1}^m$ is spanning e_iA. Also, given any $a \in A$, we can write $a = 1a = \left(\sum_{i=1}^n e_i\right) a = \sum_{i=1}^n e_ia$. So every element in A can be written as a sum of elements in e_iA, $1 \leqslant i \leqslant n$.

Now, take two distinct idempotent e_i, e_j, and note that if there exists $a \in A$ such that $e_ia = e_ja$, multiplying both sides on the left by e_i we have $a = 0$ due to orthogonality. Hence $e_iA \cap e_jA = \{0\}$.

Combining the last two paragraphs yields the result. □

Proposition 4.1.17. Let e be an idempotent in a k-algebra A. Then e is primitive if and only if eA is an indecomposable A-module.

Proof. Since eA is a right ideal of A, hence it is an A-module. Now, if e is not primitive, suppose $e = e_1 + e_2$, where e_1, e_2 are non-trivial orthogonal idempotent, we can write $eA = e_1A \oplus e_2A$. Hence eA is not indecomposable.

Conversely, let e be a primitive idempotent. Suppose by contradiction that eA is decomposable and write $eA = U \oplus V$, then there exists $u \in U, v \in V$ such that $e = u + v$. Now $u \in U \subseteq eA$ so we can write $u = ea$ for some $a \in A$. Multiply $u = ea$ on the left by e yields $eu = ea = u$. Now, multiply $e = u + v$ on the right by u, and using $eu = u$, one have $u = u^2 + vu$, so $vu = u - u^2$. This implies $vu \in U$ since vu is a difference of elements in U. But $vu \in V$ since V is a submodule of eA (so it is closed under right multiplication). Hence $vu \in U \cap V$ and $vu = 0$, by symmetry $uv = 0$, and we have $u = u^2$, $v = v^2$. So u, v are orthogonal idempotents with $e = u + v$. This raises a contradiction, so eA is indecomposable as required. □

Hence, by the Krull-Schmidt theorem, if the idempotents in (4.1) are primitive, then it is the unique decomposition into indecomposable A-modules, given A is finite-dimensional.

4.2 Quivers Algebras and Quiver Representations

In this section, we will define what quivers are and investigate their algebraic operations. Quivers are graphs with directed edges. One also can make an elegant definition using category theory:

Definition 4.2.1. A **quiver** Q is a functor from the category $\bullet \underset{f_2}{\overset{f_1}{\rightrightarrows}} \bullet$ to **Set**.

Note that this encapsulates what a quiver would have intuitively. The category picks out sets VQ, EQ, called the **vertex set** and the **edge set** and functions $EQ \underset{t}{\overset{s}{\rightrightarrows}} VQ$, and $s : EQ \to VQ$ give each edge (in EQ) where to **start** or "go out" (also called the **source** of the edge), and $t : EQ \to VQ$ give each edge where to **end** or "goes to" (also called the **target** of the edge). The pair of starting vertex and ending vertex will sometimes be referred to as **oriented edge**.

Note that unlike some definitions of graphs favoured by most combinatorialists, we make no condition on the arrows' start and end vertices. This means a quiver can have self-loops and more than one edge between any two vertices. As with graphs, some basic properties of quivers will be of constant use. We will mainly borrow notations from [ASS06, §2.1].

Definition 4.2.2. Fix a quiver Q with vertex set VQ and edge set EQ.

- We say Q is **connected** if the underlying graph is connected.
- Given an positive integer n, a **path of length** n is a sequence of n arrows $(p_1, \cdots p_n)$, where the source of p_{i+1} is the same as the target of p_i for all $1 \leqslant i \leqslant n-1$. We say the path starts at the source of p_1 and ends at the target of p_n.
- A path is called an **oriented cycle** if it starts and ends at the same vertex. A cycle of length 1 is called a **loop**. If Q has no orient cycles, then we call Q **acyclic**.

- We also define paths of length 0: for each vertex $a \in VQ$, we realize the empty sequence of paths $e_a := ()$ as a path that starts and ends at a. We will denote this path as the **constant path at** a.

Example 4.2.3. Consider the following quiver We usually label vertices with positive integers. Note that

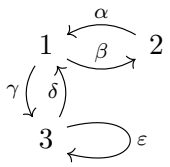


Figure 4.1: An example of a quiver.

this is a connected quiver with a loop at ε. An example of a path of length 2 is (α, β), which we write as $\alpha\beta$ if it does not cause confusion. This quiver has oriented cycles (for example the loop ε) and is connected. $\triangle$

Now, we want to associate algebra and module structures with a quiver.

Important Working Assumptions (3)

From now on, we will fix an algebraically closed field k, say $k = \mathbb{C}$. We always work with finite quivers that are connected. That is, we will assume both VQ, EQ are finite, and Q as an undirected graph is connected.

Definition 4.2.4. We denote kQ as the vector space over k with basis $\{p : p \text{ is a path}\}$.

If the underlying graph of a quiver is cyclic, then there are infinitely many paths since we can pick a loop ℓ and there is a path of length n $(\ell, \ell \cdots \ell)$ for any non-negative ℓ. In this case, kQ is an infinite dimensional vector space. In fact, kQ is not only a k-vector space, but it is also a ring if we can define multiplication and additive identity.

First, we define multiplication $kQ \times kQ \to kQ$. To do this, we can define a bilinear map by specifying the multiplication of basis elements. We do it as follows: given two paths P_1, P_2, if the endpoints of P_1 is the starting point of P_2, then define P_1P_2 to be the concatenation of path (note that it is an element of kQ), else, define $P_1P_2 = 0$.

Next, we define the identity on kQ. Given Q, we can label the vertices $v_1, \cdots v_n$, the constant paths $e_1, e_2 \cdots e_n$ as basis elements of kQ. We define $1_{kQ} := \sum_{i=1}^{n} e_i$. To check this is indeed the multiplicative identity in kQ, take any path P, then

$$P1_{kQ} = P\sum_{i=1}^{n} e_i = \sum_{i=1}^{n} Pe_i$$

by the linearity of multiplication. Note that P has one endpoint, call it e_j, then $Pe_i = \delta_{ij}P$ by how multiplication is defined on paths, then it follows $P1_{kQ} = P$, and similarly

$$1_{kQ}P = \left(\sum_{i=1}^{n} e_i\right) P = \sum_{i=1}^{n} (e_i)P = 1.$$

Hence kQ indeed is a ring.

Definition 4.2.5. The above construction makes kQ into an (k-)algebra, and it is commonly referred to as the **path algebra** of Q.

Here is a proposition that is easy to check, it highlights the reason why we want to define the constant path:

Proposition 4.2.6. The constant paths are primitive, pairwise orthogonal idempotent of any path algebras.

We are interested in the category $\mathbf{Mod}$-kQ. However, as with our assumption to Q, we will be more satisfied if kQ is actually finite-dimensional as vector spaces over k, and that will not happen if Q contains a self-loop. That does not mean it is completely irrelevant to study quivers with loops, rather, we can "eliminate redundant paths" by introducing another powerful tool in abstract algebra: quotients.

Definition 4.2.7. We denote the vector subspace of kQ with basis $\{p : p \text{ is a path of length larger than or equal to } \ell\}$ as I^ℓ. This is an ideal in kQ for $\ell \geqslant 1$.

An ideal I in kQ is **admissible** if there exists $l \geqslant 2$ such that $I^\ell \subseteq I \subseteq I^2$.

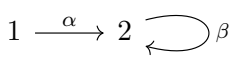
Remark.
- I^ℓ indeed is an ideal. To check this, since I^ℓ is a vector subspace, it is always closed under addition. Now given $i \in I_\ell$ and path p in Q (as a basis element of kQ), both ip and pi either is a path of no shorter length (by property of concatenation) than ℓ, or it is 0. Hence they are both in I^ℓ.
- The notation I^ℓ is not only a notation, but it is also the product of ideals of I^1 by ℓ times. Indeed, taking ℓ products of paths with lengths not smaller than 1 either yields 0 or a concatenation of ℓ length-1-or-more paths, that become a path of length ℓ or more.

$\Diamond$

Definition 4.2.8. Given a(n algebraically complete) field k, a quiver Q, and an admissible ideal I, the **quiver algebra** is the quotient algebra kQ/I.

In particular, if I is admissible, then since we assumed Q is finite, we have kQ/I is always a finite-dimensional algebra. We shall present an example here.

Example 4.2.9. Consider the following quiver Q:



Which has two vertices $1, 2$ and two arrows α, β. Then kQ is infinite-dimensional since the basis contains the paths

$$\{e_1, e_2, \alpha, \alpha\beta, \alpha\beta\beta, \alpha\beta\beta\beta \cdots, \beta, \beta\beta, \beta\beta\beta \cdots\} \tag{4.2}$$

(Note: we will denote $\beta\beta$ as β^2) et cetera. However, if we require I to be the ideal (subspace) generated by β^2. Another common way of saying this is the resulting quiver algebra kQ/I "has the relation $\beta^2 = 0$". Meaning, in the quiver algebra, elements like $\alpha\beta^2$ will be equivalent to α under the quotient. Hence, one can see the basis in (4.2) "reduces" to the basis $\{e_1, e_2, \alpha, \beta, \alpha\beta\}$. Note that they are equivalence classes of paths, but we overload the notation. That is, kQ/I is a 5-dimensional k-algebra. $\triangle$

The definition of admissible ideals might seem ad-hoc, but there is a good reason for this definition:

- The condition $I^\ell \subseteq I$ makes sure that all paths longer than a certain length are reduced to 0 in kQ/I, which ensures kQ/I to be a finite-dimensional vector space, since for a finite quiver Q, there can only be finitely many paths with bounded lengths.
- The condition $I \subseteq I^2$ makes sure we do not "forget" the structure of Q by making a directed edge immediately quotiented to 0.

It is hard to visualize what is a right kQ-module, but it turns out that adhering to the theory of quiver representations is enough for our purposes, as those are equivalent categories. We will introduce the basics below. Perhaps unsurprisingly, there is an elegant definition using category theory. Given a quiver Q, there is a category $\mathscr{C}_Q$ with the following data:

- Objects: VQ
- Given $v_1, v_2 \in VQ, , \mathrm{Hom}_Q(v_1, v_2) = \{p : p \text{ is a path from } v_1 \text{ to } v_2\}$.
- Composition: Concatenation of paths
- Identity: $\mathrm{Hom}_Q(v_1, v_1)$ is the constant path at v_1.

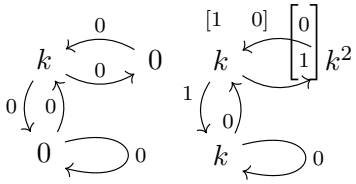
Definition 4.2.10. A **representation** M of a quiver Q is a functor from $\mathscr{C}_Q$ to $\mathbf{FDVect}_k$. The **category of representation** of a quiver Q is the functor category $[\mathscr{C}_Q, \mathbf{FDVect}_k]$, denoted as $\mathbf{Rep}_k^Q$.

Because we would need some notation, let us rewrite Definition 4.2.10 in a more colloquial way, following [ASS06, Definition III.1.1]. Since a representation M is a functor, the composition in Q (i.e. concatenation of paths) M respects the composition in vector space. So, to specify a representation, it is enough to specify the linear maps on every oriented edge of Q. We can therefore define

Definition 4.2.11. Given a quiver Q, A **representation** $(K_v, \phi_\alpha)_{v \in VQ, \alpha \in EQ}$ of Q consist of the following data:

- For every $v \in VQ$, a vector space K_v over k, and
- for every oriented edge α from v to w in Q, a k-linear map $\phi_\alpha : K_v \to K_w$.

Example 4.2.12. Here are some quiver representations:



Note the use of column and row vectors. We assume k and k^2 are given the standard basis, hence a linear map can be specified as a matrix. △

Note that we introduced the quiver algebra kQ/I in Definition 4.2.8 with an admissible ideal I. We can define a representation *satisfying the relation* in I. This follows [ASS06, Definition III.1.4].

Definition 4.2.13. Given any representation (K_v, ϕ_α) of a quiver Q and a path $p = (p_1, \cdots p_n)$, we can compose ϕ_{p_i} to obtain a linear map $\phi_p := \phi_{p_n} \circ \phi_{p_{n-1}} \cdots \circ \phi_{p_1}$. Since ϕ_p is now defined on all paths p, extend it linearly to give a definition for ϕ_ρ for all $\rho \in kQ$.

We say a representation is **bound by** an admissible ideal I if $\phi_\rho = 0$ for all $\rho \in I$.

Definition 4.2.14. Given a quiver Q and an admissible ideal I, we define the category $\mathbf{Rep}_k^Q/I$ to be the full subcategory of $\mathbf{Rep}_k^Q$ with objects being the representations bounded by I.

Theorem 4.2.15. $\mathbf{Mod}$-kQ/I and $\mathbf{Rep}_k^Q/I$ are equivalent categories.

Proof. Recall definition 2.2.1, given a quiver Q with n vertices, we need to construct functors $F : \mathbf{Mod}$-$kQ/I \to \mathbf{Rep}_k^Q/I$ and $G : \mathbf{Rep}_k^Q/I \to \mathbf{Mod}$-$kQ/I$ with desired properties. Our proof closely follows

[ASS06, Theorem III.1.6], we first define the two functors, and then we point out all the technicalities we need to check.

We first construct the functor F. Given an A-module M, $F(M)$ is the following representation, denoted by $\widetilde{M}$: at each vertex i of Q, attach the vector space Me_i. And for each directed edge $i \xrightarrow{\alpha} j$, we define $\phi_\alpha : Me_i \to Me_j$ to be the map mapping an element me_i of Me_i to $m(e_i \alpha e_j)$. Hence we have defined $\widetilde{M} := (K_i, \phi_\alpha)_{i \in VQ, \alpha \in EQ}$. Now, for each A-module homomorphism $f : M \to N$, for each $1 \leqslant i \leqslant n$, we denote $\widetilde{f}(i) : Me_i \to Ne_i$ as the restriction of f to Me_i. We claim that the collection $\{\widetilde{f}(i)\}_{i \in VQ}$ is a morphism of representation $\widetilde{M} \to \widetilde{N}$.

Now we construct the functor G. Given a a representation $V = (V_i, \phi_\alpha)_{i \in VQ, \alpha \in EQ}$ bounded by I, we define $G(V)$ to be the following A-module. As a vector space, it is the (finite) direct sum $\oplus_{i \in VQ} V_i$. To give the A-module structure, we first define right multiplication on $\oplus_{i \in VQ} V_i$ with basis elements of kQ. Namely, given $(v_i)_{i \in VQ}$ an element of $\oplus_{i \in VQ} V_i$ and p a path that starts at vertex i and ends at vertex j, we define

$$(v_i)_{i \in VQ} \cdot p := (0, \cdots, 0, \phi_p(v_i), 0 \cdots, 0) \tag{4.3}$$

with $\phi_p(v_i)$ at the jth coordinate, and other coordinate having 0. Now, given two representation $V = (V_i, \phi_\alpha)_{i \in VQ, \alpha \in EQ}$ and $W = (W_i, \psi_\alpha)_{i \in VQ, \alpha \in EQ}$ and a morphism of representation $\{f_i\}_{i \in VQ}$, we define an A-module homomorphism $G(\{f_i\}) : \oplus_i V_i \to \oplus_i W_i$ by just sending $(v_i)_{i \in VQ}$ to $(\phi_i(v_i))_{i \in VQ}$.

Now we present the required checks to complete the proof. We justify one of them, we invite the reader to read the rest at the reference for further details.

- $\widetilde{M}$ defined above is bounded by I.
- $\widetilde{f}(i)$ is well-defined for each i, that is, it is k-linear and the image is contained in Ne_i
- $\{\widetilde{f}(i)\}_{i \in VQ}$ is indeed a morphism of quiver representations.
- The map defined in (4.3) gives $\oplus_i V_i$ an A-module structure.
  - Note that (4.3) defined right multiplication with basis kQ, hence it defines a kQ-module. Take any $(v_i)_{i \in VQ} \in \oplus_i V_i$, note that for any path $p \in I$, $\phi_p(v_i) = 0$ since we are given V bounded by I, hence $(v_i)_{i \in VQ} \cdot p = 0$. Extending linearly we have $v \cdot \rho = 0$ for all $\rho \in I$. Hence, given an element $w + I \in kQ/I$ the product

    $$(v_i)_{i \in VQ} \cdot (w + I) := (v_i)_{i \in VQ} \cdot w$$

    is well defined, this gives an A-module structure on $\oplus_i V_i$.
- The map $G(\{f_i\}) : \oplus_i V_i \to \oplus_i W_i$ is indeed an A-module homomorphism, and
- F, G are functors, and FG and GF are isomorphic to the identity in their respective category.

□

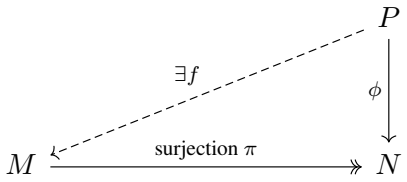
4.3 Visualizing Different Submodules of the Regular Module

Given a (right) A-module M, now we turn our focus to finding what is called *projective* and *injective* submodules of M. These submodules will play a big part in Chapter 6, where we look at its relation with the derived category.

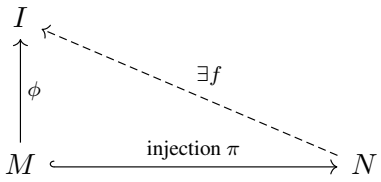
Definition 4.3.1. An A-module P is called **projective** if given a surjective A-module homomorphism $\pi : M \to N$ and a A-module homomorphism $\phi : P \to N$, there is an A-module homomorphism $f : P \to M$ such that $f \circ \pi = \phi$.

Definition 4.3.2. An A-module P is called **injective** if given a injective A-module homomorphism $\pi : M \to N$ and a A-module homomorphism $\phi : M \to I$, there is an A-module homomorphism $f : N \to I$ such that $f \circ \pi = \phi$.

A commutative diagram representation for the above two properties is:



Projective P



Injective I

Let us construct some examples of projective A-modules.

Proposition 4.3.3. Let e be an idempotent of a k-algebra A, then the ideal eA is projective.

Proof. Given A-module M, N and homomorphism $\phi : eA \to N$ and surjective homomorphism $\pi : M \to N$. From subjectivity of π, we find $m \in M$ such that $\pi(m) = \phi(e)$.

Note that

$$\pi(me) = \pi(m)e = \phi(e)e = \phi(e^2) = \phi(e),$$

where the first equality is true since π is an A-module homomorphism. Now we define the required f as follows: take an element $x \in eA$ and write $x = ea$ for some $a \in A$. We send $f(x) = (me)a$. It is obvious that this is well-defined. And $\pi(f(x)) = \pi(mea) = \pi(me)a = \phi(e)a = \phi(ea) = \phi(x)$, so the required diagram commutes. □

As seen from the commutative diagram one might expect there is some "symmetry" between them by "flipping the arrows". We now attempt to make this precise.

Recall in Example 4.1.10 there are two ways to “dualize” an A-module M. In category theory terms, they define contravariant functors (e.g. for the dual space, a homomorphism $M \to N$ induces homomorphism $N^* \to M^*$), so putting those two in succession provides an ordinary covariant functor from the category of modules to itself:

Definition 4.3.4. Denote the functor $D : A\text{-}\mathbf{Mod} \to \mathbf{Mod}\text{-}A$ the contravariant functor that assign an A-module to its dual A^*. The **Nakayama functor** is $\nu : A\text{-}\mathbf{Mod} \to A\text{-}\mathbf{Mod}$ defined by $\nu := D \circ \mathrm{Hom}_A(-, A)$.

Definition 4.3.5. Fix an algebra A, The injective (resp. projective) A-modules form a full subcategory of A-$\mathbf{Mod}$, denoted by $\mathbf{Inj}$-A (resp. $\mathbf{Proj}$-A).

Theorem 4.3.6. The Nakayama functor is also a functor from $\mathbf{Proj}$-A to $\mathbf{Inj}$-A. Moreover, it is an equivalence.

Now we want to investigate what the injective and projective modules look like as a representation. The following theorem provides an answer for the decomposition of $A = kQ/I$, viewing A as a right A-module itself. The following theorem heavily uses the notation in [Sch14, Definition 2.2]

Theorem 4.3.7. Given a quiver Q with n vertices and I an admissible ideal. Label the vertices from 1 to n and consider the following collection of representations of Q bounded by I for each integer $1 \leqslant i \leqslant n$:

- The class $S(i)$, at vertex j, equip the vector space k if and only if $i = j$, and equip 0 otherwise. For any oriented edge in Q, assign the zero map.
- The class $P(i)$, at vertex j, equip the vector space with the set of equivalence classes

  $$\{p + I : p \text{ is a path from } i \text{ to } j\}$$

  as basis. For an oriented edge α from j to k, we assign the linear map by specifying a path p will be sent to $p\alpha$ by concatenation.
- The class $I(i)$, at vertex j, equip the vector space with the set of equivalence classes

  $$\{p + I : p \text{ is a path from } j \text{ to } i\}$$

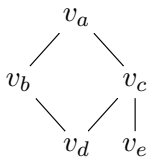
  as basis. For an oriented edge α from j to k, we assign the linear map by specifying a path p will be sent to the following:
  - If the path p is in the form of $\alpha p'$ (i.e. it starts with the entry α and have remaining part as a path p'), send it to p'.
  - Otherwise, send the path p to 0.

Then, under the equivalence in Theorem 4.2.15, if $A = kQ/I$, then $S(i)$ are isomorphic to simple kQ/I-modules for any i, $P(i)$ are isomorphic the indecomposable projective kQ/I-modules e_iA, and $I(i)$ are the indecomposable injective modules $\nu(e_iA)$.

Indeed, in Theorem 4.2.15 we constructed functors $G : \mathbf{Rep}_k^Q/I \to A\text{-}\mathbf{Mod}$. Note that $G(S(i))$ will be sent to a 1-dimensional vector space, hence it must be simple. Similarly, $G(P(i))$ is sent to the vector space with basis as the collection of all (equivalence classes of) paths that starts at vertex i, which is the set of basis in e_iA.

Now, given a quiver Q with n vertices, we present a way to write down n graphs so that it specifies an ideal I and recover the n indecomposable projectives in the regular right module of $A := kQ/I$. This method is used in [GH96], and will reappear in Chapter 6.

We use the quiver presented in Figure 4.2.3 with three vertices. We first present and name the following 3 diagrams, labelled with the vertex of Q: We first define a kQ-module structure on $P(1)$. To do this, we draw the diagram again with new symbols:



And define $P(1)$ to be the following quiver representation of Q according to this recipe:

Step 1. At each vertices i labelled by the positive integers, define K_i to be

$$\text{span}\langle\text{symbols that have label } i \text{ in the original picture}\rangle$$

. (If there are no such labels, assign $K_i = \{0\}$.)

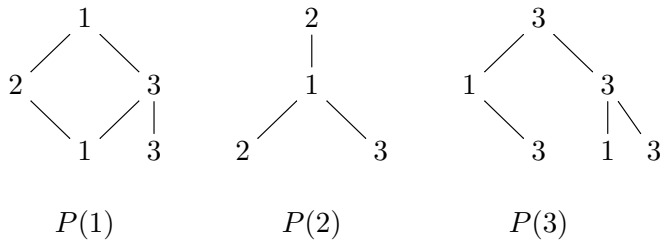
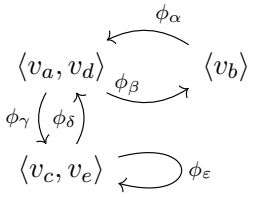


Figure 4.2: A picture of three modules $P(1), P(2), P(3)$, understood via the quiver Q in Figure 4.2.3.

Step 2. The linear map assigned to the edges in the representation can be assigned as follows. Read $P(1)$ from top to bottom, a line from i to j that goes *down* in the diagram means the linear map of the edge from i to j send the top basis element to the bottom one. Do this for all the edges in the diagram.

Step 3. If there are basis elements where we have not defined the linear map, then it is sent to 0.

We go through building the quiver representation of $P(1)$. Step 1 yields the quiver representation



Now it remains to define the linear maps $\phi_\alpha \dots \phi_\varepsilon$. By step 2, seeing the line from 1 to 2 (top to bottom) means v_a is sent to v_b via ϕ_β, similarly v_b is sent to v_a via ϕ_α. Do this for the remaining three lines seen on $P(1)$.

Finally, note that we still have not fully defined ϕ_α, since v_d is not connected to anything downwards. Hence, by Step 3, we define $\phi_\alpha(v_d) = 0$, similarly, $\phi_\delta(v_e) = \phi_\varepsilon(v_e) = 0$, this will complete the definition of the quiver representation for $P(1)$. Note that this recovers $P(1)$ as a kQ-module.

And we build the ideal I as follows. Firstly, if the diagram has "depth" at most n, we declare I contains all paths of length n or above. Next, for any cycle in the diagram, we read off all the paths from top to bottom (according to the label of the vertices). Then, for the paths p, q in the cycle, we declare the element $p - \lambda q$ in kQ is in our ideal I, where we do not specify $\lambda \in k$. After that, we look at the vertices "layer by layer". If we see a vertex where an arrow going out of the label in the quiver is missing, we trace the path from the top of the diagram and declare that I contains this path.

In Figure 4.2, it is of "depth" three, and there is only one cycle and the cycle reads off paths $\beta\alpha$ and $\gamma\delta$. Moreover, note that at $P(3)$, "layer" two, there is an edge missing connecting 1 at the top to 2 at the bottom. This corresponds to the path $3 - 1 - 2$, i.e. $\delta\beta$. So the ideal I is:

$$I = \langle \text{all paths of length 3 or more}, \beta\alpha - \lambda\gamma\delta, \delta\beta \rangle,$$

where $\lambda \in k$ is an undetermined constant.

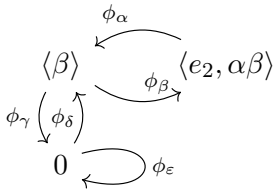
Remark.

- This method works very generally to any quiver Q where there is at most one oriented edge from any vertex to any vertex. Even when Q does not satisfy this property, we can decorate the diagram with lines with another style.

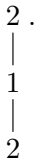
- In fact, using the identication $P(1) \cong e_1 A$, the basis of $P(1)$ corresponds to the path starting at vertex 1 as expected. For example, the vector v_d is the equivalence class of the path $\beta\alpha$ in Figure 4.2.3.
- We defines the ideal I with a remaining variable λ, and this actually produce non-isomorphic algebras A_λ for different λs. All results in this report will still hold true regardless of the value λ, so we will drop the subscript.

$\Diamond$

From here, since we have the whole algebra $A = kQ/I$ one can recover the injective indecomposable modules $I(1), I(2), I(3)$ from Theorem 4.3.7. Let us work out $I(2)$ for example. Note that in Figure 4.2 we have three vertex labelled 2, which means we have three basis elements of A that end at 2. By the remark above, they are $\beta, e_2, \alpha\beta$. So by Theorem 4.2.15 the quiver representation for $I(2)$ is



Note that $\alpha\beta$ get sent to β by ϕ_α, β get sent to e_2 by ϕ_β, and e_2 get sent to zero. So in picture we introduced, $I(2)$ can be represented by



Similarly, we can recover the diagram for $I(1), I(2)$, and$I(3)$ below.

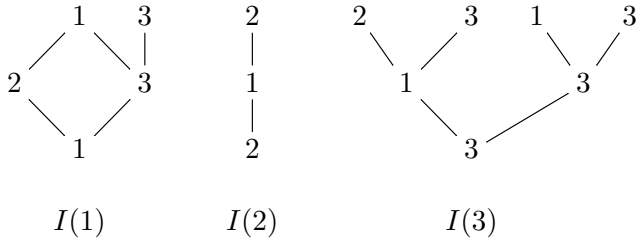
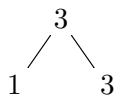
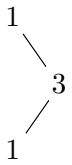


Figure 4.3: A picture of three modules $I(1), I(2), I(3)$.

Note that, given $I(1), I(2), I(3)$, one can carry out a similar procedure to recover I. Instead of looking at vertices from top to bottom, we look at vertices *from bottom to top*. Now we present some facts about this pictorial way of presenting modules of quiver algebra. Note that a submodule is closed under right multiplication, and note that in Figure 4.2 (and the functor G in the equivalence in Theorem 4.2.15) right multiplication of the module is defined by looking downwards on the diagram. So a submodule can be visualized as a subgraph of the diagram "closed under going downwards". For example, in $P(1)$, we note that



is a submodule of $P(1)$, but



is not (since, for example, the edge going down from 1 to 2 is missing). In particular, this provides an example of a module that is indecomposable (since $P(1) \cong e_1 A$ and is indecomposable by Proposition) but not simple.

In the case where the subgraph (corresponds to a module) N is a submodule of the original module M (presented in pictures), note that M/N is isomorphic to the vector space spanned by the remaining basis elements of M that are not in N, with the same right multiplication (but now on equivalence classes), hence the quotient submodule M/N can be represented by the diagram with the remaining vertices. In the example of $P(1)$, the quotient module of $P(1)$ over



is (isomorphic to)



Note that this is not a submodule of $P(1)$.

Chapter 5

Homological Algebra and the Derived Category

5.1 Cochain Complexes

We now introduce another powerful tool for studying A-modules: cochain complexes. We will fix an arbitrary ring R. This Chapter has [GM03, Chapter III], [Wei94, Chapter 1, 10] as the main reference.

Definition 5.1.1. A **cochain complex of (right) R-modules** is a sequence $\{C^n\}_{n\in\mathbb{Z}}$ of (right) R-modules with a sequence $\{d_C^n : C^n \to C^{n+1}\}_{n\in\mathbb{Z}}$ of R-module homomorphisms such that $d^{i+1}\circ d^i = 0$ for all $i \in \mathbb{Z}$, we will denote the collection $\{C^n, d_C^n\}_{n\in\mathbb{Z}}$ as $C^\bullet$. We will often refer to the module C^n and the map d_C^n as the **module** and the **differential at degree** n.

We will often write a cochain complex pictorially as:

$$C^\bullet : \qquad \cdots \xrightarrow{d_C^{n-2}} C^{n-1} \xrightarrow{d_C^{n-1}} C^n \xrightarrow{d_C^n} C^{n+1} \xrightarrow{d_C^{n+1}} C^{n+2} \xrightarrow{d_C^{n+2}} \cdots .$$

Definition 5.1.2. Given two cochain complex $C^\bullet = (C^n, d_C^i)$, $D^\bullet = (D^n, d_D^i)$, a **cochain map** $u : C^\bullet \to D^\bullet$ is a collection of R-modules homomorphism $u^i : C^i \to D^i$ such that $u^{i+1}\circ d_C^i = d_D^i \circ u^i$ for all $i \in \mathbb{Z}$.

Notation

We shall also use the word "**complex**" to denote a cochain complex.

In other words, one can think of the commutative diagram:

$$\begin{array}{lccccccccccc}
C^\bullet : & \cdots & \xrightarrow{d_C^{n-2}} & C^{n-1} & \xrightarrow{d_C^{n-1}} & C^n & \xrightarrow{d_C^n} & C^{n+1} & \xrightarrow{d_C^{n+1}} & C^{n+2} & \xrightarrow{d_C^{n+2}} & \cdots \\
 & & & \downarrow{\scriptstyle u^{n-1}} & & \downarrow{\scriptstyle u^n} & & \downarrow{\scriptstyle u^{n+1}} & & \downarrow{\scriptstyle u^{n+2}} & & \\
D^\bullet : & \cdots & \xrightarrow{d_D^{n-2}} & D^{n-1} & \xrightarrow{d_D^{n-1}} & D^n & \xrightarrow{d_D^n} & D^{n+1} & \xrightarrow{d_D^{n+1}} & D^{n+2} & \xrightarrow{d_D^{n+2}} & \cdots
\end{array}$$

This immediately proves:

Proposition 5.1.3. The collection of all cochain complexes of (right) R-modules is a category, we will call it $\mathrm{Ch}(R\text{-mod})$.

Since identity and composition can be defined degree-wise, it is trivial to check that they satisfy all the properties in Definition 1.1.1.

Definition 5.1.4. Given $C^\bullet$ a cochain complex of R-modules:

- The n**-cocycle** of $C^\bullet$ is the R-module $Z^n(C^\bullet) := \ker d_C^n$.
- The n**-coboundary** of $C^\bullet$ is the R-module $B^n(C^\bullet) := \operatorname{im} d_C^{n-1}$.
- The nth**-cohomology module** of $C^\bullet$ is the quotient module $H^n(C^\bullet) := Z^n(C^\bullet)/B^n(C^\bullet)$.
- $C^\bullet$ is **exact at degree** n (or **exact at** C^n) if $H^n(C^\bullet) = 0$, we also say $C^\bullet$ is **exact** if it is exact at degree i for all $i \in \mathbb{Z}$.

Remark. We will write $Z^n(C^\bullet)$ as Z^n if there is no confusion arising without specifying the cochain complex $C^\bullet$. ◊

Recall that $\mathbb{Z}$-modules correspond exactly to abelian groups, and k-modules for a fixed field k is a vector space, so we can construct a lot of examples of cochain complexes.

Remark. Note that there is only one module homomorphism

- from the zero module to any module $0 \to A$,
- from any module to zero module $A \to 0$, and
- from any module to any module, sending everything to zero. $0 : A \to B$

we will write down sequences and complexes without naming them. ◊

Example 5.1.5. Let $f : A \to B$ be a homomorphism. Consider cochain complex

$$\cdots \longrightarrow 0 \longrightarrow A \xrightarrow{f} B \longrightarrow 0 \longrightarrow \cdots$$

where A is at degree 0. Note that it is exact at A when $\ker f = \{0\}$, i.e. f is injective, and exact at B when $\operatorname{im} f = B$, that is f is surjective. So This cochain complex is exact if and only if f is an isomorphism. △

To denote the exactness at A and B respectively, we would sometimes write the about example as $0 \longrightarrow A \xrightarrow{f} B$ exact if and only if f is injective, $A \xrightarrow{f} B \longrightarrow 0$ exact if and only if f is surjective. This motivates us to define

Definition 5.1.6. A **short exact sequence** of R-modules is a quintuple (A, B, C, f, g) such that the cochain complex

$$\cdots \longrightarrow 0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0 \longrightarrow \cdots$$

is exact, where $\cdots$ denotes the zero module and zero maps. We will usually denote the short exact sequence as $0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$

Combining with the previous example, to check $0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$ is indeed a short exact sequence, it is both necessary and sufficient to check the following three conditions:

- f is injective ($\ker f = \{0\}$),
- g is surjective ($\operatorname{im} g = C$), and
- $\operatorname{im} f = \ker g$.

Example 5.1.7. Given a R-module homomorphism $f : A \to B$, we always have the short exact sequence

$$0 \longrightarrow \ker f \hookrightarrow A \twoheadrightarrow A/\ker f \cong \operatorname{im} f \longrightarrow 0 \tag{5.1}$$

where the maps from $\ker f \to A$ are the inclusion and $A \to A/\ker f$ are the canonical maps sending elements to the coset with that element as the representative. One can check the three conditions above are indeed satisfied. In particular, if N is a submodule of M, then one has the canonical map $f : M \to M/N$ with kernel N, so the sequence $0 \to N \to M \to M/N \to 0$ will be short exact. △

Example 5.1.8. Here is a non-trivial example we will use later. Recall in Example 3.2.19 we constructed the direct limit of modules from a direct system $M_0 \xrightarrow{\alpha_0} M_1 \xrightarrow{\alpha_1} M_2 \xrightarrow{\alpha_2} \dots$, which we will denote as (unfortunately) $\varinjlim M_i$, overloading the notation for colimit. There is an exact sequence

$$0 \longrightarrow \bigoplus_{i=0}^{\infty} M_i \xrightarrow{f} \bigoplus_{i=0}^{\infty} M_i \twoheadrightarrow \varinjlim M_i \longrightarrow 0$$

where

- The map $\bigoplus_{i=0}^{\infty} M_i \to \varinjlim M_i$ is the canonical quotient map as $\varinjlim M_i$ is a quotient of $\bigoplus_{i=0}^{\infty} M_i$.
- The map f, however, is defined by

$$x_i \in M_i, \qquad x_i \mapsto \alpha_i(x_i) - x_i$$

  where α_i is the morphism in the direct system from M_i to M_{i+1}.

One can check they are well-defined and the above three conditions hold. In particular, to check that f is injective, we check that $\ker f = \{0\}$ as follows. Suppose $x := (x_i)_{i=0}^{\infty}$ with $f(x) = 0$, we compare $f(x)$ and 0 coordinate by coordinate. Note that $f(x)$ has the first coordinate as x_0, and the second coordinate $x_1 - \alpha_0(x_0)$, etc. The ith coordinate of $f(x)$ is $\alpha(x_{i-1}) - x_{i-2}$ for $i \geqslant 2$. Now the first coordinate is 0 so $x_0 = 0$, the second coordinate is zero so $x_1 - \alpha_0(x_0) = x_1 = 0$ (first equality follows from $x_0 = 0$). Continue inductively we have $x_i = 0$ for all non-negative integer i. Hence $x = 0$ and f is injective as required. △

Here we introduce *splicing*, which is a common technique we will use to form new exact complexes.

Proposition 5.1.9. If $0 \longrightarrow A \xrightarrow{\alpha_1} B \xrightarrow{\alpha_2} C \longrightarrow 0$, $0 \longrightarrow C \xrightarrow{\beta_1} D \xrightarrow{\beta_2} A \longrightarrow 0$ are short exact sequences, then the following two complexes are exact:

$$\dots \xrightarrow{\beta_1\alpha_2} D \xrightarrow{\alpha_1\beta_2} B \xrightarrow{\beta_1\alpha_2} D \xrightarrow{\alpha_1\beta_2} B \xrightarrow{\alpha_2} C \longrightarrow 0 \tag{5.2}$$

$$\dots \xrightarrow{\alpha_1\beta_2} B \xrightarrow{\beta_1\alpha_2} D \xrightarrow{\alpha_1\beta_2} B \xrightarrow{\beta_1\alpha_2} D \xrightarrow{\beta_2} A \longrightarrow 0 \tag{5.3}$$

Remark. In Proposition 5.1.9, it is irrelevant where is the degree 0 part of this complex, when this happens, sometimes we will refer to them as *long exact sequences*. ◊

Proof. We only show the exactness in (5.2), it is similar for (5.3).

There are two key facts we will use: If f is surjective, then $\operatorname{im}(gf) = \operatorname{im}(g)$. Similarly, if f is injective, then $\ker(fg) = \ker(g)$. Now we check the exactness of all the terms in (5.2). Exactness at C is obvious since α_2 is surjective. For exactness at D, note that $\operatorname{im}(\beta_1\alpha_2) = \operatorname{im}(\beta_1)$ since α_2 is surjective, $\operatorname{im}(\beta_1) = \ker(\beta_2)$

by exactness of $0 \to C \to D \to A \to 0$, and $\ker(\beta_2) = \ker(\alpha_1\beta_2)$ since α_1 is injective. So $\mathrm{im}(\beta_1\alpha_2) = \ker(\alpha_1\beta_2)$ and the exactness at D is shown as required. Similarly,

$$\ker(\beta_1\alpha_2) = \ker(\alpha_2) = \mathrm{im}(\alpha_1) = \mathrm{im}(\alpha_1\beta_2)$$

So the exactness at B is checked. □

Since $\mathrm{Ch}(R\text{-}\mathbf{Mod})$ is a category, we immediately have the definition of isomorphism and functors available for us to use.

Example 5.1.10. Let $n \in \mathbb{Z}$, there is a functor $H^n : \mathrm{Ch}(R\text{-}\mathbf{Mod}) \to R\text{-}\mathbf{Mod}$ by sending A cochain complex $C^\bullet$ to $H^n(C^\bullet)$, and any cochain map $u : C^\bullet \to D^\bullet$ to a map $H^n(u) : H^n(C^\bullet) \to H^n(D^\bullet)$. Where $H^n(u)$ is the map u^{n-1} on the equivalence classes. One might want to check that the resultant map is well-defined. △

Definition 5.1.11. If $f : C^\bullet \to D^\bullet$ is a cochain map such that $H^n(f) : C^n \to D^n$ is an isomorphism for all $n \in \mathbb{Z}$, then we say f is a **quasi-isomorphism**.

From a complex or a cochain map, one can form various other complexes, below are the most useful examples. Note that we will double-underline the degree zero term in the complex and skip the formal definition (of specifying the degree n part and its differential) in easy case, in the hope that readers can write down the formal definition if needed.

Definition 5.1.12. Given a complex $C^\bullet$, the **subcomplex** is the complex

$$\cdots \longrightarrow \underline{\underline{Z^0}} \xrightarrow{d_C^0} Z^1 \xrightarrow{d_C^1} Z^2 \xrightarrow{d_C^2} Z^3 \longrightarrow \cdots$$

where Z^n are the cocycles. Similarly, the **quotient complex** is the complex:

$$\cdots \longrightarrow \underline{\underline{B^0}} \xrightarrow{d_C^0} B^1 \xrightarrow{d_C^1} B^2 \xrightarrow{d_C^2} B^3 \longrightarrow \cdots$$

where the B^n are the coboundaries.

Definition 5.1.13. Given a complex $C^\bullet$, the **"translate by n"** complex $C[n]$ is the complex with $(C[n])^i = C^{i+n}$, and the differentials at degree i as $d_{C[n],i} = (-1)^n d_{n+i}$. The sign is to make our later computations easier.

$$\cdots \longrightarrow C^0 \xrightarrow{(-1)^n d_C^0} C^1 \xrightarrow{(-1)^n d_C^1} \cdots \xrightarrow{(-1)^n d_C^{n-1}} \underline{\underline{C^n}} \xrightarrow{(-1)^n d_C^n} \cdots$$

In particular, note that we can define a functor $[n] : \mathrm{Ch}(R\text{-}\mathbf{Mod}) \to \mathrm{Ch}(R\text{-}\mathbf{Mod})$ by sending a cochain complex to its translate by n, and a cochain map to the same cochain map.

Definition 5.1.14. Given a complex $C^\bullet$ the nth **brutal truncation** $\sigma^{\geqslant n}(C^\bullet)$ with:

$$(\sigma_{\geqslant n}(C^\bullet))^i = \begin{cases} 0 & \text{if } I < n \\ C^i & \text{otherwise} \end{cases}$$

In picture:

$$\cdots \longrightarrow 0 \longrightarrow 0 \longrightarrow \underline{\underline{0}} \longrightarrow \cdots \longrightarrow C^n \xrightarrow{d_C^n} C^{n+1} \xrightarrow{d_C^{n+1}} \cdots$$

Definition 5.1.15. A cochain complex $C^\bullet$ is **bounded above** if there exists an $N \in \mathbb{Z}$ such that $C^n = 0$ for all $n > N$.

Proposition 5.1.16. Given a bounded above cochain complex $C^\bullet$, say $C^n = 0$ for all $n > N$. Then there is a direct system $\sigma_{\geqslant N}(C^\bullet) \xrightarrow{\alpha_0} \sigma_{\geqslant N-1}(C^\bullet) \xrightarrow{\alpha_1} \sigma_{\geqslant N-2}(C^\bullet) \xrightarrow{\alpha_2} \cdots$, and $C^\bullet$ is the direct limit of the system.

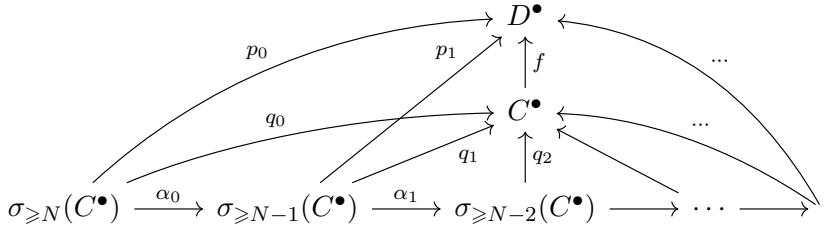
Proof. The maps α_i for $i \geqslant 0$ are given by the identity or zero maps at any term. Here is a diagram depicting the situation.

$$\begin{array}{ccccccccccc}
\sigma_{\geqslant N}(C^\bullet) & & \cdots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & C^N & \longrightarrow & 0 & \longrightarrow & \cdots \\
\downarrow{\scriptstyle \alpha_0} & & & & \downarrow{\scriptstyle \alpha_0^{N-2}} & & \downarrow{\scriptstyle \alpha_0^{N-1}} & & \downarrow{\scriptstyle \alpha_0^{N}} & & \downarrow & & \\
\sigma_{\geqslant N-1}(C^\bullet) & & \cdots & \longrightarrow & 0 & \longrightarrow & C^{N-1} & \longrightarrow & C^N & \longrightarrow & 0 & \longrightarrow & \cdots \\
\downarrow{\scriptstyle \alpha_1} & & & & \downarrow{\scriptstyle \alpha_1^{N-2}} & & \downarrow{\scriptstyle \alpha_1^{N-1}} & & \downarrow{\scriptstyle \alpha_1^{N}} & & \downarrow & & \\
\sigma_{\geqslant N-2}(C^\bullet) & & \cdots & \longrightarrow & C^{N-2} & \longrightarrow & C^{N-1} & \longrightarrow & C^N & \longrightarrow & 0 & \longrightarrow & \cdots
\end{array}$$

It is obvious that the diagram on the right commutes, so α_i are indeed cochain maps. Recall the colimit is given by the object (which we claim to be $C^\bullet$) with the coprojections $\{q_i\}_{i=0}^\infty$, $q_i : \sigma_{N-i}(C^\bullet) \to C^\bullet$. We claim the coprojections are given at degree $j \in \mathbb{Z}$ by

$$q_i^j(a) = \begin{cases} a & \text{if } j \geqslant N - i \\ 0 & \text{otherwise} \end{cases}$$

. It is obvious that $q_{i+1}\alpha_i = q_i$ for all $i \in \mathbb{Z}$, so $(C^\bullet, q_i)$ is indeed a cocone. To show that it is the limit cocone, Suppose we have another cocone $\{p_i\}_{i=0}^\infty$, where $p_i : \sigma_{\geqslant N-i}(C^\bullet) \to D^\bullet$. So the following diagram commutes:



Since the diagram commutes, this implies, $p_0 = p_1\alpha_0$, which implies p_0, p_1 agrees on degree N. Similarly, $p_1 = p_2\alpha_1$ implies p_1, p_2 agrees on both degree $N, N-1$. One can inductively see that $p_0, \ldots p_i$ agrees on degree $N, N-1, \ldots N-i$ for any non-negative integer i. Now note $p_i = fq_i$ for any such i, so f is p_i at degree $N, N-1, \ldots N-i$. Since this holds for any non-negative integer i, it uniquely determines f at degree not larger than N, which uniquely determines f since $C^\bullet$ is bounded above. □

Definition 5.1.17. Given a cochain map $f : C^\bullet \to D^\bullet$, the **kernel** of f is the complex $\ker f$ by:

$$\cdots \longrightarrow \underline{\underline{\ker f^0}} \xrightarrow{d_C^0} \ker f^1 \xrightarrow{d_C^1} \ker f^2 \xrightarrow{d_C^2} \ker f^3 \xrightarrow{d_C^n} \cdots$$

Similarly one can define the **image** of f

$$\cdots \longrightarrow \underline{\underline{\operatorname{im} f^0}} \xrightarrow{d_D^0} \operatorname{im} f^1 \xrightarrow{d_D^1} \operatorname{im} f^2 \xrightarrow{d_D^2} \operatorname{im} f^3 \xrightarrow{d_D^n} \cdots$$

and the **cokernel** of f

$$\cdots \longrightarrow \underline{\underline{\operatorname{coker} f^0}} \xrightarrow{d_D^0} \operatorname{coker} f^1 \xrightarrow{d_D^1} \operatorname{coker} f^2 \xrightarrow{d_D^2} \operatorname{coker} f^3 \xrightarrow{d_D^n} \cdots$$

With this definition, one can talk about short exact sequences of cochain complexes.

Definition 5.1.18. A sequence $0 \longrightarrow A^\bullet \xrightarrow{f} B^\bullet \xrightarrow{g} C^\bullet \longrightarrow 0$ of cochain complexes is **short exact** if

- $\ker f = \{0\}$, where 0 is the complex where every degree (and map) is 0.
- $\operatorname{im} g = C^\bullet$, and
- $\operatorname{im} f = \ker g$.

There are a few famous lemmas and theorems which we will now discuss.

Lemma 5.1.19 (The Five Lemma)**.** If we have a commutative diagram of the following 10 modules and 13 morphisms:

$$\begin{array}{ccccccccc} A & \xrightarrow{\alpha_1} & B & \xrightarrow{\alpha_2} & C & \xrightarrow{\alpha_3} & D & \xrightarrow{\alpha_4} & E \\ {\scriptstyle f}\downarrow & & {\scriptstyle g}\downarrow & & {\scriptstyle h}\downarrow & & {\scriptstyle i}\downarrow & & {\scriptstyle j}\downarrow \\ A' & \xrightarrow{\beta_1} & B' & \xrightarrow{\beta_2} & C' & \xrightarrow{\beta_3} & D' & \xrightarrow{\beta_4} & E' \end{array}$$

and the two rows of the diagram are exact (meaning exactness at B, C, D, B', C', D' in the rows), then

- If f is surjective, g, i are injective, then h is injective.
- If g, i are surjective, j is injective, then h is surjective

That is, if f is surjective, j is injective, and j, i are isomorphisms, then h is also an isomorphism.

Proof. We will proof only the first item. Suppose $h(c) = 0$, to show injectivity we show this implies $c = 0$. Now $\beta_3 h(c) = i\alpha_3(c) = 0$. Now i is injective, so $\alpha_3(c) = 0$. $c \in \ker \alpha_3$, so by exactness $c \in \operatorname{im} \alpha_2$ and take $b \in B$ such that $\alpha_2(b) = c$. Now $h\alpha_2(b) = \beta_2 g(b) = h(c) = 0$, so $\beta_2 g(b) = 0$. $g(b) \in \ker \beta_2$, by exactness $g(b) \in \operatorname{im} \beta_1$. Take $a' \in A'$ such that $\beta_1(a') = g(b)$. Now f is surjective, so take $a \in A$ such that $f(a) = a'$. Now $\beta_1 f(a) = g\alpha_1(a)$ at the same time $\beta_1 f(a) = \beta_1(a') = g(b)$. So $g\alpha_1(a) = g(b)$ and by injectivity of g, $\alpha_1(a) = b$ so $b \in \operatorname{im} \alpha_1$, by exactness $b \in \ker \alpha_2$, so $\alpha_2(b) = c = 0$. □

Lemma 5.1.20 (The Snake Lemma)**.** Consider the following commutative diagram of 6 modules and 7 morphisms:

$$\begin{array}{ccccccccc} & & A & \xrightarrow{\alpha_1} & B & \xrightarrow{\alpha_2} & C & \longrightarrow & 0 \\ & & {\scriptstyle f}\downarrow & & {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle h} & & \\ 0 & \longrightarrow & A' & \xrightarrow{\beta_1} & B' & \xrightarrow{\beta_2} & C' & & \end{array}$$

If the two rows are exact (at B, C, A', B'), then there is an exact sequence:

$$\ker f \longrightarrow \ker g \longrightarrow \ker h \xrightarrow{\exists \delta} \operatorname{coker} f \longrightarrow \operatorname{coker} g \longrightarrow \operatorname{coker} h.$$

Recall that $\operatorname{coker} f = A'/\operatorname{im} f$, etc.

On proving the Snake Lemma, we refer to a quote from Aluffi's book [Alu09, p. 180]:

> "PROVING the snake lemma is something that should not be done in public, and it is notoriously useless to write down the details of the verifications for others to read: the details are all essentially obvious, but they lead quickly to a notational quagmire. Such proofs are collectively known as the sport of *diagram chase*, best executed by pointing several fingers at different parts of a diagram on a blackboard, while enunciating the elements one is manipulating and stating their fate."

The proof of the Five Lemma is also most conveniently proved via a diagram chase, as readers might find the proof presented above hard to follow for the exact reasons stated in the quote.

Theorem 5.1.21 (Long exact sequence of Cohomology)**.** If $0 \longrightarrow C^\bullet \xrightarrow{f} D^\bullet \xrightarrow{g} E^\bullet \longrightarrow 0$ is a short exact sequence of complexes, then there is an exact complex:

$$\cdots \xrightarrow{\exists\delta^{-1}} H^0(C^\bullet) \xrightarrow{f^0} H^0(D^\bullet) \xrightarrow{g^0} H^0(E^\bullet) \xrightarrow{\exists\delta^0} H^1(C^\bullet) \xrightarrow{f^1} H^1(D^\bullet) \xrightarrow{g^1} H^1(E^\bullet) \xrightarrow{\exists\delta^1} \cdots .$$

Proof. (Sketch.) From the diagram

$$\begin{array}{ccccccc} & & C^n & \xrightarrow{f^n} & D^n & \xrightarrow{g^n} & E^n \longrightarrow 0 \\ & & \downarrow{\scriptstyle d_C^n} & & \downarrow{\scriptstyle d_D^n} & & \downarrow{\scriptstyle d_E^n} \\ 0 & \longrightarrow & C^{n+1} & \xrightarrow{f^{n+1}} & D^{n+1} & \xrightarrow{g^{n+1}} & E^{n+1} \end{array}$$

The two rows are exact since $0 \longrightarrow C^\bullet \xrightarrow{f} D^\bullet \xrightarrow{g} E^\bullet \longrightarrow 0$ is short exact. By snake lemma, one gets the exact sequence

$$\ker d_C^n \to \ker d_D^n \to \ker d_E^n \xrightarrow{\delta^n} \operatorname{coker} d_C^n \to \operatorname{coker} d_D^n \to \operatorname{coker} d_E^n.$$

That is, an exact sequence in the form of

$$Z^n(C^\bullet) \to Z^n(D^\bullet) \to Z^n(E^\bullet) \to C^{n+1}/B^{n+1}(C^\bullet) \to D^{n+1}/B^{n+1}(D^\bullet) \to C^{n+1}/E^{n+1}(E^\bullet),$$

One can put this again into a commutative diagram

$$\begin{array}{ccccccc} & & C^n/B^n(C^\bullet) & \xrightarrow{\widetilde{f^n}} & D^n/B^n(D^\bullet) & \xrightarrow{\widetilde{g^n}} & E^n/B^n(E^\bullet) \longrightarrow 0 \\ & & \downarrow{\scriptstyle \widetilde{d_C^n}} & & \downarrow{\scriptstyle \widetilde{d_D^n}} & & \downarrow{\scriptstyle \widetilde{d_E^n}} \\ 0 & \longrightarrow & Z^{n+1}(C^\bullet) & \xrightarrow{f^{n+1}} & Z^{n+1}(D^\bullet) & \xrightarrow{g^{n+1}} & Z^{n+1}(E^\bullet) \end{array},$$

where we put a tilde on maps that are induced on equivalence classes. Now we can invoke the snake lemma again. Note that $\ker \widetilde{d^n)C} = Z^nC^\bullet/B^n(C^\bullet) = H^n(C^\bullet)$, and $\operatorname{coker}(C^\bullet) = Z^{n+1}(C^\bullet)/\operatorname{im} \widetilde{d_C^n} = Z^{n+1}(C^\bullet)/B^{n+1}(C^\bullet) = H^{n+1}(C^\bullet)$, hence once has exact sequence:

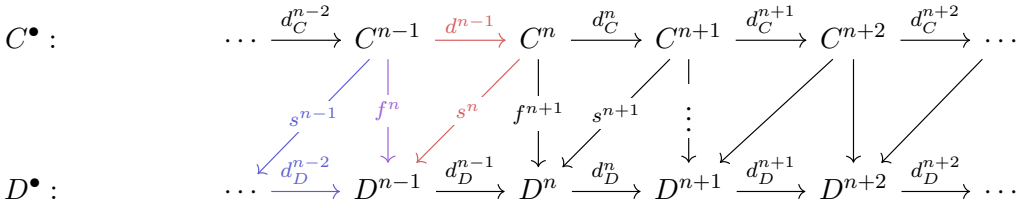
$$H^n(C^\bullet) \to H^n(D^\bullet) \to H^n(E^\bullet) \to H^{n+1}(C^\bullet) \to H^{n+1}(D^\bullet) \to H^{n+1}(E^\bullet).$$

then the long exact sequence is produced by pasting these sequences together. □

Given two R-module homomorphisms $f, g : A \to B$, we can define $f + g, f - g : A \to B$ easily.

Definition 5.1.22. Given two cochain maps $f, g : C^\bullet \to D^\bullet$, we say f is **null-homotopic** and write $f \simeq 0$ if there is a collection of maps $s^n : C^n \to D^{n-1}$ such that $f^n = s^{n+1}d_C^n + d_D^{n-1}s^n$. We say f is **homotopic** to g if $f - g$ is null-homotopic.

We can also visualize the situation with a diagram but note that it is not commutative:



but we have the purple part being the sum of the red and blue parts.

Definition 5.1.23. Let $C^\bullet$ be a complex, it is **split** if there maps $S^n : C^n \to C^{n-1}$ for all $n \in \mathbb{Z}$ such that $d_C^n = d_C^n \circ S^{n+1} \circ d_C^n$ for all $n \in \mathbb{Z}$. If in addition $C^\bullet$ is an exact complex, we will say $C^\bullet$ is **split exact**.

Proposition 5.1.24. Let $f, g : C^\bullet \to D^\bullet$ be cochain maps, then the relation

$$f \sim g \qquad \text{if and only if } f - g \simeq 0$$

(i.e. if and only if f is homotopic to g) is an equivalence relation.

Proof. Clearly, the relation is reflexive as 0 is homotopic to 0. If $f - g \simeq 0$ by the maps $\{s_n\}_{n\in\mathbb{Z}}$, then it is obvious that $g - f \simeq 0$ by the maps $\{-s_n\}_{n\in\mathbb{Z}}$. If $f - g \simeq 0$ by the maps $\{s_n\}_{n\in\mathbb{Z}}$, $g - h \simeq 0$ by the maps $\{t_n\}_{n\in\mathbb{Z}}$, then $f - h = (f - g) + (g - h) \simeq 0$ by the maps $\{s_n + t_n\}_{n\in\mathbb{Z}}$. □

We will call any representative $[f]$ under this relation the **homotopy class** of f. From this, one creates a new category, called the homotopy category.

Definition 5.1.25. The **homotopy category**, denoted by $\mathcal{K}(R)$ is the category with objects as R-modules cochain complexes, and the morphisms between cochain complexes $\mathrm{Hom}_{\mathcal{K}(R)}(C^\bullet, D^\bullet)$ is the set of equivalence class of all cochain maps from $C^\bullet$ to $D^\bullet$.

Remark. To check this indeed is a category, one needs to establish various theorems on well-definedness. Such as checking the composition of two functions is invariant under homotopy. ◊

Definition 5.1.26. Given two cochain complexes of R-modules $C^\bullet$ and $D^\bullet$, we say they are **homotopically equivalent** if they are isomorphic objects in $\mathcal{K}(R)$. In other words, there are cochain maps $f : C^\bullet \to D^\bullet$, $g : D^\bullet \to C^\bullet$ such that fg and gf are homotopic to the identity map (of $C^\bullet, D^\bullet$ respectively).

The word "homotopic" is borrowed from algebraic topology, as one might realize the definition of two functions being homotopic is very similar to this, as seen in any introductory algebraic topology textbook, say [Hat02].

Proposition 5.1.27. The cohomology functor $H^n(-)$ is a well-defined functor on $\mathcal{K}(R)$.

Proof. Given $f, g : C^\bullet \to D^\bullet$ cochain maps. We need to show $f \simeq g$ implies $H^n(f) = H^n(g)$. It suffices to show $f \simeq 0$ implies $H^n(f) = 0$. Note that $H^n(f)$ is a module homomorphism from $H^n(C^\bullet)$ to $H^n(D^\bullet)$, so pick $x \in H^n(C^\bullet)$. Since $f \simeq 0$ we assume there are maps $\{s_n\}_{n\in\mathbb{Z}}$ such that $f^n(x) = d_D^{n-1}s^n(x) + s^{n+1}d_C^n(x)$, but $x \in H^n(C^\bullet)$ so $d_C(x) = 0$, so $f^n(x) = d_D^{n-1}s^n(x)$, so $f^n(x) \in \operatorname{im} d_D^{n-1}$. But since $f^n(x) \in H^n(D^\bullet)$, this is 0 because $H^n(D^\bullet) = \ker d_D^n / \operatorname{im} d_D^{n-1}$. □

Corollary 5.1.28. Given a cochain complex $C^\bullet$, if the identity map $id_C : C^\bullet \to C^\bullet$ is null-homotopic, then it is exact.

Proof. This follows from the fact that for every $m \in \mathbb{Z}$, $H^n(-)$ is a well-defined functor on $\mathcal{K}(R)$ (Proposition 5.1.27). Hence in $\mathcal{K}(R)$, $H^n(id_C) = H^n(0) = 0$, it follows that $H^n(C^\bullet) = 0$ for all $n \in \mathbb{Z}$ as required. □

Definition 5.1.29. Given a cochain map $f : C^\bullet \to D^\bullet$, we can define the **cone** of f, denoted as $C(f)$, as follows:

$$\cdots \longrightarrow C^0 \oplus D^{-1} \xrightarrow{d^{-1}_{C(f)}} \underline{\underline{C^1 \oplus D^0}} \xrightarrow{d^0_{C(f)}} C^2 \oplus D^1 \longrightarrow \cdots$$

Where the differentials at degree n are defined as $d^n_{C(f)}(c^{n+1}, d^n) = (-d^{n+1}_C(c^{n+1}), f^{n+1}(c^{n+1}) + d^n_D(d^n))$. We can write it as a matrix

$$d^n_{C(f)} = \begin{pmatrix} -d^{n+1}_C & 0 \\ f^{n+1} & d^n_D \end{pmatrix}$$

, acting on a column vector $(c^{n+1}, d^n) \in C^{n+1} \oplus D^n$. Now it is easy to check $d^{n+1}_{C(f)} \circ d^n_{C(f)} = 0$ (since $d^{n+1}_C \circ d^n_C = d^{n+1}_D \circ d^n_D = 0$ as $C^\bullet, D^\bullet$ are complexes), that is, $C(f)$ is indeed a cochain complex.

Notation

Note that in the formula

$$d^n_{C(f)}(c^{n+1}, d^n) = (-d^{n+1}_C(c^{n+1}), f^{n+1}(c^{n+1}) + d^n_D(d^n))$$

, we used lowercase English alphabet in upper indices to denote the elements. That is, $(c^{n+1}, d^n) \in C^{n+1} \oplus D^n$. We will continue to use this notation. In particular, the notation $d^n \in D^n$ will not be confused with d^n_D, the differential of the complex $D^\bullet$ at degree n.

Definition 5.1.30. Given a cochain map $f : C^\bullet \to D^\bullet$, the **mapping cylinder** $\mathrm{Cyl}(f)$ is the cochain complex:

$$\cdots \longrightarrow C^{-1} \oplus C^0 \oplus D^{-1} \xrightarrow{d^{-1}_{\mathrm{Cyl}(f)}} \underline{\underline{C^0 \oplus C^1 \oplus D^0}} \xrightarrow{d^0_{\mathrm{Cyl}(f)}} C^1 \oplus C^2 \oplus D^1 \longrightarrow \cdots$$

With differential at degree n given by the matrix:

$$d^n_{\mathrm{Cyl}(f)} = \begin{pmatrix} d^n_C & -1_{C^{n+1}} & 0 \\ 0 & -d^{n+1}_C & 0 \\ 0 & f^{n+1} & d^n_D \end{pmatrix}, \text{or } d^n_{\mathrm{Cyl}(f)}(c^n, c^{n+1}, d^n) = (d^n_C(c^n) - c^{n+1}, -d^{n+1}_C(c^{n+1}), f^{n+1}(c^{n+1}) + d^n_D(d^n)).$$

The cone and the cylinder of a cochain map are used in the following proposition, where we can get some useful exact sequences and commutating functions from a single cochain map f, which will be useful later.

Proposition 5.1.31. Given a cochain map $f : C^\bullet \to D^\bullet$, one always have two short exact sequence of complexes:

$$0 \longrightarrow D^\bullet \xrightarrow{f_1} C(f) \xrightarrow{g_1} C[1] \longrightarrow 0 \tag{5.4}$$

$$0 \longrightarrow C^\bullet \xrightarrow{f_2} \mathrm{Cyl}(f) \xrightarrow{g_2} C(f) \longrightarrow 0 \tag{5.5}$$

Where f_1, g_1, f_2, g_2 are cochain map, and at degree n they are given by:

$$f^n_1(d^n) = (0, d^n),\ g^n_1(c^{n+1}, d^n) = c^{n+1},\ f^n_2(c^n) = (c^n, 0, 0),\ g^n_2(c^n, c^{n+1}, d^n) = (c^{n+1}, d^n).$$

Moreover, they pack into a diagram

$$\begin{array}{ccccccccc} & & 0 & \longrightarrow & D^\bullet & \xrightarrow{f_1} & C(f) & \xrightarrow{f_2} & C^\bullet[1] & \longrightarrow & 0 \\ & & & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle id_{C(f)}} & & & & \\ 0 & \longrightarrow & C^\bullet & \xrightarrow{f_2} & \mathrm{Cyl}(f) & \xrightarrow{g_2} & C(f) & \longrightarrow & 0 & & \\ & & \downarrow{\scriptstyle id_{C^\bullet}} & & \beta\downarrow & & & & & & \\ & & C^\bullet & \xrightarrow{f} & D^\bullet & & & & & & \end{array} \tag{5.6}$$

With α, β are cochain maps defined at degree n by:

$$\alpha^n(d^n) = (0, 0, d^n), \beta^n(c^n, c^{n+1}, d^n) = f^n(c^n) + d^n$$

Such that (5.6) commutes. Moreover, $\beta\alpha = id_{D^\bullet}$ and $\alpha\beta \simeq id_{\mathrm{Cyl}(f)}$. In particular, this concludes that $D^\bullet$ and $\mathrm{Cyl}(f)$ are homotopically equivalent.

Proof. We list out all the things we have to check here:

- The maps $f_1, g_1, f_2, g_2, \alpha, \beta$ are indeed cochain maps.
- (5.4), (5.5) are indeed exact sequence of complexes.
- The diagram (5.6) commutes: that is, $g_2\alpha = f_1$ and $f = \beta f_2$.
- $\beta\alpha = id_{D^\bullet}$ and $\alpha\beta \simeq id_{\mathrm{Cyl}(f)}$.

Note that almost all checks are trivial by writing the definition of the 6 maps down, and one can read most of the details in [GM03, pp. 155 – 156]. Note that to check exactness, say for (5.4), one only needs to check f_1^n is injective, $\mathrm{im}\, f_1^n = \ker g_1^n$ and g_1^n is surjective for all n, which is obvious from the definition of the maps. To check $\alpha\beta \simeq id_{\mathrm{Cyl}(f)}$, we need to show there exists collection of maps $s^n : \mathrm{Cyl}(f)^n \to \mathrm{Cyl}(f)^{n-1}$ such that

$$\alpha^n\beta^n - id_{Cyl(f)} = s^{n+1}d^n_{\mathrm{Cyl}(f)} + d^{n-1}_{\mathrm{Cyl}(f)}s^n. \tag{5.7}$$

It turns out the correct definition is

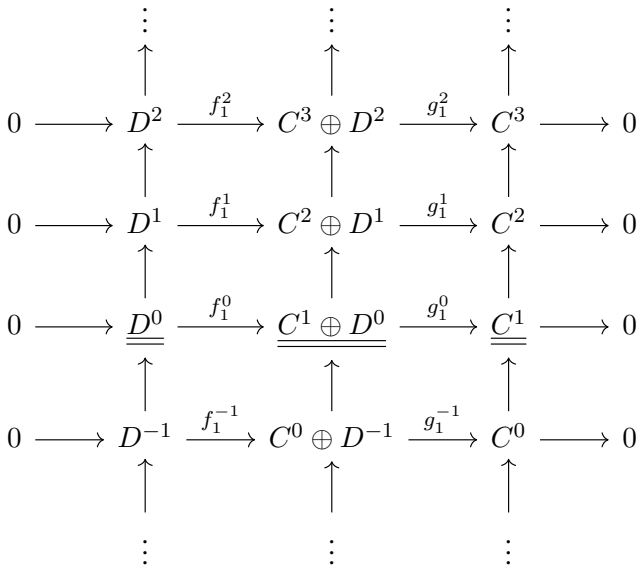
$$s^n : C^n \oplus C^{n+1} \oplus D^n \to C^{n-1} \oplus C^n \oplus D^{n-1}, \qquad (c^n, c^{n+1}, d^n) \mapsto (0, c^n, 0),$$

and one can check that (5.7) holds. □

5.2 Triangles in the Homotopy Category

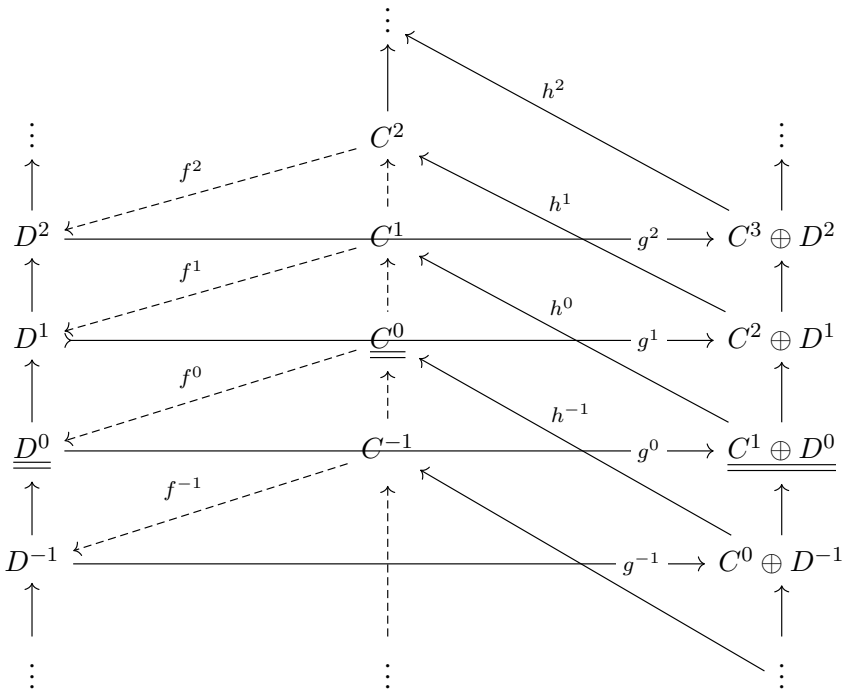
The short exact sequence (5.4) is so fundamental we present it pictorially. Recall that given a cochain map $f : C^\bullet \to D^\bullet$, we have a short exact sequence of cochain complex $0 \longrightarrow D \xrightarrow{f_1} C(f) \xrightarrow{g_1} C[1] \longrightarrow$

0:



Where the upward arrows are the differential of the complex $D, \mathrm{Cone}(f), C[1]$ respectively.

It might be surprising to note that, there is another way of drawing the same diagram which puts the map f, f_1, g_1 together. We put it in a bizarre triangular commutative prism in Figure 5.1.



The Complex D The Complex C The Complex $\mathrm{Cone}(f)$

Figure 5.1: A "triangle" arising from the short exact sequence (5.4).

A more compact way to present the picture is:

$$\begin{array}{ccccc} & & C & & \\ & \swarrow^{f} & & \nwarrow^{g_1} & \\ D & & \xrightarrow{\quad f_1 \quad} & & \mathrm{Cone}(f) \end{array} \tag{5.8}$$

Note that although drawn this way, g_1 is a cochain map from $\mathrm{Cone}(f) \to C[1]$, as defined in 5.4. We make this a definition. But instead of the category $\mathrm{Ch}(R\text{-}\mathbf{Mod})$, we work in the homotopy category $\mathcal{K}(R)$, for reasons that will be clear soon.

Definition 5.2.1. A **triangle** in $\mathcal{K}(R)$ is a ordered triple of equivalence classes of cochain maps $([f], [g], [h])$, represented by $f : X \to Y, g : Y \to Z, h : Z \to X[1]$. We will write a triangle as

$$\begin{array}{ccccc} & & X & & \\ & \swarrow^{f} & & \nwarrow^{h} & \\ Y & & \xrightarrow{\quad g \quad} & & Z \end{array}$$

or $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$.

Definition 5.2.2. Given two triangles

$$\begin{array}{ccccc} & & X & & \\ & \swarrow^{f} & & \nwarrow^{h} & \\ Y & & \xrightarrow{\quad g \quad} & & Z \end{array} \qquad \begin{array}{ccccc} & & X' & & \\ & \swarrow^{f'} & & \nwarrow^{h'} & \\ Y' & & \xrightarrow{\quad g' \quad} & & Z' \end{array}$$

, a **morphism of triangles** in $\mathcal{K}(R)$ between them is a triplet (α, β, γ) such that the diagram

$$\begin{array}{ccccccc} X & \xrightarrow{f} & Y & \xrightarrow{g} & Z & \xrightarrow{h} & X[1] \\ \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \downarrow{\scriptstyle \alpha[1]} \\ X' & \xrightarrow{f'} & Y' & \xrightarrow{g'} & Z' & \xrightarrow{h'} & X'[1] \end{array} \tag{5.9}$$

is commutative in $\mathcal{K}(R)$. We say two triangles are **isomorphic** if all α, β, γ are isomorphisms in $\mathcal{K}(R)$.

Note a diagram commutes in $\mathcal{K}(R)$, which means following the arrows from the same source to the same target gives homotopically equivalent maps. For example in (5.9), we have $\beta f \simeq f'\alpha$.

We give the triangle (and those isomorphic to it) in Figure 5.1 special attention.

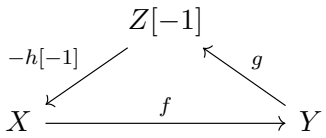
Definition 5.2.3. If a triangle in $\mathcal{K}(R)$ is isomorphic to the triangle (5.8), it is called a **distinguished triangle** in $\mathcal{K}(R)$.

Remark. Note that a distinguished triangle is determined up to isomorphism ◊

Proposition 5.2.4 (Rotation Axiom)**.** The triangle in $\mathcal{K}(R)$

$$\begin{array}{ccccc} & & X & & \\ & \swarrow^{f} & & \nwarrow^{h} & \\ Y & & \xrightarrow{\quad g \quad} & & Z \end{array}$$

is distinguished if and only if the triangle in $\mathcal{K}(R)$



is distinguished.

Proof. ($:\Rightarrow$) Assume we are given the (distinguished) triangle

$$X^\bullet \xrightarrow{f} Y^\bullet \xrightarrow{f_1} C(f) \xrightarrow{g_1} X[1]$$

, where f_1, g_1 are given in (5.6). We want to show $C(f)[-1] \xrightarrow{-g_1[-1]} X^\bullet \xrightarrow{f} Y^\bullet \xrightarrow{f_1} C(f)$ is isomorphic to a distinguished triangle. To show this, note that the sequence

$$C(f)[-1] \xrightarrow{-g_1[-1]} X^\bullet \xrightarrow{h_1} \mathrm{Cyl}(f) \xrightarrow{h_2} C(f)$$

where h_1, h_2 are given by

$$h_1(x^n) = (x^n, 0, 0), \quad h_2(x^n, x^{n+1}, y^n) = (x^{n+1}, y^n)$$

has the form of (5.4), since $\mathrm{Cyl}(f) = C(-g_1[-1])$. In particular, note that the first negative sign in $-g_1[-1]$ is required to show the differentials $d_{Cyl(f)} = d_{C(-g_1[-1])}$ are exactly equal. Hence

$$C(f)[-1] \xrightarrow{-g_1[-1]} X^\bullet \xrightarrow{h_1} \mathrm{Cyl}(f) \xrightarrow{h_2} C(f)$$

is a distinguished triangle. Now, we construct an isomorphism of triangles by claiming the diagram below commutes in $\mathcal{K}(R)$:

$$\begin{array}{ccccccc} C(f)[-1] & \xrightarrow{-g_1[-1]} & X^\bullet & \xrightarrow{h_1} & \mathrm{Cyl}(f) & \xrightarrow{h_2} & C(f) \\ \downarrow{\scriptstyle id_{C(f)[-1]}} & & \downarrow{\scriptstyle id_X} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle id_{C(f)}} \\ C(f)[-1] & \xrightarrow{-g_1[-1]} & X^\bullet & \xrightarrow{f} & Y^\bullet & \xrightarrow{f_1} & C(f) \end{array},$$

where β is given in (5.6), i.e. $\beta^n(x^n, x^{n+1}, y^n) = f^n(x^n) + y^n$. One can see the leftmost two squares commutes. We now claim that $f_1\beta \simeq h_2$, so the square commute in $\mathcal{K}(R)$.

We now show $f_1\beta - h_2 \simeq 0$, define $s^n : \mathrm{Cyl}(f)^n \to C(f)^{n-1}$ by

$$s^n : X^{n+1} \oplus Y^n \oplus X^n \to X^n \oplus Y^{n-1} \qquad (x^{n+1}, y^n, x^n) \mapsto (x^n, 0)$$

And one can compute that

$$\begin{aligned} & s^{n+1} d^n_{\mathrm{Cyl}(f)} + d^{n-1}_{C(f)} s^n(x^{n+1}, y^n, x^n) \\ &= s^{n+1}(f^{n+1}(x^{n+1}, -d^{n+1}_X(x^{n+1}), d^n_X(x^n) - x^{n+1}) + d^n_Y(y^n)) + d^{n-1}_{C(f)}(x^n, 0) \\ &= (d^n_X(x^n) - x^{n+1}, 0) + (-d^n_X(x^n), f^n(x^n)) = (-x^{n+1}, f^n(x^n)) \\ &= (0, f^n(x^n) + y^n) - (x^{n+1}, y^n) \\ &= f_1\beta - h_2(x^{n+1}, y^n, x^n) \end{aligned}$$

So $f_1\beta \simeq h_2$ as required. Now the identity maps and β are isomorphisms in $\mathcal{K}(R)$, so this is an isomorphism of triangles as required.

($\Leftarrow$:) Now suppose

$$\begin{array}{ccccc} & & Z[-1] & & \\ & \swarrow^{-h[-1]} & & \nwarrow_{g} & \\ X & & \xrightarrow{\quad f \quad} & & Y \end{array}$$

is distinguished. We can apply the implication we just proved twice. That is, one can conclude

$$\begin{array}{ccccc} & & Y[-1] & & \\ & \swarrow^{-g[-1]} & & \nwarrow_{f} & \\ Z[-1] & & \xrightarrow{\quad -h[-1] \quad} & & X \end{array}$$

is distinguished. Then conclude

$$\begin{array}{ccccc} & & X[-1] & & \\ & \swarrow^{-f[-1]} & & \nwarrow_{-h[-1]} & \\ Y[-1] & & \xrightarrow{\quad -g[-1] \quad} & & Z[-1] \end{array}$$

is distinguished. This implies

$$\begin{array}{ccccc} & & X & & \\ & \swarrow^{f} & & \nwarrow_{h} & \\ Y & & \xrightarrow{\quad g \quad} & & Z \end{array}$$

is distinguished as it can be seen as the triangle with all maps translated one degree up. □

Remark. Since one could not expect the map β to be an isomorphism of chain complexes, the rotation axiom 5.2.4 does not hold in $\mathrm{Ch}(R\text{-}\mathbf{Mod})$ in general. This is one of the reasons why we prefer to work over $\mathcal{K}(R)$ than $\mathrm{Ch}(R\text{-}\mathbf{Mod})$. ◊

We will be interested in full subcategories of $\mathcal{K}(R)$ that preserve the triangles.

Definition 5.2.5. A category $\mathscr{C}$ is a **triangulated subcategory** of $\mathcal{K}(R)$ if the following conditions holds true:

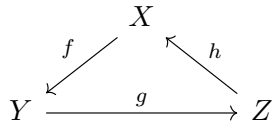
- $\mathscr{C}$ is a full subcategory of $\mathcal{K}(R)$.
- If X is complex of $\mathscr{C}$ then $X[n]$ also a complex of $\mathscr{C}$ for all $n \in \mathbb{Z}$.
- If

$$\begin{array}{ccccc} & & X & & \\ & \swarrow^{f} & & \nwarrow_{h} & \\ Y & & \xrightarrow{\quad g \quad} & & Z \end{array}$$

  is a distinguished triangle in $\mathcal{K}(R)$, and X, Y are objects in $\mathscr{C}$, then Z is also in $\mathscr{C}$.

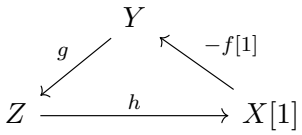
Due to the rotation axiom in $\mathcal{K}(R)$, one can immediately get a stronger condition than the last condition in Definition 5.2.5. This is sometimes used as a replacement, say in [Avr07, p.114].

Corollary 5.2.6. If



is a distinguished triangle in a triangulated subcategory $\mathscr{C}$ of $\mathcal{K}(R)$, then any two object or X, Y, Z in $\mathscr{C}$ means the remaining one is also in $\mathscr{C}$.

Proof. Assume $Y, Z \in \mathscr{C}$, apply the rotation axiom, then we have

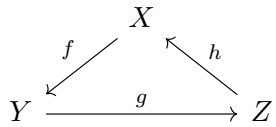


is a distinguished triangle. The condition for a triangulated subcategory for this distinguished triangle implies $X[1] \in \mathscr{C}$, so $X \in \mathscr{C}$. The proof is similar for assuming $X, Z \in \mathscr{C}$ and concluding $Y \in \mathscr{C}$ (by using the reverse implication of the rotation axiom). □

5.3 The Derived Category as a Localization

In this section, we introduce the definition of a derived category and deduce the main properties of its subcategory. The motivation to introduce triangles in Definition 5.2.2 turns out to be follows:

Proposition 5.3.1. If $0 \to X^\bullet \to Y^\bullet \to Z^\bullet \to 0$ is a short exact sequence of cochain complexes and $0 \to X^n \to Y^n \to Z^n \to 0$ is a split exact sequence of R-modules for all $n \in \mathbb{Z}$, then there is a cochain map $h : Z^\bullet \to X[1]$ such that



is a distinguished triangle in $\mathcal{K}(R)$.

We will not prove this, but we want to remove the assumption that the short exact sequence of complexes is term-wise split, it turns out the derived category allows us to do this (see Proposition 5.3.11). Recall in Definition 2.1.2 that the localization of a ring RS^{-1} of a *commutative* ring R has a universal property of "lifting" any functions that send S to units to a unique map in RS^{-1}. Is there another categorical way to generalize this? One answer is as follows: a ring R can be seen as a category on its own with elements as morphisms, as in Example 1.1.8 where the composition is the multiplication of elements. Now, a ring homomorphism is a functor. Hence the universal property that claims the existence of the category RS^{-1} can be restated as:

For any functor $F : R \to T$ such that elements of S (i.e. a set of morphisms in the category) are sent to isomorphisms T, there is a functor $F^! : RS^{-1} \to T$ such that $F = F^! \circ i$,

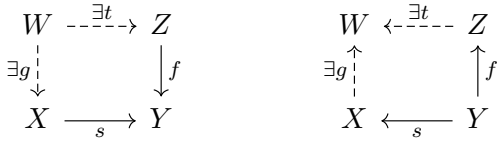
where i is the ring homomorphism (as functor) of the canonical map $R \to RS^{-1}$. The same process can be generalized to claim the existence of the *localization* arbitrary category, subject to S, now a set of *morphisms*, satisfying the respective multiplicative properties.

Definition 5.3.2. Fix a category $\mathscr{C}$. And S collection of morphisms in $\mathscr{C}$. We say S is **localizing** in $\mathscr{C}$ if the following three conditions is satisfied:

- For every $X \in \mathscr{C}$, the identity $1_X \in S$.

(MS1) S is closed under composition. That is, if $\alpha, \beta \in S$, then $\alpha\beta \in S$ whenever it make sense.

(MS2) (Ore condition.) For any morphism f, any morphism s in S, there exists morphism g and morphism t in S such that the following diagram commutes:



(MS3) (Cancellation Property.) Given $f, g : X \to Y$ are two morphisms, the following statement are equivalent:

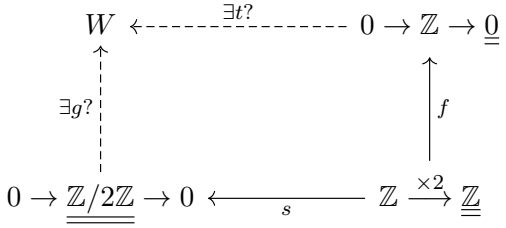
- There exists $s \in S$ such that $sf = sg$.
- There exists $t \in S$ such that $ft = gt$.

Example 5.3.3. If S is a multiplicative subset of a commutative integral domain R, seen as a collection of morphisms in a one-object category, then S is localizing in R. Indeed, (MS1) is simply the definition of a multiplicative subset; For (MS2), one can take $g = f, t = s$ for the diagrams above, as the ring is commutative, the diagrams commutes as well; (MS3) is obvious since R is commutative. △

This property also shows the convenience of working over $\mathcal{K}(R)$ than $\mathrm{Ch}(R\text{-}\mathbf{Mod})$.

Example 5.3.4. The collection of (homotopy classes of) quasi-isomorphisms is localizing in the homotopy category $\mathcal{K}(R)$. See [GM03, pp. 161–162] for a proof. △

Example 5.3.5. The collection of quasi-isomorphisms is *not* localizing in the homotopy category $\mathrm{Ch}(R\text{-}\mathbf{Mod})$. Consider the following diagram in $\mathrm{Ch}(R\text{-}\mathbf{Mod})$ with given f, s, with degree zero terms highlighted. We define f that sends $\mathbb{Z}$ to itself via the identity at degree -1 (and the zero map everywhere else). We also define s sends $\mathbb{Z}$ to $\mathbb{Z}/2\mathbb{Z}$ via the canonical map at degree 0 (and the zero map everywhere else).



s is a quasi-isomorphism as both complexes have zeroth cohomology group $\mathbb{Z}/2\mathbb{Z}$ and 0 for other cohomology groups. We claim there is no cochain map g and quasi-isomorphism t such that the above commutes, violating (MS2). If there are, note that gs must be the zero map at degrees other than 0, and tf must be the zero map for degrees other than -1. Since the diagram commutes, it follows that both gs, tf must be zero maps. In particular, since f is identity at degree -1, it follows that t is the zero map at degree -1, hence t is the zero map (because this $\mathbb{Z}$ is the only non-zero entry in the complex $0 \to \mathbb{Z} \to \underline{0}$). Hence W is isomorphic to the zero complex (0 at every term), but t is a quasi-isomorphism, so this implies $0 \to \mathbb{Z} \to \underline{0}$ is an exact complex, but this is not the case. △

The derived category of R-modules, $\mathcal{D}(R)$ is the localization of quasi-isomorphisms in $\mathcal{K}(R)$, which we now define.

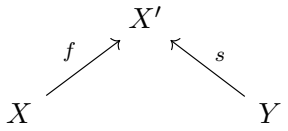
Definition 5.3.6. Given a category $\mathscr{C}$ and a collection of morphism S localizing in $\mathscr{C}$, the **localization category** $\mathscr{C}S^{-1}$ is the category satisfying the universal property: there is a faithful functor $i : \mathscr{C} \to \mathscr{C}S^{-1}$ such that

- $i(s)$ is an isomorphism for all $s \in S$, and
- For all functors $F : \mathscr{C} \to \mathscr{D}$ such that elements in S are sent to the isomorphisms in $\mathscr{D}$, there exists $F^! : \mathscr{C}S^{-1} \to \mathscr{D}$ such that $F^! \circ i = F$.

The **derived category** $\mathcal{D}(R)$ or a ring R is $\mathcal{K}(R)S^{-1}$, where S is the collection of all quasi-isomoprhisms.

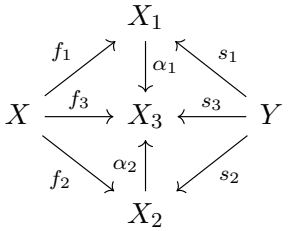
As with any definition using universal properties, we need to give a construction of the category $\mathscr{C}S^{-1}$ as below.

The object of $\mathscr{C}S^{-1}$ is just the object of $\mathscr{C}$. Given X, Y two objects in $\mathscr{C}$. The set $\mathrm{Hom}_{\mathscr{C}S^{-1}}(X, Y)$ is defined as follows: Consider all the triplets (X', f, s), where $f : X \to X', s : Y \to X'$, which is better represented by the diagram



(5.10)

, then $\mathrm{Hom}_{\mathscr{C}S^{-1}}(X, Y)$ is the collection of all such diagrams (i.e. triplets) modulo the following relation. We say $(X_1, f_1, s_1) \sim (X_2, f_2, s_2)$ if and only if there exists (X_3, f_3, s_3), and two maps α_1, α_2 such that

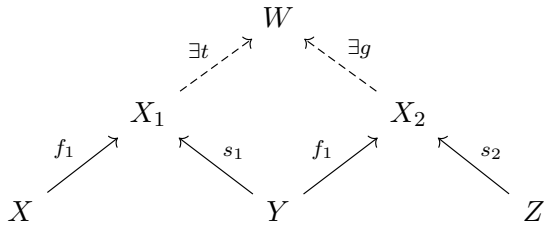


commutes. One might want to check that it is an equivalence relation. In particular, checking the fact that the relation is transitive requires the use of (MS2).

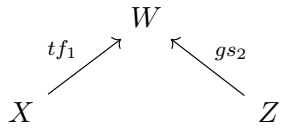
The composition of two maps is defined as follows: given two roofs, and pick a representative:



, one notice we can put them together and form the following diagram:

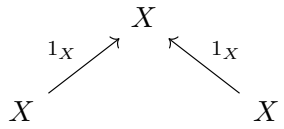


where t, g are found by (MS2). Then the composition of the two roofs is simply the roof



. One needs to check that this composition definition is well-defined. That is if we choose another representative of the two roofs, the composition will be equivalent to this one.

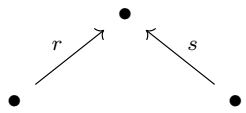
The identity morphism from an object X of $\mathscr{C}$ to itself is the equivalence class of



An element (representing the equivalence class) in $\mathrm{Hom}_{\mathscr{C}S^{-1}}(X, Y)$, as depicted in (5.10) is usually call a **roof** from X to Y. Intuitively, one should think of a roof as pairs (f, s) as "fraction" $s^{-1}f$.

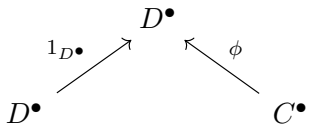
A check that $\mathscr{C}S^{-1}$ is indeed a category and satisfies the universal property can be found at [Jäd22, pp. 14–21]. Note that their definition has the arrows flipped, but the proof is similar.

Remark. In the case where $\mathscr{C}$ is a commutative integral domain with S a multiplicative set, then a roof



is exactly the (formal symbol) fraction $\dfrac{r}{s}$. One can go through the about definition and recover the addition and multiplication rule of the localization RS^{-1}. In fact, when R is not a commutative ring, the localization is defined in this abstract way. ◊

We now re-focus ourselves to $\mathcal{D}(R) = \mathcal{K}(R)S^{-1}$ where S are the collection of all quasi-isomorphisms between two cochain complexes. As motivated by the view of "fractions", given a quasi-isomorphism $\phi : C^\bullet \to D^\bullet$ between two cochain complexes, we denote ϕ^{-1} to be a morphism of $\mathrm{Hom}_{\mathcal{D}(R)}(D^\bullet, C^\bullet)$ defined by the representative class of



And one can check that $\phi\phi^{-1} = 1_{D^\bullet}, \phi^{-1}\phi = 1_{C^\bullet}$ when we compose in $\mathcal{D}(R)$ using the method described above. This agrees with the intuition that we can "invert" quasi-isomorphisms. In other words:

Proposition 5.3.7. Given $C^\bullet, D^\bullet$ two cochain complex, if there is a quasi-isomorphism $\Phi : C^\bullet \to D^\bullet$, then $C^\bullet, D^\bullet$ are isomorphic objects in $\mathcal{D}(R)$.

Now note that one can similarly define triangles and their morphisms on $\mathcal{D}(R)$ by essentially copying the definitions in Section 5.2.

Definition 5.3.8. A **triangle** in $\mathcal{D}(R)$ is an ordered triple of morphisms (f, g, h) in $\mathcal{D}(R)$, where $f : X \to Y, g : Y \to Z, h : Z \to X[1]$.

Note that (f, g, h) need not be cochain maps anymore, as the category $\mathcal{D}(R)$ has morphisms other than that.

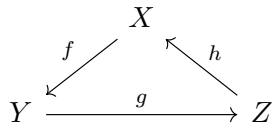
Definition 5.3.9. Given two triangles



in $\mathcal{D}(R)$, a **morphism of triangles** in $\mathcal{D}(R)$ between them is a triplet (α, β, γ) such that the diagram (5.9) is commutative in $\mathcal{D}(R)$. We say two triangles are **isomorphic** if all α, β, γ are isomorphisms in $\mathcal{D}(R)$. A triangle is called a **distinguished triangle** if it is isomorphic to the triangle (5.8).

One can then copy the definition to reach the same definition of a triangulated subcategory of $\mathcal{D}(R)$, and the rotation axiom also holds. We record it here for convenience.

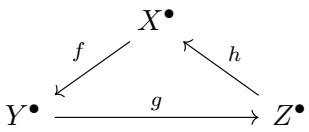
Corollary 5.3.10. If



is a distinguished triangle in a triangulated subcategory $\mathscr{C}$ of $\mathcal{D}(R)$, then any two object or X, Y, Z in $\mathscr{C}$ means the remaining one is also in $\mathscr{C}$.

As promised at the start of the section, we now prove:

Proposition 5.3.11. If $0 \to X^\bullet \xrightarrow{f} Y^\bullet \xrightarrow{g} Z^\bullet \to 0$ is a short exact sequence of cochain complexes, then there is $h : Z \to X[1]$ a morphism in $\mathcal{D}(R)$ such that



is (isomorphic to) a distinguished triangle in $\mathcal{D}(R)$.

Proof. We first claim there is a quasi-isomorphism $\gamma : C(f) \to Z^\bullet$. We define γ at degree n by

$$\gamma^n(x^{n+1}, y^n) := g^n(y^n).$$

We can verify γ is a cochain map as:

$$d_Z^n \gamma(x^{n+1}, y^n) = d_Z^n(g^n(y^n))$$

, and

$$\gamma d_{C(f)}^n(x_i, y_i) = \gamma(-d_X^{n+1}(x^{n+1}), f^{n+1}(x^{n+1}) + d_Y^n(y^n)) = g^n(f^{n+1}(x^{n+1}) + d_Y^n(y^n)).$$

By short exact sequence $gf = 0$, and by property of module homomorphism this evaluates to $g(d_Y^n(y^n)) = d_Z^n g^n(y^n)$ since g is a cochain map, so $d_Z^n \gamma = \gamma d_{C(f)}^n$ as required. Now note that the complex $\ker \gamma$ as a subcomplex of $C(f)$ is the complex formed by taking the kernel at each degree. In other words, at degree n, $\ker \gamma^n = X^{n+1} \oplus (\ker g^n) = X^{n+1} \oplus (\operatorname{im} f^n)$ by the exactness $0 \to X^\bullet \xrightarrow{f} Y^\bullet \xrightarrow{g} Z^\bullet \to 0$.

We first claim $\ker\gamma$ is an exact complex. To show this, we claim that $id_{\ker\gamma}$ is null-homotopic, hence by Corollary 5.1.28 it is exact. We define the maps $s^n : X^{n+1} \oplus (\operatorname{im} f^n) \to X^n \oplus (\operatorname{im} f^{n-1})$ by $s^n(x^{n+1}, f^n(x^n)) = (x^n, 0)$. One can then compute

$$\begin{aligned}
s^{n+1}d^n_{C(f)} + d^n_{C(f)}s^n(x^{n+1}, f^n(x^n)) &= s^{n+1}(-d^{n+1}_X(x^{n+1}), f^{n+1}(x^{n+1}) + d^n_Y(f^n(x^n))) + d^n_{C(f)}(x^n, 0) \\
\text{Since } f \text{ is a cochain map,} \quad &= s^{n+1}(-d^{n+1}_X(x^{n+1}), f^{n+1}(x^{n+1}) + f^n(d^n_X(x^n))) + (-d^n_X(x^n), f^n(x^n)) \\
&= s^{n+1}(-d^{n+1}_X(x^{n+1}), f^{n+1}(x^{n+1} + d^n_X(x^n))) + (-d^n_X(x^n), f^n(x^n)) \\
&= (x^{n+1} + d^n_X(x^n), 0) + (-d^n_X(x^n), f^n(x^n)) = (x^{n+1}, f^n(x^n))
\end{aligned}$$

So $s^{n+1}d^n_{C(f)} + d^n_{C(f)}s^n = id_{\ker\gamma}$ as required. Now, there is a short exact sequence of chain complexes

$$0 \to \ker\gamma \overset{inc}{\to} C(f) \overset{\gamma}{\to} Z^\bullet \to 0$$

where inc is the inclusion map from $\ker\gamma \to C(f)$ at every degree since $\ker\gamma$ is a subcomplex of $C(f)$. One can see $\operatorname{im}\gamma = Z^\bullet$ as g^n is surjective for all n by exactness, hence the sequence is indeed short exact. Now by the long exact sequence of cohomology, one gets an exact sequence

$$\cdots \to H^0(\ker\gamma) \to H^0(C(f)) \to H^0(Z^\bullet) \to H^1(\ker\gamma) \to H^1(C(f)) \to H^1(Z^\bullet) \to \cdots$$

But $\ker\gamma$ is exact, so the exact sequence is

$$\cdots \to 0 \to H^0(C(f)) \to H^0(Z^\bullet) \to 0 \to H^1(C(f)) \to H^1(Z^\bullet) \to \cdots$$

that continues indefinitely to both sides, by the exactness of the sequence we conclude $H^n(C(f)) \to H^n(Z^\bullet)$ for all $n \in \mathbb{Z}$, so γ is a quasi-isomorphism as required.

Now, given $f : X^\bullet \to Y^\bullet$, we know there is a distinguished triangle of the form $X^\bullet \overset{f}{\to} Y^\bullet \overset{f_1}{\to} C(f) \overset{g_1}{\to} X[1]$. We now create an isomorphism of triangles by declaring the required $h := g\gamma^{-1}$, (note that $\gamma^{-1} : Z^\bullet \to C(f)$ exists as a morphism in $\mathcal{D}(R)$. Then, one has the following commutative diagram:

$$\begin{array}{ccccccc}
X^\bullet & \xrightarrow{f} & Y^\bullet & \xrightarrow{f_1} & C(f) & \xrightarrow{g_1} & X[1] \\
\downarrow{\scriptstyle id_X} & & \downarrow{\scriptstyle id_Y} & & \downarrow{\scriptstyle \gamma} & & \downarrow{\scriptstyle id_{X[1]}} \\
X^\bullet & \xrightarrow{f} & Y^\bullet & \xrightarrow{g} & Z^\bullet & \xrightarrow{h:=g_1\gamma^{-1}} & X[1]
\end{array}$$

and γ is a quasi-isomorphism, so it is an isomorphism in $\mathcal{D}(R)$ (Proposition 5.3.7), so

$$\begin{array}{ccccc}
 & & X^\bullet & & \\
 & {\scriptstyle f}\swarrow & & \nwarrow{\scriptstyle h} & \\
Y^\bullet & & \xrightarrow{\quad g \quad} & & Z^\bullet
\end{array}$$

is distinguished as required. □

Corollary 5.3.12 (Two-out-of-three Property)**.** If $\mathscr{C}$ is a triangulated subcategory of $\mathcal{D}(R)$ and $0 \to X^\bullet \to Y^\bullet \to Z^\bullet \to 0$ is a short exact sequence of cochain complexes, then any two of $X^\bullet, Y^\bullet, Z^\bullet$ in $\mathscr{C}$ implies the remaining one is also in $\mathscr{C}$.

Proof. This follows from Corollary 5.3.10 and Proposition 5.3.11. □

Chapter 6

The "Injective Generation" Condition

6.1 The Algebra A

Fix a ring R. In this Chapter, we want to investigate the notion of a "generating set" in the category $\mathcal{D}(R)$. Firstly we note that $\mathcal{D}(R)$ is closed under arbitrary coproducts as the coproduct of cochain complexes in $\mathrm{Ch}(R\text{-}\mathbf{Mod})$ is formed by taking coproducts degree-wise.

Definition 6.1.1. Given a collection S of cochain complexes of R-modules, we define the category **generated** by S, denoted by $\langle \mathcal{S} \rangle$, as the smallest triangulated subcategory of $\mathcal{D}(R)$ that is closed under arbitrary coproduct and contains $\mathcal{S}$ as a subset of $\mathrm{Ob}(\langle \mathcal{S} \rangle)$. We also write $\mathcal{S}$ **generates** $\langle \mathcal{S} \rangle$.

The important special case is as follows:

Definition 6.1.2. Let S be the collection of all injective R-modules, considered as complexes (with each module placed on degree 0 with other terms being the zero module). We will denote the category generated by this collection as $\langle \mathrm{Inj}\text{-}R \rangle$. If $\langle \mathrm{Inj}\text{-}R \rangle = \mathcal{D}(R)$ we say **injective generates for** R.

This section is devoted to showing the injectives generate a particular algebra A. To show this condition, there are some key properties we shall use.

Proposition 6.1.3 ([Ric19], Lemma 6.1)**.** Given an algebra A, if every simple A-module is in $\langle \mathrm{Inj}\text{-}A \rangle$, then injective generates for A.

We will omit the proof here.

Proposition 6.1.4 ([Ric19], Proposition 2.1)**.** Consider the following bounded above exact complex of R-modules:

$$\cdots \longrightarrow C_n \cdots \longrightarrow C_{n-1} \longrightarrow \cdots \longrightarrow C_0 \longrightarrow X \longrightarrow 0 \longrightarrow 0 \longrightarrow \cdots \tag{6.1}$$

and suppose $C_n, C_{n-1} \cdots, C_0$ are all modules in $\langle \mathrm{Inj}\text{-}R \rangle$, then X is also in $\langle \mathrm{Inj}\text{-}R \rangle$.

Proof. Consider the cochain complex (with degree 0 underlined):

$$X^\bullet : \qquad \cdots \longrightarrow C_n \cdots \longrightarrow C_{n-1} \longrightarrow \cdots \longrightarrow \underline{C_0} \longrightarrow 0 \longrightarrow 0 \longrightarrow \cdots$$

and consider their brutal truncation $X_i := \sigma_{\geqslant -i}$ for each non-negative integer i. We claim that $X_i \in \langle \mathrm{Inj}\text{-}R \rangle$ for all i, and we prove this by induction. X_0 is just the module C_0, so the base case is checked. Now assume $X_k \in \langle \mathrm{Inj}\text{-}R \rangle$ for some non-negative integer k, now note that one has exact sequence

$$0 \longrightarrow X_k \longrightarrow X_{k+1} \longrightarrow C_k[k+1] \longrightarrow 0$$

. The cochain maps in the sequence are given by either identity or zero in each degree. See the figure below.

$$\begin{array}{lccccccccccc}
X_k & \cdots & \longrightarrow & 0 & \longrightarrow & C_k & \longrightarrow & C_{k-1} & \longrightarrow \cdots \longrightarrow & C_0 & \longrightarrow & 0 \longrightarrow \cdots \\
 & & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
X_{k+1} & \cdots & \longrightarrow & C_{k+1} & \longrightarrow & C_k & \longrightarrow & C_{k-1} & \longrightarrow \cdots \longrightarrow & C_0 & \longrightarrow & 0 \longrightarrow \cdots \\
 & & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
C_{k+1}[k+1] & \cdots & \longrightarrow & C_{k+1} & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow \cdots \longrightarrow & 0 & \longrightarrow & 0 \longrightarrow \cdots
\end{array}$$

The sequence is clearly exact. Since $C_k \in \langle \text{Inj-}R \rangle$ by assumption, hence $C_k[k+1] \in \langle \text{Inj-}R \rangle$ since $\langle \text{Inj-}R \rangle$ is a triangulated subcategory (see Definition 5.2.5). Now $X_k \in \langle \text{Inj-}R \rangle$ by assumption, so by the two-out-of-three property Corollary 5.3.12 $X_{k+1} \in \langle \text{Inj-}R \rangle$, as required. Now, note that $X^\bullet$ is the direct limit of $X_0 \to X_1 \to \cdots$ by Proposition 5.1.16. Write $X^\bullet = \varinjlim X_i$ and note that we have a short exact sequence (in Example 5.1.8)

$$0 \longrightarrow \bigoplus_{i=0}^{\infty} X_i \longrightarrow \bigoplus_{i=0}^{\infty} X_i \longrightarrow \varinjlim X_i \longrightarrow 0$$

. Since $\langle \text{Inj-}R \rangle$ is closed under coproduct and $X_i \in \langle \text{Inj-}R \rangle$ for all non-negative integer i, the first two terms of this short exact sequence are in $\langle \text{Inj-}R \rangle$, hence $X^\bullet \in \langle \text{Inj-}R \rangle$. Now we claim that there is a quasi-isomorphism from $X^\bullet \to X$ (where X is a complex with only non-zero term X at degree 0) as in the following figure:

$$\begin{array}{ccccccccccc}
\cdots & \longrightarrow & C_{k+1} & \longrightarrow & C_k & \longrightarrow & C_{k-1} & \longrightarrow \cdots \longrightarrow & \underline{\underline{C_0}} & \longrightarrow & 0 \longrightarrow \cdots \\
 & & \downarrow & & \downarrow & & \downarrow & & \downarrow \phi & & \downarrow \\
\cdots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow \cdots \longrightarrow & \underline{\underline{X}} & \longrightarrow & 0 \longrightarrow \cdots
\end{array}$$

where the blue arrows represent the morphisms in the exact complex (6.1). Since it is an exact complex, $\phi : C_0 \to X$ is surjective. Hence

$$H^n(X^\bullet) = \begin{cases} C_0/\ker\phi \cong \text{im}(\phi) \cong X & \text{if } n = 0 \\ 0 & \text{otherwise} \end{cases}$$

by first isomorphism theorem. Also, $H^n(X)$ is X when $n = 0$, and 0 otherwise. So it is a quasi-isomorphism as required. In $\mathcal{D}(R)$, any two objects with a quasi-isomorphism between them are isomorphic, hence $X \in \langle \text{Inj-}R \rangle$ as subcategories are closed under isomorphisms, as required. □

Now we present our example of the quiver algebra A and show that injective generates for this algebra. The algebra is taken from [GH96]. Note that in their paper, all modules are left modules. It turns out the injective module will be the *mirror image* of the projective module owning to this difference. The quiver Q we will consider is the quiver in figure 6.1 with 6 vertices. We define A to be kQ/I with I admissible. We present the indecomposable injective modules $I(1), I(2) \cdots I(6)$ (of the right regular A-module A_A) as a diagram in Figure 6.2. As discussed in Section 4.3, it specifies the admissible ideal I up to a collection of scalars in the field k (which will not be relevant to our purposes). Note that there are more cycles in Figure 6.2 than the example in Figure 4.2.3, so there might be more than one scalar which is not determined.

Now we show injective generates for A. By Proposition 6.1.4, we only need to show $S(1), \ldots S(6) \in \langle \text{Inj-}A \rangle$, where $S(1), \ldots, S(6)$ are the simple module introduced in 4.2.15. To start, we notice there are

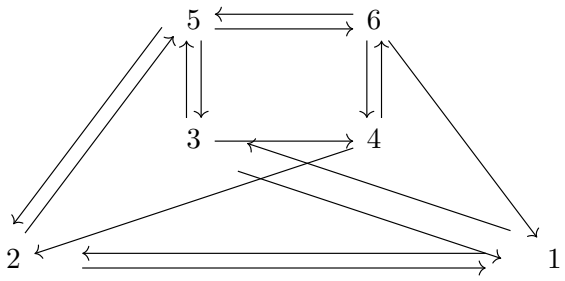


Figure 6.1: The Quiver Q.

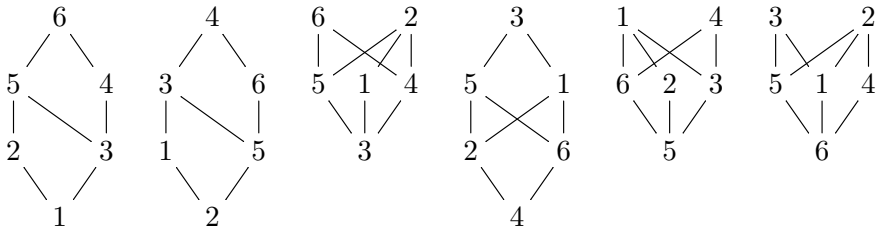


Figure 6.2: The six indecomposable injective modules $I(1), I(2), ...I(6)$ of A_A.

short exact sequences:

$$0 \to \begin{smallmatrix} 2 \\ | \\ 1 \end{smallmatrix} \to \begin{smallmatrix} & 6 & \\ 5 & & 4 \\ | & & | \\ 2 & & 3 \\ & 1 & \end{smallmatrix} \to \begin{smallmatrix} 6 & \\ & \\ 5 & 4 \\ & \\ & 3 \end{smallmatrix} \to 0 \tag{6.2}$$

$$0 \to \begin{smallmatrix} & 6 & \\ & & \\ 5 & & 4 \\ & & \\ & 3 & \end{smallmatrix} \to \begin{smallmatrix} 6 & & 2 \\ | & & | \\ 5 & 1 & 4 \\ & | & \\ & 3 & \end{smallmatrix} \to \begin{smallmatrix} 2 \\ | \\ 1 \end{smallmatrix} \to 0 \tag{6.3}$$

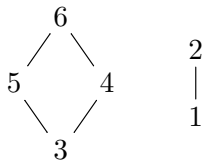
They are short exact since they are of the form of $0 \to N \to M \to M/N \to 0$ (see Example 5.1.7), as discussed in Section 4.3. Splicing copies of (6.2), (6.3) by Proposition 5.1.9, and recognizing the middle term in the two sequences is $I(1), I(3)$ we get exact sequences

$$\cdots \to I(3) \to I(1) \to I(3) \to I(1) \to \begin{smallmatrix} & 6 & \\ & & \\ 5 & & 4 \\ & & \\ & 3 & \end{smallmatrix} \to 0$$

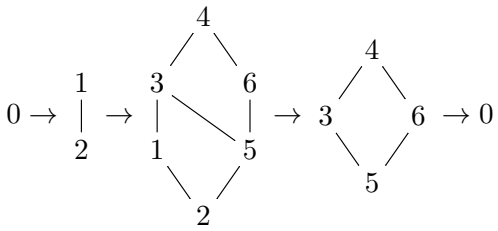
and

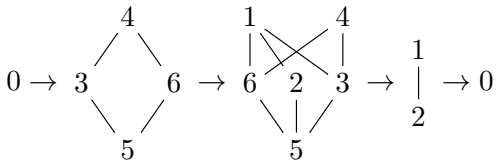
$$\cdots \to I(1) \to I(3) \to I(1) \to I(3) \to \begin{smallmatrix} 2 \\ | \\ 1 \end{smallmatrix} \to 0,$$

So by Proposition 6.1.4

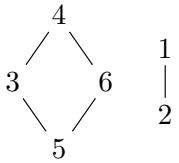


are in $\langle \mathrm{Inj}\text{-}A \rangle$. We discover more short exact sequences:

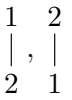




By splicing again, the middle term are $I(2), I(5)$:



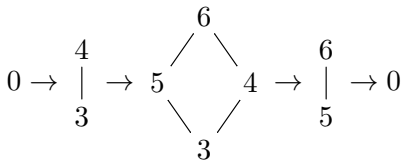
are in $\langle \mathrm{Inj}\, A \rangle$. This actually means $S(1), S(2)$ are in $\langle \mathrm{Inj}\, A \rangle$ since both

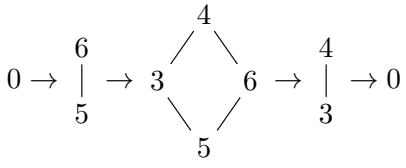


are in $\langle \mathrm{Inj}\text{-}A \rangle$ and one can do splicing on

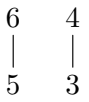
$$0 \to 1 \cong S(1) \to \begin{matrix} 2 \\ | \\ 1 \end{matrix} \to 2 \cong S(2) \to 0, \text{ and } 0 \to 2 \to \begin{matrix} 1 \\ | \\ 2 \end{matrix} \to 1 \to 0$$

. Now we proceed similarly by writing down short exact sequences and splicing.

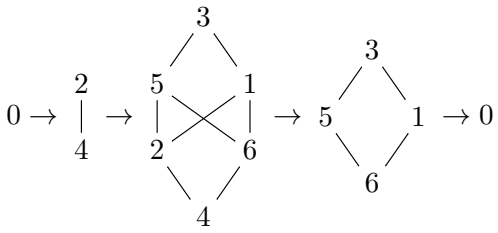


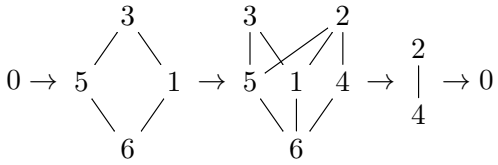


Yet again by splicing, since the middle terms are in $\langle \mathrm{Inj}\text{-}A \rangle$ by arguments before, the modules:

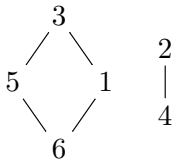


are in $\langle \mathrm{Inj}\, A \rangle$. More short exact sequences:





By splicing again, the middle term are $I(4), I(6)$, so the modules:



are in $\langle \mathrm{Inj}\text{-}A \rangle$. Now

$$0 \to S(4) \to \begin{array}{c} 2 \\ | \\ 4 \end{array} \to S(2) \to 0$$

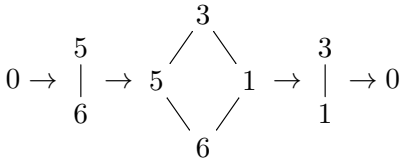
is a short exact sequence, and by the two-out-of-three property (Corollary 5.3.12) Gives $S(4) \in \langle \mathrm{Inj}\text{-}A \rangle$. Similarly, the short exact sequence

$$0 \to S(3) \to \begin{array}{c} 4 \\ | \\ 3 \end{array} \to S(4) \to 0$$

Gives $S(3) \in \langle \mathrm{Inj}\text{-}A \rangle$,

$$0 \to S(1) \to \begin{array}{c} 3 \\ | \\ 1 \end{array} \to S(3) \to 0$$

Gives $\begin{array}{c} 3 \\ | \\ 1 \end{array} \in \langle \mathrm{Inj}\text{-}A \rangle$, now



Gives $\begin{matrix}5\\|\\6\end{matrix} \in \langle \text{Inj-}A \rangle$, so splicing the modules

$$\begin{matrix}5\\|\\6\end{matrix} \quad \begin{matrix}6\\|\\5\end{matrix}$$

which are already in $\langle \text{Inj-}A \rangle$, gives $S(5), S(6) \in \langle \text{Inj-}A \rangle$. Now we are done since we show $S(1) \cdots S(6)$ are all in $\langle \text{Inj-}A \rangle$.

6.2 Conclusion and Further Directions

There are a few sections where common literature will present more general definitions than in this report. We will mention a few here.

In Chapter 4, we were given a quiver Q, an admissible ideal I and defined the quiver algebra $A = kQ/I$. Theorem 4.2.15 established that every A-module, as an abstract algebraic object, can be visualized by a quiver representation. This result (commonly attributed to Gabriel [Gab73]) highlights the importance of the quiver algebras:

Theorem 6.2.1. Let A be a finite-dimensional k-algebra, then there exists a quiver Q and an admissible ideal I such that the category kQ/I-$\mathbf{Mod}$ and A-$\mathbf{Mod}$ are equivalent.

So "categorical properties" such as indecomposable injective, projective, simple modules of any finite-dimensional algebra A can be understood through methods in Chapter 4.

In Chapter 5, we work with cochain complexes of R-modules. In fact, one can work with objects of a category $\mathcal{A}$ instead of (right) R-modules. However, there are some conditions $\mathcal{A}$ must satisfy. For instance, given a morphism f in $\mathcal{A}$, $\ker f$, $\operatorname{coker} f$ should exists. In other words, both the equalizer and coequalizer must exist in $\mathcal{A}$, which we defined abstractly as a limit in Definition 3.2.7. There are other conditions that $\mathcal{A}$ needs to satisfy, and it is referred to as an *abelian category*. One example would be the category of *sheaves* (of abelian groups) associated with a topological space X, which is under great interest of algebraic geometers. Various methods in homological algebra are used to investigate them. However, not much is lost as we have the following theorem by Mitchell [Mit64]:

Theorem 6.2.2. If $\mathcal{A}$ is an abelian category, then there is a ring R and a full, faithful, exact functor $F : A \to R$-$\mathbf{mod}$.

A functor is *exact* if $0 \to A \to B \to C \to 0$ is a short exact sequence implies $0 \to F(A) \to F(B) \to F(C) \to 0$ is a short exact sequence too.

In Chapter 5, we also showed that the homotopy category $\mathcal{K}(R)$ satisfied the rotation axiom 5.2.6. This, with the other three axioms led to a definition of a *triangulated category*, which is introduced in [GM03, Chapter IV]. This is a type of category where we mainly work with triangles in Definition 5.2.2. As hinted in Proposition 5.3.11, researchers found that working with triangles in some cases would be more convenient than working with short exact sequences. An introduction to triangulated categories can be found at [Nee01].

We focused on the injective generation condition in Chapter 6. It has the following implication proved by Rickard.

Theorem 6.2.3 ([Ric19])**.** Let A be a finite-dimensional algebra (over an algebraically complete field k). If injective generates for A, then the Finitistic Dimension Conjecture holds for A.

The Finitistic Dimension Conjecture states all finite-dimensional algebras A satisfy a particular property and is a longstanding problem in representation theory. See [Hui95] for a short exposition of the subject. Theorem 6.2.3 allows us to prove the Finitistic Dimension Conjecture for some algebras where it is easy to see injective generates them. Although it may be harder to deduce from the definition that the conjecture holds for this algebra.

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